Omnisoot: a process design package for gas-phase synthesis of carbonaceous nanoparticles

by

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I would like to dedicate this thesis to my loving parents …

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements. This dissertation contains fewer than 65,000 words including appendices, bibliography, footnotes, tables and equations and has fewer than 150 figures.

Mohammad Adib
2026

Acknowledgements

And I would like to acknowledge ...

Abstract

The formation of carbonaceous nanoparticles such as soot and Carbon Black (CB) is a complex phenomenon that involves heat transfer, fluid dynamics, chemical kinetics, and particle dynamics. Understanding the effect of process parameters on the morphology, internal structure, and composition of carbonaceous nanoparticles is essential for assessing the environmental impact of soot as a major air pollutant and for producing specific grades of CB, a key reinforcing ingredient in rubber and tire industries. However, the wide span of time and length scales involved in the process, together with strong temperature and pressure sensitivities, complicates accurate prediction of CB and soot properties.

This project is aimed at developing a robust computational package, titled *Omnisoot*, to describe the formation of carbonaceous nanoparticles, such as soot and CB, from the reactions of gaseous hydrocarbons. Omnisoot extends the chemical-kinetics capabilities of Cantera for gas-phase chemistry to the simulation of solid particles based on well-justified assumptions. It integrates constant volume, imposed pressure, perfectly stirred, and plug flow reactor models with three widely used inception models, as well as a new inception model based on E-Bridge bond formation. The particle dynamics are described using two population balance models, a monodisperse model and a fixed sectional model, that account for the evolving fractal-like structure of soot agglomerates. The implementation captures soot inception, surface growth, and oxidation coupled with detailed gas-phase chemistry to close the mass and energy balances of the gas–particle system. A step-by-step validation approach was followed to ensure the reliability of all sub-models, enabling the package to serve as an integrated process design tool to predict soot mass, morphology, and composition under varying process conditions.

The developed tool was then used to investigate the temperature dependence of the implemented inception models and its impact on predicted soot mass, morphology, and size distribution for various use-cases. The importance of combined soot yield and morphology data as calibration targets was emphasized for constraining the inception flux and surface growth rate. Special attention was paid to methane pyrolysis because of its practical significance in plasma reactors for co-generation of hydrogen and CB.

Shock-tube pyrolysis of methane was studied at high fuel loadings in post-reflected-shock temperature and pressure ranges relevant to CB synthesis. Detailed comparison of model predictions with time-resolved species mole fractions and soot volume fraction (f_v) recorded in collaboration with Stanford University enabled evaluation of several kinetic mechanisms and highlighted the major carbon flux pathways from small hydrocarbons to soot precursors. The impact of fuel loading on gas chemistry and soot formation was interpreted in terms of the endothermicity of pyrolysis. Omnisoot successfully predicted the time history of f_v and the final primary particle diameter (d_p) in agreement with measurements. The different stages of soot formation were identified by analyzing the time histories of predicted f_v, inception flux, and surface growth rates. The evolution of the hydrogen-to-carbon ratio (H/C) predicted using Omnisoot showed strong correspondence with the maturity index (β) inferred from extinction ratios at multiple wavelengths, highlighting its potential as a quantitative, time-resolved probe of soot maturity.

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Nomenclature

Acronyms

CFD Computational Fluid Dynamics

DEM Discrete Element Modelling

HACA H-abstraction-C_2H_2/Carbon-addition

MPBM Monodisperse Population Balance Model

SPBM Sectional Population Balance Model

Constants

Av	Avogadro's number	$6.02214076 \times 10^{23}$ 1/mol
k_B	Boltzmann constant	$1.3806488 \times 10^{-23}$ J/K
R	Global gas constant	8.314 J/mol

English symbols

A	Surface area	m^2
c_p	Heat capacity at constant pressure	J/kgK
c_v	Heat capacity at constant volume	J/kgK
c_{soot}	Heat capacity of soot	J/kgK
C_{tot}	Total carbon content of soot particles (per section)	mol/kg
D	Diffusion coefficient of particles	m^2/s

d	Diameter	m
d_c	Collision diameter	m
d_g	Gyration diameter	m
D_H	Duct hydraulic diameter	m
d_m	Mobility diameter	m
d_p	Primary particle diameter	m
e	Internal energy	J/kg
$E(m)$	Absorption function of soot	–
f	Friction factor	–
f_v	Soot volume fraction based on gas volume	m^3/m^3
h	Entalpy	J/kg
H_{tot}	Total hydrogen content of soot particles (per section)	mol/kg
I	Partial source terms for soot variables	$mol/(kg \cdot s)$
k	Reaction rate constant	$m^3/(mol \cdot s)$
k_{dep}	Deposition velocity	m/s
Kn	Knudsen number	–
m	Mass	kg
n_p	Number of primary particles of per agglomerate	
N_{agg}	Number density of agglomerates	mol/kg
N_{pri}	Number density of primary particles	mol/kg
p	Pressure	Pa
P_c	Wetted perimeter of the flow reactor	m
R_H	Duct hydraulic radius	m
Re	Reynolds number	–

S	Source terms for soot variables	$\mathrm{mol}/(\mathrm{kg} \cdot \mathrm{s})$
Sc	Schmidt number	–
SF	Sectional spacing factor	–
Sh	Sherwood number	–
T	Temperature	K
u	Velocity	m/s
V	Volume	m^3
W	Molecular weight	kg/mol
Y	Mass fraction of gaseous species	–

Greek symbols

β	Collision frequency	m^3/s
δ_a	Mean distance of particles	m
ϵ	Wall roughness	m
η_{ads}	PAH adsorption adjustment factor	–
η_{inc}	Inception adjustment factor	–
λ	Mean free path	m
λ_a	Mean stopping distance of particles	m
μ	Gas dynamic viscosity	$\mathrm{Pa} \cdot \mathrm{s}$
ω	The rate of production of gaseous species	$\mathrm{mol}/\mathrm{m}^3\mathrm{s}$
ρ	Density	kg/m^3
τ_w	Wall shear stress	Pa
τ_{psr}	Nominal residence time of perfectly stirred reactor	Pa
φ	Soot volume fraction based on reactor volume	kg/m^3

ζ	Collision efficiency	–

Subscripts

ads	Adsorption
agg	Agglomerate
coag	Coagulation
cont	Continuum
f	Forward
fm	Free molecular
grow	Surface growth
inc	Inception
ox	Oxidation
pri	Primary particle
r	Reverse
reac	Reactive

Chapter 1

Introduction

1.1 Motivation

Carbonaceous nanoparticles, such as Carbon Black (CB) and soot, are widely encountered in nature and engineering. Every year, nearly 9.5 megatons of soot (black carbon) is emitted into the atmosphere from anthropogenic activities and natrual sources such as wildfires and volcanoes [1]. The wide light absorption range of soot reduces the albedo of snow-covered area and alters the radiative forcing balance in the atmosphere making soot the third strongest contributor to climate change after methane and carbon dioxide [1]. Also, exposure to combustion-generated soot could promote respiratory and cardiovascular diseases [2]. So, strict regulations are targeting combustion engines to limit environmental and health risks from soot formation [3]. Accurate and affordable models are essential for the prediction of soot composition and morphology in combustion devices and the reduction of emissions in engines. They can also provide better understanding of soot optical properties that are major indicator of soot environmental effects.

On the other hand, CB with a similar synthesis process and structure to soot but higher elemental carbon to hydrogen ratios (>97%) [4] is commercially produced and sold in large scales. In fact, CB is the largest industrially produced nanomaterial by value and volume (∼15 megatons per year with a value of $17B) with applications as a reinforcing agent in rubber and tire industries [5], and conductive additive in lithium-ion batteries [6]. CB is primarily manufactured by the so-called furnace process in which about 50% of heavy fuel oil is partially combusted to convert the rest of it into CB [7]. This process suffers from low mass yield and excessive emission, generating 4 tons of CO_2 per each ton of product on average [8]. Plasma reactor is an emerging alternative production method with distinct advantages over flame-based methods: They can achieve 100% carbon yields with no direct CO_2 emission or other pollutants [9], and the energy required for pyrolysis is supplied by an electric arc that does not depend on the feedstock composition. Controlling CB properties such as its specific surface area (or primary particle diameter), hard agglomerate size (or gyration diameter of agglomerates with primary particles connected to each other by strong chemical bonds), and composition (or particle carbon to hydrogen ratio) is important to make the process economical and to achieve specific grades of CB for different target applications. However, this is a challenging task because of the complexity of CB formation and mass growth processes and its coupling with gas phase chemistry, dependence on local temperature and pressure. This requires accurate process design and optimization tools built on physics of CB formation and evolution to inform manufacturers' decisions for production of CB with desired grades [10].

The term *"soot"* usually refers to the unwanted particulate matter formed during incomplete combustion of any carbon-containing material from jet and diesel fuel to wood, heavy oil, and plastics with variable organic content and a large variation in C/H ratio [4], but this research focuses on soot particles generated under controlled laboratory conditions from fuels with known compositions. The mature soot formed in methane and ethylene premixed flame can reach 95% elemental C/H ratio [11], which is close to CB composition. The comparison of transmission electron microscopy (TEM) images of industrially produced CB [12] with soot sampled from diesel fuel [13, 14] shown in Fig.1.1 indicates similarity of their morphology and structure. Hereafter, soot will be used to collectively refer to carbonaceous nanoparticles produced in flame/reactor during combustion/pyrolysis processes.

1.2 Background

1.2.1 Soot inception and surface growth

Soot formation involves concurrent processes with different time and length scales starting from the breakdown of hydrocarbon molecules to formation of intermediate species and radicals that transition into soot particles and grow via heterogeneous reactions on the surface as well as coagulation [15]. The TEM analysis of soot sampled from flames [14], reactors [16], and engines [17] with different fuels and process conditions revealed a common fractal-like morphology, often characterized as agglomerates of primary particles. High resolution transmission electron microscopy (HRTEM) of soot primary particles showed clusters of precondensed Polycyclic Aromatic

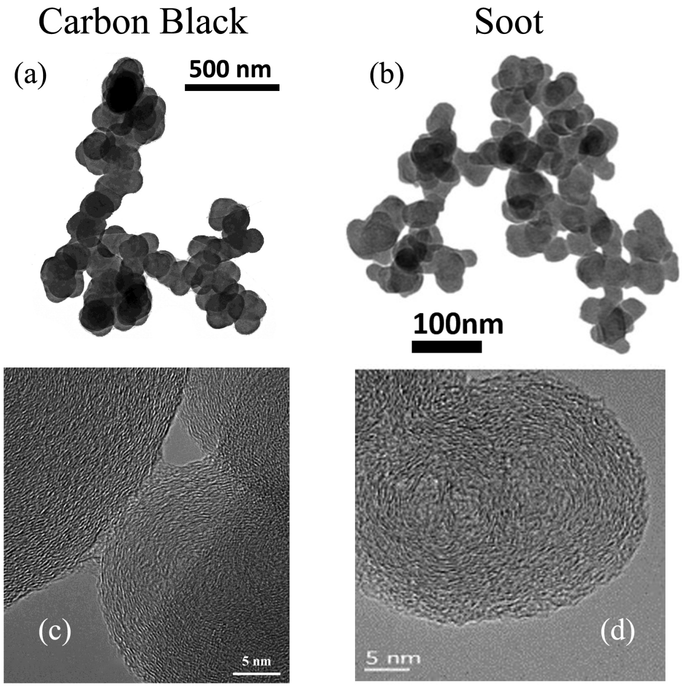


Fig. 1.1 The TEM images of Carbon black (a & c) [12] and soot (b & d) [13, 14] that shows soot and CB has similar morphology and structure

Hydrocarbons (PAHs) [18] pointing to PAHs as main soot precursors. This hypothesis was also supported by thermodynamic stability of PAHs enabling them to resist dissociation at high flame temperatures [19].

However, the transition of PAHs (soot precursors) to incipient soot, known as *soot inception* has not been well understood at the level of pathways and elementary reactions [20] primarily due to uncertainties in PAH (precursor) chemistry. The formation of carbonaceous particles is kinetically controlled and known to be an entropy-driven process [20]. Wang [20] uses the pyrolysis of propane at 1600 K to highlight how thermodynamics of soot formation is distinct from synthesis of inorganic nanomaterials. As shown in Fig. 1.2, acetylene is highly endothermic, but it is accompanied with drastic increase in entropy due to hydrogen release resulting in overall decrease of Gibbs free energy, which explains acetylene (C_2H_2) being the dominant hydrocarbon species. The growth of PAHs from acetylene does not undergo a significant enthalpy release or entropy increase, so the path is driven by a gradual reduction in Gibbs free energy. As a result, the kinetics of PAH growth into subsequent soot particles can be highly reversible hence sensitive to local temperature, pressure and intermediate species concentration.

The growth of PAHs beyond first ring (benzene) is predominantly driven by so-called hydrogen abstraction carbon (acetylene) addition (HACA) mechanism [21] where a hydrogen radical abstracts an hydrogen atom at the edge of PAH, providing a reactive site for acetylene addition. The kinetic reversibility of HACA opens the door for competing pathways such as chain reactions of resonance-stabilized radical (RSR). Propargyl is a prominent example of these radicals, whose combination is known as a major contributor to benzene formation [22]. Built on this hypothesis, Johansson et al. [23] proposed a radical-driven growth mechanism by addition of vinyl (C_2H_3) starting from cyclopentadienyl (C_5H_5) to larger hydrocarbon radicals that can survive long enough in high temperatures to react with other radicals, PAHs and unsaturated aliphatic species, through radical chain reactions. However, low concentrations of these radicals limit the growth rate through RSR pathways [24]. Moreover, some of the intermediate steps for radical regeneration such as formation of vinylcyclopentadienyl were shown to be kinetically unfavorable [24] compared to HACA. Regardless of their mechanistics, these sequential growth mechanisms, termed as *chemical growth* cannot account for rapid soot formation [21] and its nanostructure [25].

There is abundant but mostly indirect experimental and computational evidence pointing to an alternative collision-based mechanism for soot inception. HRTEM images of nascent and mature soot shows disordered PAH clusters highlighting the role of PAH collisions. The bimodality of particle size distribution (PSD) of nascent soot particles in premixed flames [26] indicates that the kinetics of inception is second order in precursor concentration [27]. The time-of-flight mass spectrometry (TOFMS) experiments in a 13 kPa acetylene-oxygen flame showed a series of peaks with a periodicity of 500 amu [28].

It is not clear how PAH clusters form and what forces allow the binding occur and resist dissociation at flames temperatures ($T \geq 1600$ K). Frenklach [21] described clustering of PAHs as an irreversible process where two PAHs collide and stick upon collision forming dimers held together by Van der Waals (vdW). Herdman and Miller [29] found that the binding energy of PAHs dimers due to dispersive and electorstatic forces increases linearly with molecular mass and reaches the limit of exfoliation energy for graphite. However, the entropy barrier of dimerization increases with PAH size making them unfavorable under equilibrium conditions. So, PAHs as large as circumcoronene ($C_{54}H_{18}$) can only form dimer to survive flame temperatures [20]. However, the concentration of large PAHs are too low to account for the observed number density of soot particles in flames [30].

There is also the possibility of PAH clustering governed by non-equilibrium kinetics. PAH molecules can form rovibrationally excited dimer with long enough life-time [31] to react with H atoms forming covalent bonds. There is another possibility for these clusters to be joined by aliphatic linkages. Micro-FT-IR spectroscopy analysis by Cain et al. [32] showed the ratio of aliphatic-to-aromatic C–H bonds can exceed unity at the flame temperature. Using these findings, they suggested that the aliphatic components in the form of alkyl and alkenyl can covalently bound to aromatic units in soot particles. Although such a mechanism is viable near the flame region, it cannot explain persistent soot inception in the post-flame zone [33] where the H atom concentration is too low to initiate H abstraction reactions that produce those radicals.

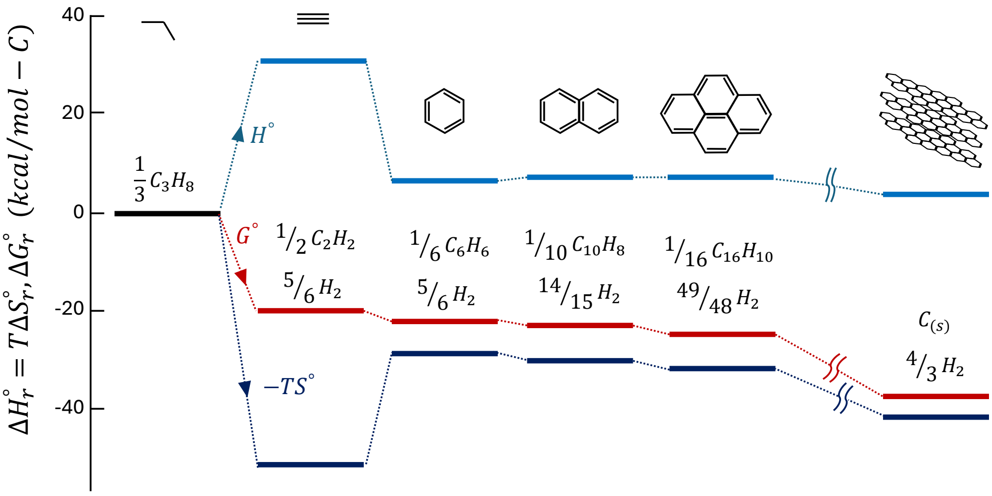


Fig. 1.2 Standard enthalpy (ΔH) and entropy ($T\Delta S$) contributions to Gibbs function of reaction ΔG at 1600 K for carbon formation from propane (reprinted from Ref. [20])

There are other complicating factors that hinders the fundamental understanding of soot inception such as lack of a decisive criterion to distinguish gaseous molecules from particles [15], and the overlap of inception with surface growth and agglomeration [34]. Measurement techniques have limited capability in soot detection due to short time and length scales of soot inception and surface growth ranging from pico- to milliseconds and micro- to millimeter [35]. Fig. 1.3 demonstrates the length and time scales relevant to different stages of soot formation from PAH precursors to incipient, nascent and mature soot in flames. Moreover, the collected data from measurements might not be true representatives of soot formed at the probed location of the studied process. For example, intrusive diagnostic methods based on thermophoresis or dilution have a sampling probe that can perturb flow dynamics or alter structure and composition of soot [36]. Non-intrusive techniques such as optical methods also rely on assumptions about light absorption and scattering of soot [37] and its morphology [38].

Despite the gaps in fundamental understanding of soot formation and limitations of diagnostics methods, models have been developed describe the soot inception and growth. These models have been formulated as a set of clear pathways that explain soot inception based on collisions of PAH molecules. They have to be consistent with current knowledge of soot inception, feasible to be coupled with chemistry and particle dynamics models, and able to predict soot mass, PSD and morphology observed in flames and reactors.

The classic description of soot inception relies on PAH dimerization where collision of two PAH molecules (monomers in this context) forms a dimer held together by Van der Wales forces [39]. The dimerization is a irreversible process with an efficiency that accounts for the reversibility or dissociation of dimers. The theory postulates that PAH growth continues by sequential addition of a monomer (PAH molecule) forming stacks of dimers, trimers, tetramers and so on to reach a certain mass threshold that marks the emergence of incipient soot [39], but for practical purposes, a dimer is usually considered as an incipient soot. Here, we call this model *Irreversible Dimerization*. Irreversible Dimerization has been used to predict soot formation in burner-stabilized premixed [40, 41], counterflow diffusion flames [42, 43], coflow diffusion flames [44, 45]. A collision efficiency factor ranging between 10^{-6} to 1 is also employed to adjust the inception flux and PAH adsorption rates to achieve desired soot mass and size distribution. PAHs of moderate sizes such as pyrene (4 rings) to coronene (7 rings) have been considered as the starting point of inception due to their thermodynamic instability that justifies the irreversibility at high temperatures [39]. However, the theoretical calculations [46] and experiments [47] indicated that PAH dimerization is highly reversible in flame conditions.

The inception flux of irreversible dimerization is determined from collision of PAHs, which is proportional to $\sqrt{T}$, so new particles can be formed at low temperatures (even less than 500 K) [48] despite experimental evidence for termination of inception below 1200 K [49, 50]. Moreover, the arbitrary selection of efficiency factors alters the distribution of mass between inception of surface growth could significantly change soot mass, PSD, and morphology [51]. Miller [52] used equilibrium constant for PAH dimerization to calculate the net dimerization rate and demonstrated that the collision of PAHs larger than circumovalene (~800 amu) could last long enough grow into incipient soot. However, the concentration of PAHs drops rapidly with size [20]. The entropy barrier of dimerization is significant for larger PAHs [53].

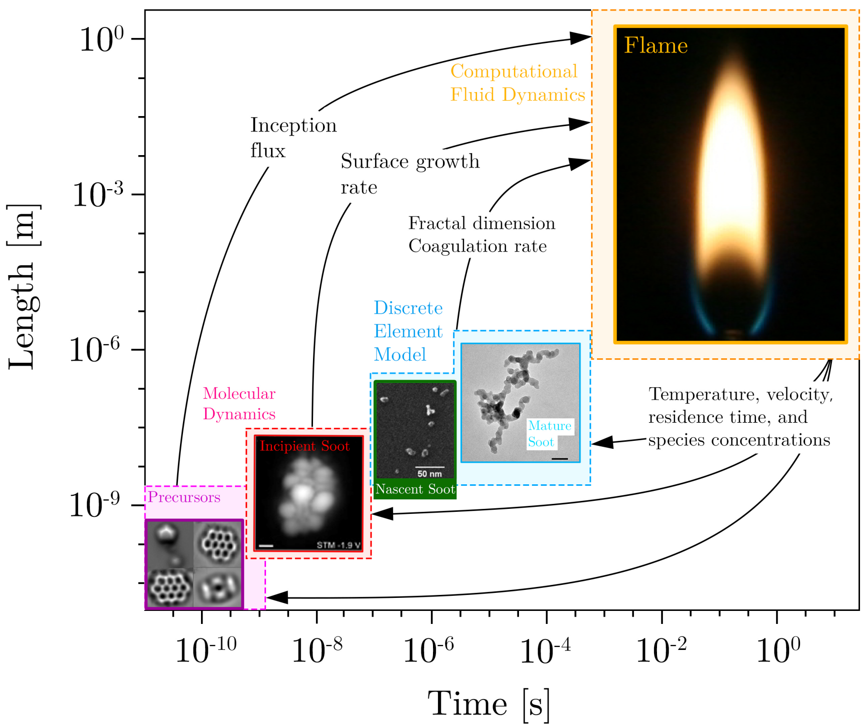


Fig. 1.3 The range of time and length scales of the processes involved in soot formation from molecular reactions to particle-fluid interaction in flames

Eaves et al. [54] relaxed the irreversibility assumption, and developed a reversible clustering model to simulate inception using an array of PAHs from naphthalene to benzo-pyrene. Building on that work, Kholghy et al. [55] emphasized on the necessity of chemical bond formation after physical PAH clustering for accurate prediction of volume fraction, primary particle diameter and PSD in ethylene coflow diffusion flames. Later, Kholghy et al. [56] proposed the *"Reactive Dimerization"* model which starts with reversible collision of PAHs leading to physical dimers held with vdW forces that are graphitized and form chemically-bonded dimers that serve as soot nuclei grow via surface reactions. They also performed a systematic analysis on the contribution of different PAHs, and concluded that one- and two-ring aromatics account for almost all of inception flux in the so-called *"sooting flame"* [41]. However, Frenklach and Mebel [24] pointed out that an inception model that initiated with a highly reversible step similar to Reactive Dimerization [56] cannot produce sufficient flux of particles to match measurements of the benchmark burner-stabilized stagnation flame [27]. Instead, they proposed a HACA-driven mechanism where addition of monomer molecule to its radical activated by hydrogen abstraction for a stable dimer via an E-Bridge bond formation, and this sequential process continues to form trimers, tetramers, and larger PAH clusters.

The gas-phase chemistry of aromatics can be extended to account for chemical growth of incipient soot via surface reactions [21]. This hypothesis, known as "chemical similarity" postulates that the reactions occurring on the soot surface are similar to those involving large molecules of PAHs in the gas phase. It also provides means to describe the rates of surface growth and particle oxidation in terms of elementary chemical reactions. In other words, it is assumed that the surface of soot particles is made up of lateral faces of larges PAHs covered with C-H bonds. This is the basis for HACA mechanism [39, 57] that assumes the soot surface to consist of hydrogenated sites with a predefined density. Mass growth on soot surface requires H-abstraction to form a radical site, followed by acetylene attack similar to growth of PAH molecules in the gas-phase. The reactivity of these sites changes with time and temperature [58, 59], described as soot aging. For modelling purposes, a temperature-dependent multiplier, usually represented by α, was introduced to account for these effects. Appel et al. [57] showed α changes with temperature and particle size.

Adsorption of PAHs on the surface of soot particles is also a viable growth mechanism [39], more specifically called physiorption or chemisorption depending on the mechanisms driving the adsorption process [60]. There is still debate over the stability of adsorbed PAH molecules on soot surface [61]. Following the hypothesis that PAHs are building blocks of soot particles, a mechanism similar to inception is often used to describe PAH-soot growth.

1.3 Coagulation and agglomeration

In typical soot formation processes such as those occurring in flames and reactor, soot particles are formed at high number concentrations (10^{12} 1/cm^3), and inception and surface growth are relatively short compared to the total residence time of soot particles. As a result, coagulation becomes dominant rapidly attaining both [62] self-preserving size distribution (SPSD) [63] and asymptotic fractal-like structure [64]. The evolving fractal-like structure of agglomerates quantified by their mobility diameter normalized by primary particle, d_m/d_p, and gyration, d_m/d_g, diameters can be described with power laws derived from mesoscale simulations [65]. The collision frequency of agglomerates depends on their evolving fractal-like morphology. Also, polydisperse agglomerates collide more frequently than monodisperse ones. The enhancement in their collision frequency reaches an asymptotic value of 35% [62] or 82% [66] in the free molecular or transition regimes, respectively at SPSD regardless of the polydispersity in their constituent primary particles. Particle morphology formed by inception, surface growth and agglomeration can be tracked precisely by mesoscale simulations, such as Discrete Element Modeling (DEM) [67] where each agglomerate is tacked separately. However, they are computationally expensive and interfacing them with chemical kinetics in computational fluid dynamics (CFD) is not trivial [68]. This limits their application in simulation of realistic flames and reactors. So, sectional population balance models (SPBM) are often used to track agglomerate and primary particle size distribution [69], morphology [70], and composition [44] in complex laminar [44] and turbulent flows [71]. Using the SPBMs coupled with relations for agglomerate fractal-like structure [72] and collision frequency [36], particle size distribution, morphology and composition can be tracked accurately. However, the computational cost of SPBMs increases exponentially with the number of sections [69] and particle properties [44] tracked. Thus, one property (e.g. agglomerate mass) is typically tracked with SPBMs to reduce computational cost. This does not allow to account for agglomerate fractal-like structure [73, 74] which limits SPBM accuracy in predicting surface growth and coagulation rate of agglomerates and their size distribution.

Alternatively, particle dynamics can be tracked by the method of moments (MOM) [75] or monodisperse population balance models (MPBM) [76]. Such models only track average particle properties (e.g. moment ratios) and their accuracy could be limited if unrealistic assumptions (e.g. approximating agglomerates as monodisperse and perfect spheres) are used. However, when inception and surface growth are short [77] and high particle (number) concentrations are formed [65], they lead to rapid attainment of self-preserving size distributions (SPSD) and agglomerates having asymptotic structure [62]. In this case a MPBM or MOM can be assembled on a firm scientific basis with accuracy on par with DEM [67], SPBM [78] and experimental data [79–81]. Such models can be readily interfaced with CFD simulations [82] without significant computational cost, making them ideal for three-dimensional and even turbulent flame simulations.

The MOM tracks moments of the PSD and estimates average particle properties such as mass [83], surface area [84], the number of constituent primary particles per agglomerate, n_p [75], or even particle composition [85] using the ratio of the moments. The MOM with four equations was used to describe synthesis of optical fibers by simultaneous reaction, diffusion, coagulation and thermophoresis of SiO_2 in laminar flow reactors assuming a lognormal PSD [86]. The MOM with interpolative closure (MOMIC) was developed to predict simultaneous nucleation, surface growth and coagulation of soot agglomerates and estimate its PSD with six equations [75]. To calculate source terms of the transported moments, additional moments that are not tracked are needed preventing the closure of the system of differential equations with the MOM [83, 87]. Thus, often the PSD shape is assumed a priori [83] or extra equations are solved to estimate it [76].

The MPBMs do not have the closure problem and calculate average particle properties by tracking their total concentration, mass [76] and area [88, 89]. Leung et al. [90] used a 2-equation MPBM to track soot concentration and mass in (non-premixed) flames assuming spherical particles. Good agreement was achieved for measured soot mass. However, the specific surface area [89] and coagulation frequency of spheres are significantly smaller compared to that of agglomerates with the same mass underestimating their oxidation rate [78] and overestimating their concentration [76]. Kruis et al. [76] proposed a 3-equation MPBM to account for the fractal-like structure of nanoparticle agglomerates during coagulation and sintering. Agglomerate volume and area were used to obtain their equivalent primary particle diameter, d_p, and n_p. Then, agglomerate collision diameter, i.e. d_g, was calculated by D_f, d_p and n_p to account for their fractal-like structure that affects their collision frequency. Tsantilis and Pratsinis [88] extended the MPBM to predict hard-(chemically-bonded) and soft-(physically-bonded) agglomerates during synthesis of SiO_2 and TiO_2 [91] nanoparticles with simultaneous reaction, surface growth, coagulation and sintering. Such a MPBM applies best at high concentrations when inception and surface growth are short [77] resulting in the dominance of coagulation where particles rapidly reach their SPSD and asymptotic fractal-like structure. This is often the case for soot emitted from a variety of combustion devices or CB reactors where inception and surface growth are limited to only a few milliseconds when temperature is very high (i.e. T$\geq$1500K) [56].

1.4 Soot maturity and its optical properties

The maturity level of soot is described as the evolution of physical and chemical properties from incipient to graphite-like mature soot [92]. It involves the growth of the graphitic crystallite fine structure of soot within, and perpendicular to, the aromatic layers. [93, 94], known as graphitization, followed by increase in the size of the crystallite-layer planes and decrease of the interlayer spacing [18]. This process is accompanied by the pyrolytic conversion of hydrocarbon species and substituted hydrocarbons toward elemental carbon and increase in the carbon-to-hydrogen (C/H) ratio, known as carbonization. Soot maturity is closely associated with reactivity of surface sites [95]. Fig. 1.4 compares nanostructure of nascent soot composed of disoriented PAH clusters with that of mature soot with a core-shell pattern where disordered core is surrounded by concentrically oriented graphitic layers [28]. The core-shell nanostructure of mature soot strongly depends on process conditions such as pressure, fuel identity, temperature and residence time of the particles.

Evolution of soot maturity and morphology impacts its optical properties. Incipient and nascent soot absorb shorter wave lengths (λ <600 μm) as opposed to mature soot particles that are broad-band light absorbers [96]. Non-intrusive optical diagnostic methods such as light extinction (absorption and scattering) [97] and Laser Induced Incandescence (LII) [37] are widely utilized to measure soot volume fraction, f_v using extinction coefficient and the absorption function, E(m) that depends on soot refractive index, m, soot composition and morphology [98].

However, the effects of soot composition, and morphology on its refractive index, and absorption function are not fully understood yet. E(m) of soot at λ =1064 nm measured by two-color LII measurements increases from 0.193 to 0.349, and 0.226 to 0.340 for ethylene premixed flames with equivalence ratio of ϕ=2.1 and ϕ=2.3, respectively when height above burner (HAB) changes from 8 to 14 mm [100]. During acetylene pyrolysis in a shock tube, E(m) of soot particles increases from 0.05 to 0.25 as their primary particle diameter, d_p, grows to 20 nm within 1.6 ms [101]. Using m and E(m) of mature spherical soot [102] at different HABs neglecting the impact of soot morphology and composition on m [103] can overestimate soot volume fraction

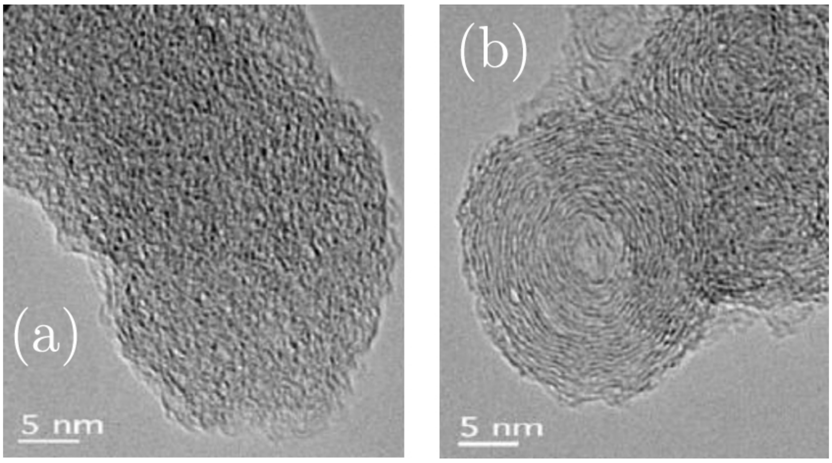


Fig. 1.4 The HRTEM image of (a) a nascent soot primary particle with disordered internal nanostructure juxtaposed with that of (b) a mature soot primary particle with well-organized cluster near the shell sampled from soot generated by pyrolysis of ethanol at 1250 and 1650, respectively. Reprinted from Ref.[99]

by 100% [104]. So, accurate estimation of the evolving optical properties of soot is essential to close the carbon mass balance in the measurements [104], and analyze reaction kinetics for soot inception [56] surface growth [57] and oxidation [105] that are essential for development and validation of aerosol dynamics models coupled with chemistry to predict soot formation.

The dependence of soot m on its morphology and composition can be quantified by soot optical band gap, E_g [106]. The optical band gap concept was originally proposed by Tauc et al. [107] for semi-conductors and later developed for amorphous carbon [108], and it can be used to describe the crystalline character of PAH clusters in soot [108]. Incipient flame-made soot with diameters less than 20 nm exhibit quantum dot behavior [109] with $0.7< E_g <2$ eV that is larger than E_g of graphitized soot, 0.12 eV [109]. Optical band gap of organic carbon coated on soot remains nearly constant (1.8-1.9 eV) over soot evolution [110]. In contrast, Russo et al. [111] measured soot optical band gap using ex-situ and in-situ methods and showed that E_g drops from 0.7 eV at HAB= 8 mm to 0.2 eV at HAB=14 mm in an ethylene premixed flame with ϕ=2.3 as particles grow and become more mature by carbonization. Kelesidis and Pratsinis [112] correlated the evolving refractive index of soot with its E_g obtained from quantum confinement theory (QCT) [109] for wavelengths of λ= 532 and 1064 nm [112] by linear interpolations between that of nascent and mature soot. The proposed relations were then used with soot morphologies obtained from Discrete Element Modeling (DEM) simulations to obtain the evolving Mass Absorption Cross-section (MAC), E(m), and the absorption function ratio of soot (E(λ=532nm)/ E(λ=1024nm)) and compared with measurements [113–115]. Employing the relation in laser diagnostics to reprocess light extinction measurement data resulted in accurate prediction of f_v in moderate (ϕ=2.34) [104] and rich (ϕ >3) ethylene premixed flames [116], and fv along the centerline of a coflow ethylene diffusion flame [117] which is necessary to close the carbon mass balance in soot generating processes.

1.5 Oxidation

Soot oxidation is described as the removal of soot mass by reaction of molecular oxygen (O_2), oxygen radical (O), and hydroxyl radical (OH) from soot surface. Because of its heterogeneous reaction kinetics and mechanisms, oxidation is expected to be sensitive to surface structure and composition. So, oxidation mechanisms depend on soot maturity and temperature-time history. In near stoichiometric and fuel-rich conditions, the contribution of OH radical to soot oxidation is predominant [118] compared to that of O radicals [119]. Fenimore and Jones [120] investigated soot oxidation rate for low oxygen partial pressures and temperatures from 1530 to 1890 K, highlighted the importance of OH as a major oxidation agent, and attributed the faster rates compared to those predicted by Lee et al. [121], to OH oxidation. One approach for describing OH oxidation is by a introducing collision efficiency representing the fraction of collisions of OH with soot particles that resulted in the removal of a carbon atom [118]. Scanning Mobility Particle Sizer (SMPS) with a two-stage burner also showed a collision efficiency of 0.13 [122].

The empirical relation of Nagle and Strickland-Constable (NSC) [123] originally developed for oxidation of pyrolytic graphite have been widely used to describe soot oxidation by O_2. It describes O_2 oxidation rates based on partial pressure of O_2 and the fraction of reactive edge sites to less reactive basal planes using the graphite analogy for soot. The dominance of edge sites in typical combustion conditions accounts for the relatively small reactivity of soot and basal planes become reactive at high temperature (>2500 K) [119]. O_2 oxidation kinetics was also described with a power-law kinetics [121], and shown to be first order in oxygen concentration [124]. Alternatively, soot oxidation can be explained based on chemical similarity using HACA mechanism assuming that active site are attacked by O_2 and OH leading to loss ot carbon and release of CO.

A different oxidation regime has been identified for soot at low temperatures (T<1000 K) where O_2 diffuses and reacts with bulk soot accounting for most of soot mass consumption [80]. Internal oxidation compacts the pore network of soot resulting in hollow CB [125], diesel [126] and biodiesel [127] soot particles and increases their SSA up to a factor of four [126]. Oxidation can cause fragmentation of soot particles affecting their morphology. The increase in number concentration and the change of soot morphology in lean premixed flames [128] and in the oxidation region of diffusion flames [129] was attributed to fragmentation. Such a change in agglomerate morphology has not been observed in fuel-rich conditions, so fragmentation was linked to O_2 oxidation.

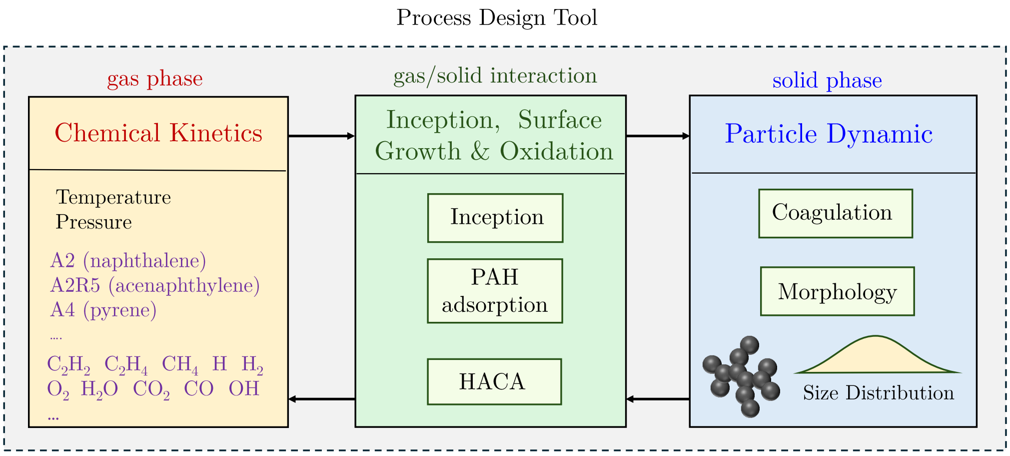


Fig. 1.5 The conceptual structure of developed process design tool that account for gas mixture properties, and soot chemistry involving inception, surface growth and oxidation as well as its coagulation leading to the fractal-like morphology of soot particles

1.6 Research objectives

The purpose of this project is developing a process design tool to predict the yield, morphology, composition and size distribution of soot particles in industrial flames and reactors under different temperatures, pressures and residence times. Integrating gas chemistry, all existing inception and surface growth models and particle dynamics in a single package provides a fully coupled soot description from breakdown of fuel to agglomeration and oxidation of particles that enables sensitivity analysis to shed light on key pathways for soot inception and surface growth in various targets. This tool can also be used for rate constant optimization and mechanism reduction under soot forming conditions. The conceptual structure of this tool is shown in Fig 1.5 that includes a chemical kinetics unit coupled with Cantera [130] to compute thermal and physical properties of gas mixture such as temperature, pressure, density, and enthalpy based on ideal equation of state for the gas mixture. The chemistry of gas mixture is quantified using detail reaction mechanisms that enables tracking concentration of fuel, intermediate species such as C_2H_2 and H_2 essential to surface growth, and PAHs such as benzene (A1), naphthalene (A2) and pyrene (A4) known as building blocks of soot. The second unit accounts for soot inception and surface growth via PAH adsorption and HACA as well as surface oxidation. It accommodates the four inception models widely used in literature. The third unit deals with particle dynamics using a MPBM and SPBM to describe the coagulation of soot particles that results in the fractal-like structure of particle with an evolving size distribution. A multi-step validation procedure is followed to assess reliability of each sub-model that entails the comparison of collision frequency kernels and agglomerate morphology of different particle dynamics models with those of detailed DEM simulations and establishing energy and elemental carbon, and hydrogen balances for all combinations of particle dynamics and inception models with reactors and flames. This tool can be applied to a variety of experimental targets with different temperature- and pressure-time-histories, residence times, and fuel compositions.

As the first step, methane pyrolysis in shock-tube will be simulated using a constant volume reactor. The species measurements will be used to assess the reaction mechanism and examine carbon conversion flux from fuel to smaller intermediates and larger hydrocarbons. The inception flux and surface growth rates of the soot model strongly depend on the reaction mechanism employed to track the concentration of soot precursors. The performance of inception models will be analyzed by comparing the predicted soot volume fraction, f_v and primary particle diameter, d_p, with data collected from extinction and TEM measurements. A sensitivity analysis will be conducted on model parameters to identify the determining factors that control both f_v and d_p in the framework of shock-tube simulations with short residence time ($t \approx 2$ ms) and high temperatures (≥ 2000K). Then, the optimization of inception models will be performed by adjusting the rate constants within their physical limit (maximum collision rate of molecules) to minimize the prediction error for yield and morphology. This shows the range of inception flux that ensures the accurate prediction of soot yield and morphology represented by f_v and d_p, respectively in good agreement with measurements. Then, the optimized rates will be applied to shock-tube in a wide temperature range, and the f_v predictions will be compared with data. The measurement on shock-tube pryolysis of methane with different H_2 concentrations will be used to investigate the effect of hydrogen on pyrolysis chemistry as well as soot yield and morphology. A main focus will be finding the reaction mechanism that describes methane decomposition rate and acetylene and PAH concentrations in good agreement with the data from absorbance measurements. The inception flux and surface growth rate that results in accurate prediction of soot yield and morphology will be estimated and compared with cases without hydrogen. A similar investigation will be conducted for flow reactors using plug flow reactor model of omnisoot at lower temperatures and longer residence times and then the range of expected inception flux with be compared with those of shock-tubes.

In the next step, a reactor network model based on omnisoot will be developed to enable building a series of plug flow reactors (PFR) and perfectly stirred reactor (PSR) models with multiple gas sources to more accurately represent complex industrial reactors. A set of tests will be developed to ensure gas and solid mass and energy balance during the development process. The connections between elements will be developed in a way that recognizes upstream and downstream flows. Moreover, safeguards will be put in place for the network to avoid faulty designs that can result in cyclical flows and detached networks. The performance of the reactor network will be evaluated using benchmark data from literature [131].

The flame solver of omnisoot will be developed from its current experimental stage to a functional level. After validation of the coagulation and surface growth sub-models and ensuring the conservation of carbon and hydrogen mass, and energy balances, the flame solver will be applied to a variety of counterflow diffusion [132, 133] and burner-stabilized premixed flames[27] with stagnation plate [81] at atmospheric and elevated pressures. The reaction mechanism study will be performed to reduce the uncertainty in tracking gas chemistry by comparing the model predictions of species concentration with measurements. The performance of the soot model will be evaluated by comparing the predicted soot volume fraction, agglomerate number concentration, PSD, mobility and primary particle diameters with those obtained from light extinction measurement, scanning mobility particle sizer (SMPS), and TEM samples, and the comparisons will be use to estimated the range of inception flux and surface growth rates that can predict soot mass, morphology and size distribution with minimum error compared to data.

Chapter 2

Theoretical Model

The mathematical basis for *Omnisoot* is explained in the top-to-bottom hierarchical order. First, the governing equations for constant volume, constant pressure, perfectly stirred and plug flow reactor models are reviewed in Sections 2.2.1-2.2.4. These equations differ from pure gas-phase transport formulation as they consider the solid phase. The transport equations of “soot variables" include the source terms that account for change in each soot variable due to inception, surface growth, oxidation and coagulation. Then, particle dynamics models are explained in Sections 2.3 that entails description of size distribution and morphology of soot particles and their collision rate, which is used to determine the coagulation source term. Section 2.4 focuses on the *“PAH growth model"*s that take care of inception and adsorption from designated precursors and calculates the corresponding source terms. Similarly, the *“surface reactions"* model are detailed in Section 2.5 that elaborates on the surface growth and oxidation rates based on HACA mechanism. Figure 2.1 illustrates the general structure of Omnisoot and the sub-models.

2.1 Assumptions and conventions

Here, the main conventions and assumptions used in the derivation of the mathematical models are listed below.

1. f_v and φ denote the volume of soot particles normalized by the gas volume and reactor volume, respectively. Their relationship can be expressed as:

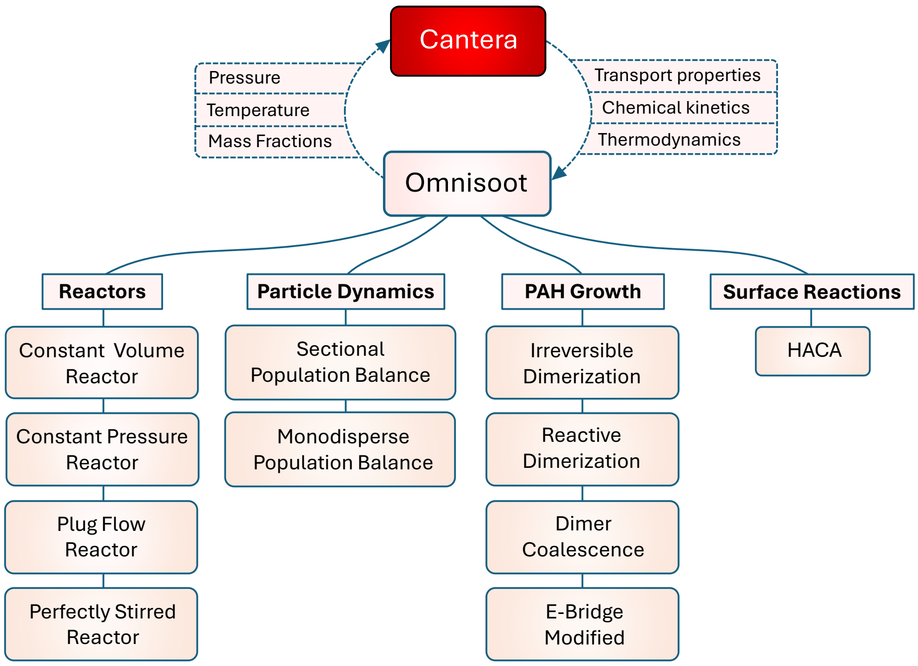


Fig. 2.1 The structure of Omnisoot that illustrates the coupling with Cantera and sub-models including reactors, particle dynamic models, PAH growth models and surface reactions models

$$
\begin{aligned}
f_v &= \frac{V_{soot}}{V_{gas}},\\
\varphi &= \frac{V_{soot}}{V_{gas} + V_{soot}},\\
\varphi &= \frac{f_v}{1 + f_v}.
\end{aligned} \tag{2.1}
$$

2. $\dot{s}_k$ denotes the rate production/consumption of k_{th} gaseous species (i.e., gas scrubbing) due to soot inception, surface growth and oxidation. It is positive when the species is released to the gas mixture.

3. Each soot agglomerate consists of monodisperse spherical primary particles, which are in point contact. So, the model does not consider formation of necking (or sintering) in soot agglomerates by surface growth.

4. The word *"particle"* refers to soot both in spherical and agglomerate shape.

5. The density of soot is assumed constant at the value of 1800 kg/m^3. Soot density changes with its maturity level, which is often linked to the elemental C/H ratio of soot particles [134]. Here, the considered value represents an average between density of mature soot with high C/H ratios ($\rho = 2000$ kg/m^3) and that of nascent soot with low C/H ratios ($\rho = 1600$ kg/m^3) [135, 134].

6. The incipient soot particles are 2 nm in diameter, so the model does not allow particles with a primary particle diameter smaller than 2 nm. The number of carbon atoms in the incipient soot particle, $n_{c,min}$, is calculated from the mass of a sphere with a diameter of $d_{p,min} = 2$ nm assuming pure carbon content, which results in:

$$n_{c,min} = \frac{\pi}{6} \rho_{soot} d_{p,min}^3 \frac{1}{W_{carbon}} \approx 378. \tag{2.2}$$

7. The specific heat, internal energy and enthalpy of soot are approximated by those of pure graphite, and are employed to close the energy balance in the system [136].

8. Soot particles and gas are in thermal equilibrium during soot formation processes, and there is no temperature gradient within each agglomerate.

9. ψ denotes a *soot variable* that represents a mean property of soot particles in each section tracked in Omnisoot by solving transport equations for the total concentration of agglomerates, N_{agg} and primary particles, N_{pri}, and the total carbon, C_{tot} and hydrogen content, H_{tot} of soot. S_ψ is the source term of the soot variable, ψ, which appears in soot equations.

10. *PAH growth* is a sub-model of Omnisoot with a set of pathways that determine the rate of inception and adsorption from PAHs (designated as soot precursors) in the gas mixture.

11. *Surface reactions* is a sub-model of Omnisoot that describes the addition of acetylene to soot surface, and removal of carbon via oxidation by OH and O_2 following the HACA scheme. The model does not consider soot oxidation with CO_2, H_2O and NO_x.

12. The single superscript i denotes the section number of a soot variable or a derived property. For example, d_p^i represents the primary particle diameter of section i. The double superscript ij indicates a property related to two sections; for

Table 2.1 The names, symbols, chemical formula and molecular weight of the soot precursors used by Omnisoot.

Species name	Symbol	Chemical formula	W [kg/mol]
Naphthalene	A2	$C_{10}H_8$	0.128
Phenanthrene	A3	$C_{14}H_{10}$	0.178
Pyrene	A4	$C_{16}H_{10}$	0.202
Acenaphthylene	A2R5	$C_{12}H_8$	0.152
Acephenanthrylene	A3R5	$C_{16}H_{10}$	0.202
Cyclopentapyrene	A4R5	$C_{18}H_{10}$	0.226

example, β^{ij} denotes the collision frequency between sections i and j. In the case of the monodisperse model, the section number can be omitted because it is equivalent to the sectional model with only one section.

13. The computation of morphological parameters is done similarly in both particle dynamics models, but they are explained separately in Section 2.3.1.

14. *precursors* refer to the PAHs larger than naphthalene used for inception and PAH adsorption. The list of precursors with their chemical formula and molecular mass is provided in Table 2.1. It should be noted that the precursors can be dynamically changed by Omnisoot's user interface.

2.2 Reactors

The governing equations of reactor models implemented in Omnisoot are briefly presented in the following sections. The control volume encompasses the gas mixture and soot particles, as illustrated in Figure 2.2. The equations ensure conservation of total mass and energy of the gas and particle system, which can also receive or lose heat through the reactor walls.

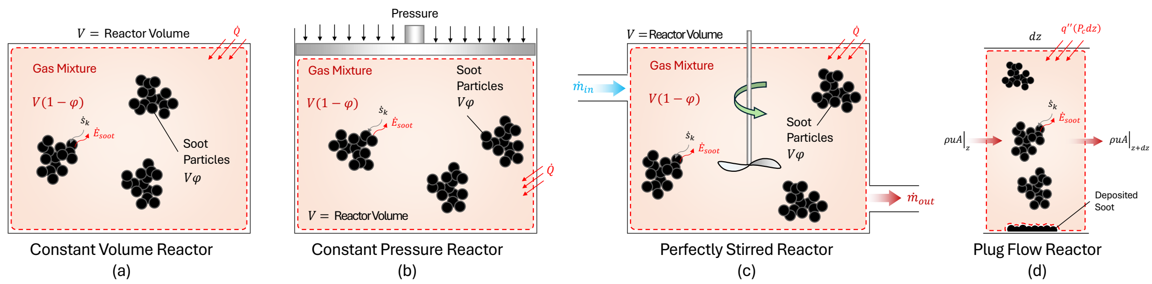


Fig. 2.2 The schematics of constant volume reactor (a), constant pressure reactor (b) perfectly stirred reactor (c), and plug flow reactor (d)

2.2.1 Constant Volume Reactor (CVR)

As shown in Figure 2.2a, CVR represents a closed system constant volume where chemical reactions converts part of the gas mixture to soot particles. The mass balance equation is written as:

$$\frac{\mathrm{d}}{\mathrm{d}t}(m) = (1-\varphi)V\sum_i \dot{s}_i W_i, \tag{2.3}$$

where m is the mass of the gas mixture. The rate of change of m is equal to the rate of production of soot mass. Similarly, the species equation for species k is expressed as:

$$\frac{\mathrm{d}Y_k}{\mathrm{d}t} = \frac{1}{\rho}\left(\dot{\omega}_k + \dot{s}_k\right) W_k - \frac{1}{\rho} Y_k \sum_i \dot{s}_i W_i. \tag{2.4}$$

The transport equation for a generic soot variable, ψ, can be written as:

$$\frac{\mathrm{d}\psi}{\mathrm{d}t} = S_\psi - \frac{\psi}{\rho}\sum_i \dot{s}_i W_i. \tag{2.5}$$

The second term on RHS of Equations (2.4) and (2.5) denotes the change in Y_k and ψ, respectively, due to the removal or addition of gas mass by soot-related processes. The energy conservation for the gas mixture is written in terms of the rate of change of temperature. An external heat source of $\dot{\mathrm{Q}}$ is considered to account for possible heat loss/gain of the reactor.

$$\begin{aligned}\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{1}{\rho c_v + \rho_{soot} f_v c_{soot}} \Bigg[&-\sum_k e_k \left(\dot{\omega}_k + \dot{s}_k\right) W_k \\ &+ e_{soot}\sum_k \dot{s}_k W_k + \frac{\dot{Q}}{V(1-\varphi)} \Bigg],\end{aligned} \tag{2.6}$$

where $\rho_{soot} f_v c_{soot}$ and $e_{soot} \sum_k \dot{s}_k W_k$ represent the formation and sensible energy of soot, respectively. We investigated the effect of considering soot formation and sensible energy on gas and soot properties by simulating the pyrolysis of 30% CH_4-Ar with and without considering the above mentioned term. As shown in Figure 2.3a, neglecting soot sensible energy results in the overprediction of temperature by nearly 150 K and mobility diameter by a factor of 3 during the 80 ms of the simulation. The overprediction of temperature changes gas chemistry, leading to a noticeable decrease in the residual methane and benzene (Figure 2.3b).

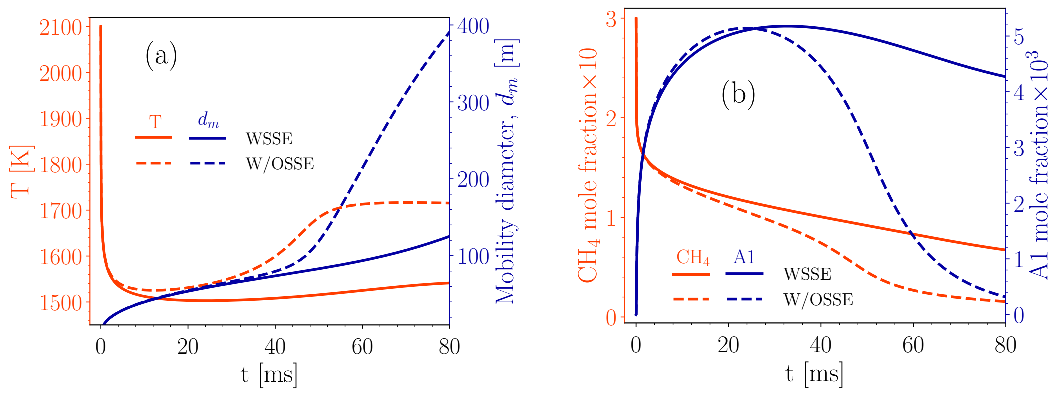


Fig. 2.3 The comparison of temperature and soot mobility diameter, d_m (a) and the mole fraction of methane, CH_4, and benzene, A1 in the simulation of the pyrolysis of 30%CH_4-Ar when soot sensible energy is considered (labeled as “WSSE") and neglected (labeled as “W/OSSE"). The CVR is used along with Caltech mechanism, Reactive Dimerization and MPBM

2.2.2 Imposed Pressure Reactor (IPR)

As show in Figure 2.2b, this reactor is a closed system similar to CVR with a main difference that the boundary can move and change the volume of the system due to an external pressure that can stay constant or vary with time. In either case, the pressure is known a priori in this model. The heat transfer can occur through reactor walls, which changes the internal energy of the gas-particle system.

The equations of mass, species, and soot variables in IPR are the same as those in CVR, but the energy equation is written based on enthalpy, h, instead of internal energy, e, as:

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{1}{\rho c_p + \rho_{soot} f_v c_{soot}} \Bigg[-\sum_k h_k \left(\dot{\omega}_k + \dot{s}_k\right) W_k + h_{soot} \sum_k \dot{s}_k W_k + \frac{\mathrm{d}P}{\mathrm{d}t} + \frac{\dot{Q}}{V(1-\varphi)} \Bigg]. \tag{2.7}$$

where $\mathrm{d}P/\mathrm{d}t$ represents the effect of variable pressure on the gas temperature, which is zero when the pressure is constant throughout the process.

2.2.3 Perfectly Stirred Reactor (PSR)

In this reactor, gas enters with a mass flow rate, composition, and temperature of $\dot{m}_{in}$, Y_{in}, and T_{in}, respectively, and homogeneously reacts with the mixture inside the reactor. The reacting gas reaches a spatially uniform temperature and composition described by T and Y. It is assumed that temperature, composition and soot properties of the outflow are the same as the mixture inside reactor. Without soot formation, the inlet and outlet mass flow rates are equal (i.e. $\dot{m}_{in} = \dot{m}_{out}$), but when soot is formed, $\dot{m}_{out}$ is slightly less than $\dot{m}_{in}$. The nominal residence time of PSR is calculated as:

$$\tau_{psr} = \frac{\rho V}{\dot{m}_{in}}. \tag{2.8}$$

The conservation of mass of PSR can be written by considering the mass flux of in- and outflow, and the removal of mass due to soot formation as:

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \dot{m}_{in} - \dot{m}_{out} + V(1-\varphi)\sum_i \dot{s}_i W_i. \tag{2.9}$$

Gas composition is obtained by solving the species transport equations as:

$$\frac{\mathrm{d}Y_k}{\mathrm{d}t} = \frac{\dot{m}_{in}}{\rho V\left(1-\varphi\right)}\left(Y_{k,in} - Y_k\right) + \frac{1}{\rho}\left[\left(\dot{\omega}_k + \dot{s}_k\right) W_k - Y_k \sum_i \dot{s}_i W_i\right]. \tag{2.10}$$

The soot transport equations can also be expressed as:

$$\frac{\mathrm{d}\psi}{\mathrm{d}t} = \frac{\dot{m}_{in}}{\rho V\left(1-\varphi\right)}\left(\psi_{in} - \psi\right) + S_\psi - \frac{1}{\rho}\psi \sum_i \dot{s}_i W_i. \tag{2.11}$$

The energy equation for this reactor is written as:

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{1}{\rho c_p + \rho_{soot} c_{p,soot} f_v} \left[\frac{\dot{m}_{in}}{V(1-\varphi)} \left(h_{in} - h\right) - \frac{\dot{m}_{in}}{V(1-\varphi)} \sum_k \left(Y_{k,in} - Y_k\right) h_k \right. \\ \left. - \sum_k \left(\dot{\omega}_k + \dot{s}_k\right) W_k h_k + \sum_i \dot{s}_i W_i h_{soot} + \frac{\dot{Q}}{V(1-\varphi)} \right]. \tag{2.12}$$

2.2.4 Plug Flow Reactor (PFR)

PFR is an ideal representation of a channel or duct where the temperature, composition, and soot properties of a steady-state one-dimensional flow evolve along the reactor. There is no spatial gradient over cross-section due to strong mixing. The diffusion along reactor is negligible.

The continuity equation for PFR is written as:

$$\frac{\mathrm{d}\dot{m}}{\mathrm{d}z} = (1-\varphi) A \sum_i \dot{s}_i W_i. \tag{2.13}$$

The momentum equation can also be established as:

$$u(1-f_v) \sum_i \dot{s}_i W_i + \rho u (1-\varphi) \frac{\mathrm{d}u}{\mathrm{d}z} = -\frac{\mathrm{d}}{\mathrm{d}z}(p(1-\varphi)) - \frac{\tau_w}{R_H}, \tag{2.14}$$

where τ_w, and R_H are the wall shear stress and hydraulic radius of the reactor. τ_w can be determined from the friction factor, f, as:

$$\tau_w = \frac{1}{2} \rho u^2 f, \tag{2.15}$$

where f can be accurately calculated over the entire range of Reynolds numbers, from laminar to turbulent flow, using the explicit formula provided by Haaland [137]:

$$\frac{1}{f^{1/2}} = -1.8 \log \left(\frac{6.9}{Re} + \left[\frac{\epsilon / D_H}{3.7} \right]^{1.11} \right), \tag{2.16}$$

where ϵ and D_H are the wall roughness and the hydraulic radius of the reactor. The species equation can be expressed as:

$$\frac{\mathrm{d}Y_k}{\mathrm{d}z} = \frac{1}{\rho u} \left[\left(\dot{\omega}_k + \dot{s}_k\right) W_k - Y_k \sum_i \dot{s}_i W_i \right]. \tag{2.17}$$

The soot transport equations can also be written as:

$$\frac{\mathrm{d}\psi}{\mathrm{d}z} = \frac{S_\psi}{u} - \frac{\psi}{\rho u}\sum_i \dot{s}_i W_i - \frac{4}{D_H}\frac{k^i_{dep}\psi}{u}, \tag{2.18}$$

where k^i_{dep} is the deposition velocity of soot particles of section i, which is calculated as:

$$k_{dep} = \frac{Sh \cdot D^i}{D_H}, \tag{2.19}$$

where $Sh = 3.66$ for laminar flows, and it calculated using the Berger and Hau correlation [138] for the turbulent flow as:

$$Sh = 0.0165 Re^{0.86} Sc^{1/3}. \tag{2.20}$$

The energy equation can be expressed as:

$$\begin{aligned}\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{1}{\rho u c_p + \rho_{soot} u f_v c_{p,soot}} \Bigg[& - \sum_k h_k \left(\dot{\omega}_k + \dot{s}_k\right) W_k \\ & + h_{soot} \sum_k \dot{s}_k W_k + q'' \frac{P_c}{A} \Bigg],\end{aligned} \tag{2.21}$$

where q'' is the wall heat flux of the reactor.

2.3 Particle Dynamics

Population balance models rely on the Eulerian description of particles where average properties of particle population such as number density, mass or surface area are treated as continuous quantities and tracked by solving scalar transport equations. These methods are computationally cheaper compared with mesoscale models such as DEM, and can be easily interfaced with chemical kinetics in CFD solvers to simulate soot formation in laminar and turbulent configurations. Here, we use two particle dynamics models: a monodisperse population balance model (MPBM), which tracks four variables and results in four transport equations, and a fixed sectional population balance model (SPBM), which tracks three variables per section. In the SPBM approach, the total number of transport equations equals the number of tracked variables multiplied by the number of sections.

Having the total number concentration of agglomerates and primary particles and total carbon content of soot enables the model to describe evolving fractal-like morphology and surface area of soot agglomerates, which is essential to compute their collision frequency [139] as well as oxidation and surface growth rates [78]. Tracking hydrogen content of soot allows capturing the soot composition, thereby its maturity [44], and surface reactivity [85].

The common features of implemented particle dynamics models are reviewed in Section 2.3.1. The composition (H/C), diffusion coefficient, and morphology of soot particles are calculated in the same manner for MPBM and each section in SPBM. However, particle dynamics models differ in terms of the calculation of coagulation, and the processing of the contributions from inception, surface growth, and coagulation to generate the source terms of soot variables, S_φ, incorporated into the transport equations.

2.3.1 Common Features

Soot Morphology

The evolving fractal-like structure of agglomerates is quantified by their mobility diameter normalized by primary particle diameter, d_m/d_p, and gyration diameter, d_m/d_g, that can be described with power-laws derived from mesoscale simulations. Incipient soot is initially a sphere formed of PAHs that grows in size by surface reactions and forms agglomerates by coagulation. The collision frequency of particles depends on their evolving fractal-like structure [139]. Mobility and gyration diameters are calculated using power-laws developed to describe the morphology of soot from premixed [79], diffusion [140] flames, and diesel engines [141]. Figure 2.4 shows the schematic of a soot agglomerate composed of 12 primary particles, with d_p, d_m, and d_g annotated.

n_p^i is the number of primary particles per agglomerate of the i^{th} section that can be obtained by dividing the number concentration of primary particles in the i^{th} section by that of agglomerates in that section as:

$$n_p^i = \frac{N_{pri}^i}{N_{agg}^i}. \tag{2.22}$$

The primary particle diameter, d_p^i, is determined from the total carbon content and the number density of primary particles using:

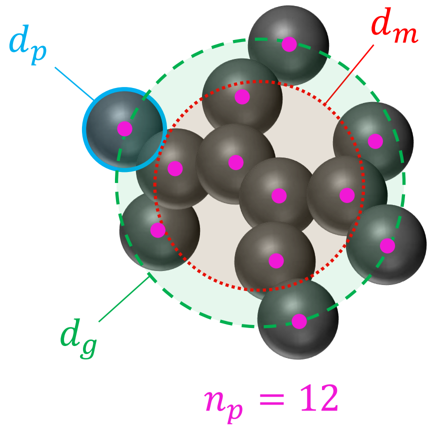


Fig. 2.4 The schematics of a soot agglomerates with 12 primary particles ($n_p = 12$). Primary particle, d_p, mobility d_m, and gyration d_g, diameters are shown.

$$d_p^i = \left(\frac{6}{\pi} \frac{C_{tot}^i \cdot W_{carbon}}{\rho_{soot}} \frac{1}{N_{pri}^i \cdot Av} \right)^{1/3}. \tag{2.23}$$

The DEM-derived power-laws [65] relate d_m^i and d_g^i to d_p^i and n_p^i as:

$$d_m^i = d_p^i \cdot {n_p^i}^{0.45}, \tag{2.24}$$

$$d_g^i = \begin{cases} d_m^i/({n_p^i}^{-0.2} + 0.4), & \text{if } n_p^i > 1.5 \\ d_m^i/1.29 & \text{if } n_p^i \leq 1.5 \end{cases} \tag{2.25}$$

The collision diameter, d_c^i, is the maximum of d_m^i and d_g^i:

$$d_c^i = \max\left(d_m^i, d_g^i\right). \tag{2.26}$$

The volume equivalent diameter, d_v^i, is the diameter of the sphere with the same mass as the agglomerate, and it is obtained as:

$$d_v^i = d_p^i \cdot {n_p^i}^{1/3}. \tag{2.27}$$

The primary particle surface area is calculated from d_p^i assuming spherical primary particles as:

$$A_p^i = \pi {d_p^i}^2. \tag{2.28}$$

A^i_{tot} (for each section) is defined as the total surface area of soot particles per unit mass of gas mixture obtained as:

$$A^i_{tot} = N^i_{pri} \cdot Av \cdot A^i_p. \tag{2.29}$$

Mass of each agglomerate, m^i_{agg} is expressed as:

$$m^i_{agg} = \frac{C^i_{tot} \cdot W_{carbon}}{N^i_{agg} \cdot Av}. \tag{2.30}$$

Soot Composition

The composition of soot characterized by their elemental carbon to hydrogen ratio (C/H) is a measure of soot maturity and increases from C/H < 2 for incipient soot [142] to 2 < C/H < 10 for nascent soot [143] and C/H > 20 for mature soot [92]. The soot agglomerates are assumed to have pure carbon graphitic core [44] with all hydrogen atoms on the surface [85]. C/H ratio can be obtained from the total carbon and the hydrogen content as:

$$\left(\frac{C}{H}\right)^i = \frac{C^i_{tot}}{H^i_{tot}}. \tag{2.31}$$

The carbon content of each agglomerate is a predefined parameter in the SPBM (depending on the section the agglomerate is placed), but it can be calculated from dividing C_{tot} by N_{agg} for the MPBM. The hydrogen content of each agglomerate is calculated for both particle dynamics models as:

$$H^i_{agg} = \frac{H^i_{tot}}{N^i_{agg}}. \tag{2.32}$$

Diffusion of soot particles

The diffusion coefficient of soot particle, D^i, is calculated as:

$$D^i = \frac{k_B T}{f^i}, \tag{2.33}$$

where f^i is the friction factor of particles in the gas mixture, and it is calculated as:

$$f^i = \frac{3\pi\mu d^i_m}{C^i(d^i_m)}, \tag{2.34}$$

where C^i is the Cunningham correction factor for the particle friction factor that accounts for non-continuum effects in the transition and free molecular regimes. C^i is calculated for a given diameter, d, as:

$$C^i(d) = 1 + \frac{2\lambda}{d}\left(1.21 + 0.4\exp(\frac{-0.78d}{\lambda})\right), \tag{2.35}$$

where λ is the mean free path of gas given as:

$$\lambda = \frac{\mu}{\rho}\sqrt{\frac{\pi W_{gas}}{2k_B A v T}}. \tag{2.36}$$

Note that λ is a property of the gas mixture that does not depend on particle mass and morphology.

2.3.2 Sectional Population Balance Model

A SPBM with fixed pivots is used to describe particle dynamics [144]. The mass range of particles is divided into discrete sections each of which includes agglomerates of the same mass. Figure 2.5 illustrates the interaction between the gas phase and mass sections, as well as the mechanisms by which particles move between sections. Inception introduces new particles to the first section with the mass corresponding to the incipient particle. The particles of each section can migrate to upper sections by gaining mass via surface growth and coagulation, and return to lower sections when they lose mass through oxidation without breaking into smaller particles (i.e., fragmentation is not considered). Particles attach at point contact without coalescence. The mass of sections is determined by a geometric progression that has an initial value equal to the mass of incipient soot particle, and a common ratio of SF, known as sectional spacing factor. The mass of each section is approximated by the carbon content of agglomerates in moles as:

$$C^i_{agg} = \frac{n_{c,min}}{Av} \cdot SF^{(i-1)}, \tag{2.37}$$

where $(i-1)$ represents the exponent of SF. The mass of hydrogen is ignored in the placement of agglomerates in the sections. The total number density of agglomerates, N^i_{agg}, and primary particles, N^i_{pri}, and the hydrogen content, H_{tot}, are tracked for each section. The mass of each section is fixed, so C_{tot}, for each section can be easily calculated by knowing the number of agglomerates; that is, $C_{tot} = N^i_{agg} \cdot Av \cdot C^i_{agg}$.

Morphological parameters for each section are determined according to the equations provided in Section 2.3.1.

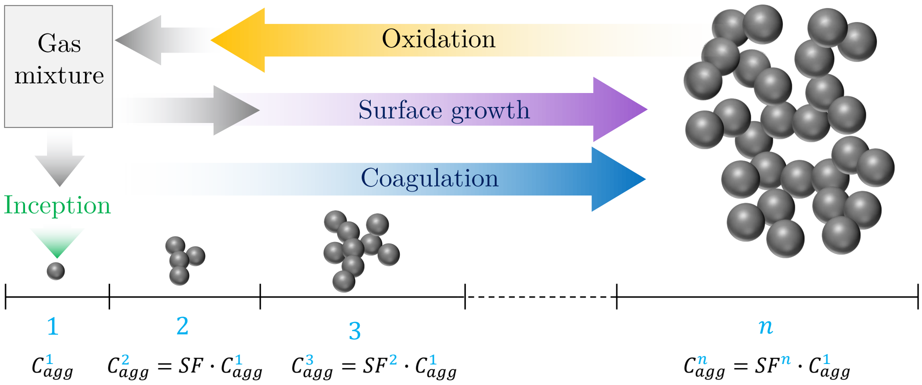


Fig. 2.5 The illustration of sections in SPBM. The mass of sections grows progressively by the scale factor of SF. Inception introduces new particles to the first section that propagate to the upper section via coagulation and surface growth and return to lower sections by oxidation. Carbon and hydrogen pass from gas to solid phase through inception and surface growth and goes back via oxidation.

New particles formed by coagulation are assigned to an upper section, with a total mass equal to the sum of the masses of the colliding particles. If the mass of resulting particle falls between two consecutive sections, the mass is distributed between them proportionally. In some cases, the mass of the newly formed particle may exceed the upper bound of the last section, causing it to fall outside the tracked mass range. This leads to potential mass loss, which is a known limitation of the fixed pivot sectional model [145]. However, this issue can be mitigated by choosing an appropriate number of sections and a suitable spacing factor to ensure the uppermost sections remain unoccupied during the simulation. In the SPBM, the source terms of tracked soot variables (which appear in Equations 2.5, 2.11, and 2.18) are split into four parts showing the contribution of inception, surface growth, oxidation and coagulation. The effect of HACA and PAH adsorption are combined (denoted by the subscript *haca,ads*) because they are similar mass-gaining mechanisms. The source terms for section i are given as:

$$S^{i}_{N_{agg}} = \left(S^{i}_{N_{agg}}\right)_{inc} + \left(S^{i}_{N_{agg}}\right)_{haca,ads} + \left(S^{i}_{N_{agg}}\right)_{ox} + \left(S^{i}_{N_{agg}}\right)_{coag}, \tag{2.38}$$

$$S^{i}_{N_{pri}} = \left(S^{i}_{N_{pri}}\right)_{inc} + \left(S^{i}_{N_{pri}}\right)_{haca,ads} + \left(S^{i}_{N_{pri}}\right)_{ox} + \left(S^{i}_{N_{pri}}\right)_{coag}, \tag{2.39}$$

$$S^i_{H_{tot}} = \left(S^i_{H_{tot}}\right)_{inc} + \left(S^i_{H_{tot}}\right)_{haca,ads} + \left(S^i_{H_{tot}}\right)_{ox} + \left(S^i_{H_{tot}}\right)_{coag}. \tag{2.40}$$

Coagulation Source Term

Coagulation redistributes the total number of agglomerates and primary particles as well as hydrogen atoms among the sections. The partial coagulation source terms for N^i_{agg}, N^i_{pri} and H^i_{tot} can be calculated as:

$$\left(S^i_{N_{agg}}\right)_{coag} = \sum_{k=1}^{n_{sec}} \sum_{j=k}^{n_{sec}} \left(1 - \frac{\delta_{jk}}{2}\right) \eta_{ijk} \zeta^{jk} \beta^{jk} N^j_{agg} N^k_{agg} - N^i_{agg} \sum_{m=1}^{n_{sec}} \zeta^{im} \beta^{im} N^m_{agg}, \tag{2.41}$$

$$\left(S^i_{N_{pri}}\right)_{coag} = \sum_{k=1}^{n_{sec}} \sum_{j=k}^{n_{sec}} \left(1 - \frac{\delta_{jk}}{2}\right) \eta_{p,ijk} \eta_{ijk} \zeta^{jk} \beta^{jk} N^j_{agg} N^k_{agg} - N^i_{pri} \sum_{m=1}^{n_{sec}} \zeta^{im} \beta^{im} N^m_{agg}, \tag{2.42}$$

$$\left(S^i_{H_{tot}}\right)_{coag} = \sum_{k=1}^{n_{sec}} \sum_{j=k}^{n_{sec}} \left(1 - \frac{\delta_{jk}}{2}\right) \eta_{h,ijk} \eta_{ijk} \zeta^{jk} \beta^{jk} N^j_{agg} N^k_{agg} - H^i_{tot} \sum_{m=1}^{n_{sec}} \zeta^{im} \beta^{im} N^m_{agg}. \tag{2.43}$$

where δ_{jk} is the Kronecker delta defined as:

$$\delta_{jk} = \begin{cases} 1, & \text{if } j = k \\ 0. & \text{if } j \neq k \end{cases} \tag{2.44}$$

The collision frequency between sections j and k (β^{jk}) can be obtained from the harmonic mean of the values in the continuum (β^{jk}_{cont}) and free molecular (β^{jk}_{fm}) regimes as:

$$\beta^{jk} = \frac{\beta^{jk}_{fm} \cdot \beta^{jk}_{cont}}{\beta^{jk}_{fm} + \beta^{jk}_{cont}}, \tag{2.45}$$

$$\beta^{jk}_{fm} = \sqrt{\frac{\pi k_B T}{2} \left(\frac{1}{m^j_{agg}} + \frac{1}{m^k_{agg}}\right)} \left(d^j_c + d^k_c\right)^2, \tag{2.46}$$

$$\beta^{ij}_{cont} = \frac{2 k_B T}{3\mu} \left(\frac{C^j}{d^j_m} + \frac{C^k}{d^k_m}\right) \left(d^j_c + d^k_c\right)^2. \tag{2.47}$$

The collision frequency can also be determined from the Fuchs interpolation as:

$$\beta^{jk} = \beta^{ij}_{cont} \left[\frac{d_c^j + d_c^k}{d_c^j + d_c^k + 2 + \delta_r^{jk}} + \frac{8\left(D^j + D^k\right)}{\bar{c}_r^{jk}\left(d_c^j + d_c^k\right)} \right]^{-1}, \tag{2.48}$$

where δ_r^{jk} and $\bar{c}_r^{jk}$ are the mean square root of mean distance and velocity of particles, respectively, and they are obtained as:

$$\delta_r^{jk} = \sqrt{{\delta_a^j}^2 + {\delta_a^k}^2}, \tag{2.49}$$

$$\bar{c}_r^{jk} = \sqrt{{c^j}^2 + {c^k}^2}. \tag{2.50}$$

The mean velocity, c^i, and mean stop distance of particles, λ_a^i, can be calculated as:

$$c^i = \sqrt{\frac{8k_BT}{\pi m^i_{agg}}}, \tag{2.51}$$

$$\delta_a^i = \frac{1}{d_c^i \lambda_a^i} \left[\left(d_c^i + \lambda_a^i\right)^3 - \left({d_c^i}^2 + {\lambda_a^i}^2\right)^{3/2} \right] - d_c^i, \tag{2.52}$$

$\lambda_a^i$is the agglomerate stopping distance defined as:

$$\lambda_a^i = \frac{8D^i}{\pi c^i}. \tag{2.53}$$

In Equation (2.41), η_{ijk} assigns newly formed agglomerates to the two consecutive sections in order to conserve mass during coagulation [70].

$$\eta_{ijk} = \begin{cases} \dfrac{C^{i+1}_{agg} - C^{jk}_{agg}}{C^{i+1}_{agg} - C^{i}_{agg}}, & \text{if } C^i_{agg} \leq C^{jk}_{agg} < C^{i+1}_{agg} \\ \dfrac{C^{i}_{agg} - C^{jk}_{agg}}{C^{i}_{agg} - C^{i-1}_{agg}}, & \text{if } C^{i-1}_{agg} \leq C^{jk}_{agg} < C^{i}_{agg} \\ 0, & \text{else} \end{cases} \tag{2.54}$$

where $C^{jk}_{agg} = C^j_{agg} + C^k_{agg}$. Similarly, $\eta_{p,ijk}$ in Equation (2.42) and $\eta_{h,ijk}$ in Equation (2.43) adjust the number of primary particles and hydrogen atoms added to consecutive sections based on their mass, respectively.

$$\eta_{p,ijk} = \frac{C^i_{agg}}{C^{jk}_{agg}} \left(n_p^j + n_p^k\right), \tag{2.55}$$

$$\eta_{h,ijk} = \frac{C^i_{agg}}{C^{jk}_{agg}} \left(H^j_{agg} + H^k_{agg} \right). \tag{2.56}$$

In Equation (2.41), ζ^{jk} is the coagulation efficiency of soot particles in sections j and k. A value of $\zeta^{jk} = 1$ indicates that every collision between two soot particles successfully results in the formation of a new agglomerate. However, numerical models [146] and experimental evidence [147] have shown that coagulation efficiency drastically decreases for particles smaller than 10 nm in the free-molecular regime ($\mathrm{Kn} \gg 10$), due to their high kinetic energy exceeding the magnitude of attractive forces [148]. The coagulation efficiency between two colliding particles can be described by [146]:

$$\zeta^{ij} = 1 - \left(1 + \frac{\Phi^{ij}_0}{k_B T} \right) \exp\left(-\frac{\Phi^{ij}_0}{k_B T} \right), \tag{2.57}$$

where Φ_0 is the potential well depth, i.e., the minimum interaction energy between two colliding particles. Hou et al. [149] calculated Φ_0 for soot particles ranging from 1 to 15 nm by considering the attractive and repulsive interactions between constituent carbon and hydrogen atoms, and proposed an equation based on the reduced diameter, d^{jk}_r, of colliding particles as:

$$\Phi^{ij}_0 = -6.6891 \times 10^{-23} (d^{jk}_r)^3 + 1.1244 \times 10^{-21} (d^{jk}_r)^2 + 1.1394 \times 10^{-20} d^{jk}_r - 5.5373 \times 10^{-21}, \tag{2.58}$$

$$d^{jk}_r = \frac{d^i_c \cdot d^j_c}{d^i_c + d^j_c}. \tag{2.59}$$

Equation (2.58) is valid for d^{jk}_r between 1 and 7 nm, and ζ^{ij} is assumed as unity for particles with a reduced diameter larger than 7 nm.

Other Source terms

Inception introduces equal number of agglomerates and primary particles to the first section.

$$\left(S^i_{N_{agg}} \right)_{inc} = \begin{cases} \dfrac{1}{Av} \dfrac{I_{N,inc}}{C^i_{agg}}, & \text{if } i = 1 \\ 0.0, & \text{else} \end{cases} \tag{2.60}$$

$$\left(S^i_{N_{pri}}\right)_{inc} = \begin{cases} \dfrac{1}{Av}\dfrac{I_{N,inc}}{C^i_{agg}}, & \text{if } i = 1 \\ 0.0, & \text{else} \end{cases} \tag{2.61}$$

$$\left(S^i_{H_{tot}}\right)_{inc} = \begin{cases} I_{H,inc}, & \text{if } i = 1 \\ 0.0. & \text{else} \end{cases} \tag{2.62}$$

Surface growth and PAH adsorption increase both the carbon mass and hydrogen content of agglomerates, transferring them to higher sections. The rate at which agglomerates are removed from the original section and added to the target section is calculated to ensure mass conservation. Specifically, it is determined by dividing the mass growth rate by the difference in mass of adjacent sections as:

$$\left(S^i_{N_{agg}}\right)_{haca,ads} = \frac{1}{Av}\begin{cases} -\dfrac{I^i_{C_{tot},haca} + I^i_{C_{tot},ads}}{C^{i+1}_{agg} - C^i_{agg}}, & \text{if } i = 1 \\ \dfrac{I^{i-1}_{C_{tot},haca} + I^{i-1}_{C_{tot},haca}}{C^i_{agg} - C^{i-1}_{agg}} - \dfrac{I^i_{C_{tot},haca} + I^i_{C_{tot},ads}}{C^{i+1}_{agg} - C^i_{agg}}, & \text{if } 1 < i < n_{sec} \\ \dfrac{I^{i-1}_{C_{tot},haca} + I^{i-1}_{C_{tot},ads}}{C^i_{agg} - C^{i-1}_{agg}}. & \text{if } i = n_{sec} \end{cases} \tag{2.63}$$

As agglomerates move up/down through sections, they carry the number of primary particles as well as hydrogen atoms, so the transfer rate of agglomerates is multiplied by n^i_p and H^i_{agg}, respectively.

$$\left(S^i_{N_{pri}}\right)_{haca,ads} = \frac{1}{Av}\begin{cases} -\dfrac{I^i_{C_{tot},haca} + I^i_{C_{tot},ads}}{C^{i+1}_{agg} - C^i_{agg}}, & \text{if } i = 1 \\ \dfrac{I^{i-1}_{C_{tot},haca} + I^{i-1}_{C_{tot},ads}}{C^i_{agg} - C^{i-1}_{agg}} n^{i-1}_p - \dfrac{I^i_{C_{tot},haca} + I^i_{C_{tot},ads}}{C^{i+1}_{agg} - C^i_{agg}} n^i_p, & \text{if } 1 < i < n_{sec} \\ \dfrac{I^{i-1}_{C_{tot},haca} + I^{i-1}_{C_{tot},ads}}{C^i_{agg} - C^{i-1}_{agg}} n^{i-1}_p, & \text{if } i = n_{sec} \end{cases} \tag{2.64}$$

$$\left(S^i_{H_{tot}}\right)_{haca,ads} = \frac{1}{Av}\begin{cases} -\dfrac{I^i_{C_{tot},haca}+I^i_{C_{tot},ads}}{C^{i+1}_{agg}-C^i_{agg}}H^i_{agg}+I^i_{H_{tot},haca}+I^i_{H_{tot},ads}, & \text{if } i = \\ \dfrac{I^{i-1}_{C_{tot},haca}+I^{i-1}_{C_{tot},ads}}{C^i_{agg}-C^{i-1}_{agg}}H^{i-1}_{agg}-\dfrac{I^i_{C_{tot},haca}+I^i_{C_{tot},ads}}{C^{i+1}_{agg}-C^i_{agg}}H^i_{agg}+I^i_{H_{tot},haca}+I^i_{H_{tot},ads}, & \text{if } 1 \\ \dfrac{I^{i-1}_{C_{tot},haca}+I^{i-1}_{C_{tot},ads}}{C^i_{agg}-C^{i-1}_{agg}}H^{i-1}_{agg}+I^i_{H_{tot},haca}+I^i_{H_{tot},ads}. & \text{if } i = \end{cases} \tag{2.65}$$

Similarly, the agglomerates lose carbon mass by oxidation, and descend to the lower sections carrying primary particle and hydrogen.

$$\left(S^i_{N_{agg}}\right)_{ox} = \frac{1}{Av}\begin{cases} \dfrac{I^{i+1}_{C_{tot},ox}}{C^{i+1}_{agg}-C^i_{agg}}-\dfrac{I^i_{C_{tot},ox}}{C^i_{agg}}, & \text{if } i=1 \\ \dfrac{I^{i+1}_{C_{tot},ox}}{C^{i+1}_{agg}-C^i_{agg}}-\dfrac{I^i_{C_{tot},ox}}{C^i_{agg}-C^{i-1}_{agg}}, & \text{if } 1<i<n_{sec} \\ -\dfrac{I^i_{C_{tot},ox}}{C^i_{agg}-C^{i-1}_{agg}}, & \text{if } i=n_{sec} \end{cases} \tag{2.66}$$

$$\left(S^i_{N_{pri}}\right)_{ox} = \frac{1}{Av}\begin{cases} \dfrac{I^{i+1}_{C_{tot},ox}}{C^{i+1}_{agg}-C^i_{agg}}n^{i+1}_p-\dfrac{I^i_{C_{tot},ox}}{C^i_{agg}}, & \text{if } i=1 \\ \dfrac{I^{i+1}_{C_{tot},ox}}{C^{i+1}_{agg}-C^i_{agg}}n^{i+1}_p-\dfrac{I^i_{C_{tot},ox}}{C^i_{agg}-C^{i-1}_{agg}}n^i_p, & \text{if } 1<i<n_{sec} \\ -\dfrac{I^i_{C_{tot},ox}}{C^i_{agg}-C^{i-1}_{agg}}n^i_p, & \text{if } i=n_{sec} \end{cases} \tag{2.67}$$

$$\left(S^i_{H_{tot}}\right)_{ox} = \frac{1}{Av}\begin{cases} \dfrac{I^{i+1}_{C_{tot},ox}}{C^{i+1}_{agg}-C^i_{agg}}H^{i+1}_{agg}-\dfrac{I^i_{C_{tot},ox}}{C^i_{agg}}H^i_{agg}+I^i_{H_{tot},ox}, & \text{if } i=1 \\ \dfrac{I^{i+1}_{C_{tot},ox}}{C^{i+1}_{agg}-C^i_{agg}}H^{i+1}_{agg}-\dfrac{I^i_{C_{tot},ox}}{C^i_{agg}-C^{i-1}_{agg}}H^i_{agg}+I^i_{H_{tot},ox}, & \text{if } 1<i<n_{sec} \\ -\dfrac{I^i_{C_{tot},ox}}{C^i_{agg}-C^{i-1}_{agg}}H^i_{agg}+I^i_{H_{tot},ox}. & \text{if } i=n_{sec} \end{cases} \tag{2.68}$$

2.3.3 Monodisperse Population Balance Model

The MPBM used in this study is based on the monodisperse model presented in [150], and it tracks N_{pri}, N_{agg}, C_{tot} and H_{tot}. The coagulation model follows the same principles as the SPBM, but the calculations are greatly simplified due to the monodispersity assumption. This approach has been shown to achieve accuracy comparable to that of DEM and SPBM when applied with well-justified assumptions [67, 78]. The rate of decay of agglomerates is simply described as:

$$I_{coag} = -\frac{1}{2}\zeta\beta N_{agg}^2, \tag{2.69}$$

where ζ is the collision efficiency of agglomerates calculated using Equation (2.57), and β is the collision frequency of agglomerates for the free-molecular (Kn > 10) to continuum regimes ($Kn < 0.1$). The value of β in the transition regime ($0.1 < Kn < 10$) can be calculated from the harmonic mean of the continuum (β_{cont}) and free-molecular (β_{fm}) regime values. Additionally, an enhancement factor of %82 is applied to take into account the effect of polydispersity [66] as:

$$\beta = 1.82\frac{\beta_{fm}\cdot\beta_{cont}}{\beta_{fm}+\beta_{cont}}, \tag{2.70}$$

$$\beta_{fm} = 4\sqrt{\frac{\pi k_B T}{m_{agg}}}d_c^2, \tag{2.71}$$

$$\beta_{cont} = 8\pi d_m D, \tag{2.72}$$

where d_c, D, and m_{agg} are calculated using Equations (2.26), (2.33), and (2.30), respectively. Alternatively, β can be obtained using Fuchs interpolation [151] as:

$$\beta = \beta_{cont}\left(\frac{d_c}{d_c + 2\sqrt{2}\delta} + \frac{8D}{\sqrt{2}cd_c}\right)^{-1}, \tag{2.73}$$

where c (mean velocity of particles) and δ (mean stop distance of particles) are calculated using Equations (2.51) and (2.52). The source terms of tracked variables combines the effect of the inception, surface growth, oxidation and coagulation.

$$S_{N_{agg}} = \frac{I_{N,inc}}{n_{c,min}} + I_{coag}, \tag{2.74}$$

$$S_{N_{pri}} = \frac{I_{N,inc}}{n_{c,min}}, \tag{2.75}$$

$$S_{C_{tot}} = I_{C_{tot},inc} + I_{C_{tot},haca} + I_{C_{tot},ads} - I_{C_{tot},ox}, \tag{2.76}$$

$$S_{H_{tot}} = I_{H_{tot},inc} + I_{H_{tot},haca} + I_{H_{tot},ads} - I_{H_{tot},ox}. \tag{2.77}$$

2.4 PAH growth models

Four different PAH growth models are implemented in Omnisoot to describe the conversion of PAHs into incipient particles and their adsorption onto existing agglomerates. In other words, these models explain the calculation of $I_{\varphi,inc}$ and $I^i_{\varphi,ads}$. The implemented models are: Irreversible Dimerization [39], Reactive Dimerization [56], Dimer Coalescence [84], and E-Bridge Modified [24]. All models are based on PAH collision, as supported by ample evidence in the literature [26–28], but they differ in terms of reversibility, temperature dependence, and the number of steps involved. The collision frequency of gaseous species including PAH molecules and polymers depends on their mass and diameter, and it is obtained as:

The collision frequency of gaseous species, including PAH molecules and polymers, depends on their mass and diameter and is calculated as:

$$\beta_{dim_{jk}} = 2.2 \cdot d_r^2 \sqrt{\frac{8\pi k_B T}{m_r}}, \tag{2.78}$$

where 2.2 is the vdW enhancement factor [56]. d_r and m_r are reduced diameter and mass for two PAH molecules/dimers, respectively, calculated as:

$$d_r = \frac{d_{PAH_k} \cdot d_{PAH_j}}{d_{PAH_k} + d_{PAH_j}}, \tag{2.79}$$

$$m_r = \frac{m_{PAH_k} \cdot m_{PAH_j}}{m_{PAH_k} + m_{PAH_j}}, \tag{2.80}$$

where m_{PAH_j} and d_{PAH_j} are the mass and equivalent diameter of the j^{th} PAH molecule, respectively, and are obtained as:

$$m_{PAH_j} = \frac{W_{PAH_j}}{Av}, \tag{2.81}$$

$$d_{PAH_j} = \left(\frac{6 \cdot m_{PAH_j}}{\pi \cdot \rho_{PAH_j}}\right)^{1/3}, \tag{2.82}$$

where ρ_{PAH_j} is the equivalent density of PAH estimated using the relation proposed by Johansson et al. [152] as:

$$\rho_{PAH_j} = 171943.5197 \frac{W_{carbon} \cdot n_{C,PAH_j} + W_{hydrogen} \cdot n_{H,PAH_j}}{n_{C,PAH_j} + n_{H,PAH_j}}, \tag{2.83}$$

where n_{C,PAH_j} and n_{H,PAH_j} denote the number of carbon and hydrogen atoms in the j^{th} PAH molecule, respectively. The collision frequency of PAH$_j$ and soot agglomerates in each section can be determined for the entire regime by harmonic mean of the collision frequencies in the free-molecular and continuum regimes as:

$$\beta^i_{ads_j} = \frac{\beta^i_{fm,ads} \cdot \beta^i_{cont,ads}}{\beta^i_{fm,ads} + \beta^i_{cont,ads}}, \tag{2.84}$$

$$\beta^i_{fm,ads_j} = 2.2 \sqrt{\frac{\pi k_B T}{2} \left(\frac{1}{m^i_{agg}} + \frac{1}{m_{PAH_j}} \right)} \left(d^i_g + d_{PAH} \right)^2, \tag{2.85}$$

$$\beta^i_{cont,ads_j} = \frac{2k_B T}{3\mu} \left[\frac{C^i\left(d_m\right)}{d^i_g} + \frac{C^i\left(d_{PAH_j}\right)}{d_{PAH_j}} \right] \left(d_g + d_{PAH_j} \right), \tag{2.86}$$

where C^i is the Cunningham correction factor calculated using Equation (2.35).

2.4.1 Irreversible Dimerization

The Irreversible Dimerization is based on the collision of a pair of PAH molecules forming a dimer. The sequential growth continues leading formation of trimers and tetramers until the PAH cluster mass reaches a threshold that can be considered a solid particle. For practical purposes, a dimer is usually considered as an incipient particle that grows by surface growth and coagulation. A single-step collision of two similar PAHs forms a new dimer as:

$$\mathrm{PAH}_j + \mathrm{PAH}_j \xrightarrow{k_{f,dim_j}} \mathrm{Dimer}_j \tag{R 2.1}$$

Similarly, the adsorption of each PAH molecule on soot particles is described by the irreversible collision of soot and PAH$_j$ as:

$$\mathrm{PAH}_j + \mathrm{Soot} \xrightarrow{k_{f,ads_j}} \mathrm{Soot{-}PAH}_j \tag{R 2.2}$$

The forward rate of dimerization, k_{f,dim_j}, and adsorption, k_{f,ads_j}, in Reactions (R 2.1) and (R 2.2) are calculated as:

$$k_{f,dim_j} = \gamma_{inc} \cdot \beta_{jj,PAH} \cdot Av, \tag{2.87}$$

$$k^i_{f,ads_j} = \gamma_{ads_j} \cdot \beta^i_{j,ads} \cdot Av, \tag{2.88}$$

where $\beta_{jk,\text{PAH}}$ and $\beta^i_{j,\text{ads}}$ are computed using Equations (2.78) and (2.84), respectively, and γ_{inc} and γ_{ads} are the collision efficiencies for dimerization and adsorption, respectively. Their values range from 10^{-7} to 1 and are typically chosen to match the predicted soot properties with experimental data. The dimerization rate of PAH$_j$ is then calculated as:

$$\omega_{dim_j} = \eta_{inc} k_{f,dim_j} [\text{PAH}_j][\text{PAH}_j]. \tag{2.89}$$

The partial source terms of inception are calculated as:

$$I_{N,inc} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH}} 2\omega_{dim_j} n_{PAH_j,C}, \tag{2.90}$$

$$I_{C_{tot},inc} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH}} 2\omega_{dim_j} n_{PAH_j,C}. \tag{2.91}$$

$$I_{H_{tot},inc} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH}} 2\omega_{dim_j} n_{PAH_j,H}. \tag{2.92}$$

The rate of PAH adsorption for each section is obtained as:

$$\omega^i_{ads_j} = \eta_{ads} k^i_{f,ads_j} [\text{soot}^i][\text{PAH}_j]. \tag{2.93}$$

The contribution of PAH adsorption to the source terms are expressed as:

$$I^i_{C_{tot},ads} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH_j}} \omega^i_{ads_j} n_{PAH_j,C}, \tag{2.94}$$

$$I^i_{H_{tot},ads} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH_j}} \omega^i_{ads_j} (n_{PAH_j,H} - 2). \tag{2.95}$$

Note that PAH adsorption is a mass growth phenomenon that only changes C_{tot} and H_{tot} but does not affect number of N_{agg} and N_{pri}. Each PAH molecule loses one H atom becoming a radical that forms bonds with a dehydrogenated site on soot surface, so two H atoms are released during adsorption that is taken into account in Equation (2.95).

The formation of a dimer consumes two PAH molecules, and during adsorption one PAH molecule is removed from the gas mixture, so the total rate of removal of PAH_j by Irreversible Dimerization is obtained as:

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{inc} = -2\omega_{dim_j}, \tag{2.96}$$

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{inc} = -\sum_{i=1}^{n_{sec}} \omega^i_{ads_j}. \tag{2.97}$$

Moreover, one H_2 is released to the gas mixture as a result of the adsorption process.

$$\left(\frac{\mathrm{d}\,[\mathrm{H}_2]}{\mathrm{d}t}\right)_{ads} = \sum_{i=1}^{n_{sec}} \omega^i_{ads_j}. \tag{2.98}$$

2.4.2 Reactive Dimerization

This model was built on Irreversible Dimerization with a main difference: The first step of dimerization and adsorption is reversible forming physically bonded dimers followed by a irreversible carbonization that leads to chemical bond formation in dimers [56]. This approach allows the formation of homo- and hetero-dimers. The dimerization of PAH_j and PAH_k is described as:

$$\mathrm{PAH}_j + \mathrm{PAH}_k \underset{k_{r,dim_{jk}}}{\overset{k_{f,dim_{jk}}}{\rightleftarrows}} \mathrm{Dimer}^*_{ij} \tag{R 2.3}$$

$$\mathrm{Dimer}^*_{jk} \xrightarrow{k_{reac}} \mathrm{Dimer}_{jk} \tag{R 2.4}$$

where $\mathrm{Dimer^*}_{jk}$ and Dimer_{jk} are physically and chemically bonded dimers, respectively, from PAH_j and PAH_k. The forward rate of physical dimerization, $k_{f,dim_{jk}}$, is calculated as:

$$k_{f,dim_{jk}} = p'' \cdot \beta_{jk,PAH} \cdot Av, \tag{2.99}$$

where $\beta_{jk,\mathrm{PAH}}$ is calculated using Equation (2.78), and $p'' = 0.1$ accounts for the probability of PAH–PAH collisions occurring in the "FACE" configuration that result in successful van der Waals (vdW) bond formation [153]. The reverse rate of physical dimerization, $k_{r,\mathrm{dim}_{jk}}$, is obtained from the dimerization equilibrium constant [52] as:

$$k_{r,dim_{jk}} = \frac{k_{f,dim_{jk}}}{K_{eq}}, \tag{2.100}$$

$$\log_{10} K_{eq} = a\frac{\epsilon_{jk}}{RT} + b, \tag{2.101}$$

$$\epsilon_{jk} = cW_{jk} - d, \tag{2.102}$$

$$W_{jk} = \frac{W_j \cdot W_k}{W_j + W_k}, \tag{2.103}$$

where $a = 0.115$ (obtained from pyrere dimerization data [47]) and $b = 1.8$ [56], $c = 933420$ J/kg, and $d = 34053$ J/mol [56].

The rate constant of chemical bond formation, k_{reac}, is defined in the Arrhenius form [48] as

$$k_{reac} = 5 \times 10^6 \cdot e^{(-96232/RT)}. \tag{2.104}$$

Assuming a steady state condition for physical dimers, i.e., $\partial[\mathrm{Dimer}^*_{\mathrm{jk}}]/\partial \mathrm{t} = 0$, the rate of formation of chemically-bonded dimers can be obtained as:

$$\omega_{dim_{jk}} = k_{reac}\frac{k_{f,dim_{jk}}[\mathrm{PAH}_j][\mathrm{PAH}_k]}{k_{r,dim_{jk}} + k_{reac}}. \tag{2.105}$$

The contribution of dimer formation to the partial source terms is evaluated by looping over all combinations of PAH precursors as:

$$I_{N,inc} = \frac{1}{\rho}\sum_{j=1}^{n_{PAH}}\sum_{k=j}^{n_{PAH}} \omega_{dim_{kj}}\left(n_{PAH_j,C} + n_{PAH_k,C}\right), \tag{2.106}$$

$$I_{C_{tot},inc} = \frac{1}{\rho}\sum_{j=1}^{n_{PAH}}\sum_{k=j}^{n_{PAH}} \omega_{dim_{kj}}\left(n_{PAH_j,C} + n_{PAH_k,C}\right), \tag{2.107}$$

$$I_{H_{tot},inc} = \frac{1}{\rho}\sum_{j=1}^{n_{PAH}}\sum_{k=j}^{n_{PAH}} \omega_{dim_{kj}}\left(n_{PAH_j,H} + n_{PAH_k,H}\right). \tag{2.108}$$

Similarly, PAH adsorption is described by a two-step process where the collision of PAH_j with soot agglomerates leads to a physically bonded $\mathrm{Soot{-}PAH}^*_j$ that is carbonized and forms a chemically bonded $\mathrm{Soot{-}PAH}_j$ on soot surface.

$$\mathrm{PAH}_j + \mathrm{Soot} \xrightleftharpoons[k_{r,ads_j}]{k_{f,ads_j}} \mathrm{Soot{-}PAH}^*_j \tag{R 2.5}$$

$$\mathrm{Soot{-}PAH}^*_j \xrightarrow{k_{c,ads}} \mathrm{Soot{-}PAH}_j \tag{R 2.6}$$

The forward and reverse rate of PAH-soot collision are calculated as:

$$k^i_{f,ads_j} = \beta^i_{j,ads} \cdot Av, \tag{2.109}$$

$$k^i_{r,ads_j} = k^i_{f,ads_j} \cdot 10^{-b} e^{-a\epsilon^i_{soot,j}\ln(10)/(RT)}, \tag{2.110}$$

$$\epsilon^i_{soot,j} = cW^i_{soot,j} - d, \tag{2.111}$$

where $\beta^i_{j,ads}$ is calculated using Equation (2.84). The values of a, b, c, and d are the same as those explained in the inception part. Computing $\epsilon^i_{soot,j}$ also requires the equivalent soot molecular weight, W^i_{soot}, which is estimated from carbon mass of each agglomerate as:

$$W^i_{soot} = \frac{C^i_{tot} W_{carbon}}{N^i_{agg}}. \tag{2.112}$$

The rate constant of carbonization of Soot$-$PAH$^*_j$ is defined as in an Arrhenius form similar to the inception formulation (Equation (2.104)). The pre-exponential factor is adjusted by matching the numercial PSD [48] with measurements in the ethylene pyrolysis in a flow reactor [154].

$$k_{c,ads} = 2 \times 10^{10} \cdot e^{(-96232/RT)}. \tag{2.113}$$

The total adsorption rate can be calculated assuming a steady-state concentration for the physically adsorbed PAH on soot, i.e., $\partial[\text{Soot}-\text{PAH}^*_j]/\partial t = 0$, calculated in a similar way to the inception flux (Equation (2.105)) as:

$$\omega^i_{ads_j} = k_{c,ads} \frac{k_{f,ads_j}[\text{soot}^i][\text{PAH}_j]}{k_{r,ads_j} + k_{c,ads_j}}, \tag{2.114}$$

The contribution of PAH adsorption rate to the partial source terms can be expressed as:

$$I^i_{C_{tot},ads} = \frac{1}{\rho} \sum_{i=1}^{n_{PAH}} \omega^i_{ads_j} n_{C,PAH_j}, \tag{2.115}$$

$$I^i_{H_{tot},ads} = \frac{1}{\rho} \sum_{i=1}^{n_{PAH}} \omega^i_{ads_j} \left(n_{H,PAH_j} - 2\right). \tag{2.116}$$

The rate of removal of PAH from the gas mixture due to inception and PAH adsorption is given as:

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{inc} = -\sum_{k=1}^{n_{PAH}} \omega_{dim_{jk}}, \tag{2.117}$$

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{ads} = -\sum_{i=1}^{n_{sec}} \omega^i_{ads_j}. \tag{2.118}$$

Additionally, during the PAH adsorption process one H_2 is released to the gas mixture changing the concentration of H_2 as:

$$\left(\frac{\mathrm{d}\,[\mathrm{H}_2]}{\mathrm{d}t}\right)_{ads} = \sum_{i=1}^{n_{sec}} \omega^i_{ads_j}. \tag{2.119}$$

2.4.3 Dimer Coalescence

The Dimer Coalescence model is a multi-step irreversible model proposed by Blanquart and Pitsch [84] where self-collision of PAH molecules forms dimers, which are an intermediate state between gaseous PAH molecules and solid soot particles. The dimers can either form incipient soot particles through self-coalescence or adsorb on the surface of existing soot particles and contribute to their surface growth. The following equations describing the inception and surface growth in Dimer Coalescence adopted from the work of Sun et al. [155].

$$\mathrm{PAH}_j + \mathrm{PAH}_j \xrightarrow{k_{dim_j}} \mathrm{Dimer}_j \tag{R 2.7}$$

$$\mathrm{Dimer}_j + \mathrm{Dimer}_j \xrightarrow{k_{inc_j}} \mathrm{Tetramer}_j \tag{R 2.8}$$

$$\mathrm{Dimer}_j + \mathrm{Soot} \xrightarrow{k_{ads_j}} \mathrm{Soot{-}PAH}_j \tag{R 2.9}$$

The rate constant of dimerization, k_{dim_j}, and inception, k_{inc_j}, are calculated from the collision rate of PAHs as:

$$k_{dim_j} = \gamma_{dim_j} \cdot \beta_{jj,PAH} \cdot Av, \tag{2.120}$$

$$k_{inc_j} = \beta_{jj,dimer} \cdot Av, \tag{2.121}$$

where $\beta_{jj,PAH}$, $\beta_{jj,dimer}$ are both calculated using Equation (2.78). γ_{dim_j} is the dimerization efficiency assumed to scale with the fourth power of the PAH molecular weight [85] as:

$$\gamma_{dim_j} = C_{N,j} \cdot W^4_{PAH_j}. \tag{2.122}$$

Blanquart and Pitsch [84] estimated the constant $C_{N,j}$ in Equation (2.122) by comparing the profiles of several PAH species with the experimental measurements in a single premixed benzene flame [156], and provide $C_{N,j}$ values for various PAHs, which are listed in Table 1 in [85]. The rate of dimer collision is expressed as:

$$\omega_{dim_j} = \eta_{inc} k_{inc_j} [\mathrm{Dimer}_j][\mathrm{Dimer}_j]. \tag{2.123}$$

Similarly, the rate of adsorption of dimers on soot particles is obtained as:

$$\omega^i_{ads_j} = \eta_{ads} k^i_{ads_j} [\mathrm{soot}^i][\mathrm{Dimer}_j], \tag{2.124}$$

$$k^i_{ads_j} = \beta^i_{ads_j} \cdot Av. \tag{2.125}$$

Assuming fast dimer consumption leads to the steady-state concentration of dimers, which can be determined by solving a quadratic equation [85] as:

$$a_{inc_j} [\mathrm{Dimer}_j]^2 + b_{ads_j} [\mathrm{Dimer}_j] = \omega_{dim,j}, \tag{2.126}$$

$$[\mathrm{Dimer}_j] = \begin{cases} \dfrac{-b_{ads_j} + \sqrt{\Delta_j}}{2a_{inc_j}}, & \text{if } \Delta_j \geq 0 \\ 0 & \text{if } \Delta_j < 0 \end{cases}, \tag{2.127}$$

$$\Delta_j = b^2_{ads_j} + 4a_{inc_j}\omega_{dim,j}, \tag{2.128}$$

where $a_{inc_j} = k_{inc_j}$ and b_{ads_j} is calculated by summing the adsorption rate of dimers for all sections as:

$$b_{ads_j} = \sum_{i=1}^{n_{sec}} k^i_{ads_j} [\mathrm{soot}^i] \tag{2.129}$$

After determining the concentration of each dimer, the contributions of inception and PAH adsorption to the source terms of the tracked soot variables can be calculated in a manner similar to previous inception models, by accounting for the number of carbon and hydrogen atoms involved in the process as:

$$I_{N,inc} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH}} 4\omega_{inc_j} n_{PAH_j,C}, \tag{2.130}$$

$$I_{C_{tot},inc} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH}} 4\omega_{inc_j} n_{PAH_j,C}, \tag{2.131}$$

$$I_{H_{tot},inc} = \frac{1}{\rho} \sum_{j=1}^{n_{PAH}} 4\omega_{inc_j} n_{PAH_j,H}, \tag{2.132}$$

$$I^i_{C_{tot},ads} = \frac{1}{\rho} \sum_{i=1}^{n_{PAH}} 2\omega^i_{ads_j} n_{C,PAH_j}, \tag{2.133}$$

$$I^i_{H_{tot},ads} = \frac{1}{\rho} \sum_{i=1}^{n_{PAH}} 2\omega^i_{ads_j} \left(n_{H,PAH_j} - 1\right). \tag{2.134}$$

The rate of removal of PAHs and release of H_2 molecule due to PAH adsorption is calculated as:

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{inc} = -4 \sum_{k=1}^{n_{PAH}} \omega_{inc_j}, \tag{2.135}$$

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{ads} = -2 \sum_{i=1}^{n_{sec}} \omega^i_{ads_j}, \tag{2.136}$$

$$\left(\frac{\mathrm{d}\,[\mathrm{H}_2]}{\mathrm{d}t}\right)_{ads} = \sum_{i=1}^{n_{sec}} \omega^i_{ads_j}. \tag{2.137}$$

2.4.4 E-Bridge Modified

The E-Bridge model was originally proposed by Frenklach and Mebel [24] to describe soot inception using a HACA-like scheme that starts with the dehydrogenation of PAH monomers, often pyrene, which forms the monomer radicals and continues with the sequential addition of the radicals that form dimers, trimers and larger polymers until the PAH structure reaches a mass threshold and the clustering process becomes irreversible [24]. Here, a modified version of this model, called E-Bridge Modified, is used where dimers are considered as incipient soot, and monomer radicals are adsorbed on soot agglomerates. This PAH growth model is described using the following set of pathways:

$$\mathrm{PAH}_j + \mathrm{H} \underset{k_{r,d_j}}{\overset{k_{f,d_j}}{\rightleftharpoons}} \dot{\mathrm{P}}\mathrm{AH}_j + \mathrm{H}_2 \tag{R 2.10}$$

$$\dot{\mathrm{P}}\mathrm{AH}_j + \mathrm{H} \xrightarrow{k_{f,h_j}} \mathrm{PAH}_j \tag{R 2.11}$$

Table 2.2 Rate coefficients for the monomer de-/hydrogenation reaction of E-bridge formation in Arrhenius form $\mathrm{k} = \mathrm{AT^n} \cdot \mathrm{e^{-E/RT}}$ [24].

Reaction		A $[\mathrm{m^3/mol \cdot s}]$	n	E/R [K]
(R 2.10)	f	$98\times\ \mathrm{n_{C,PAH_j}}$	1.8	7563.519
	r	1.6×10^{-2}	2.63	2145.346
(R 2.11)	f	4.8658×10^{7}	0.13	0.0

$$\dot{\mathrm{P}}\mathrm{AH}_j + \dot{\mathrm{P}}\mathrm{AH}_j \xrightarrow{k_{inc_j}} \mathrm{Dimer}_j \tag{R 2.12}$$

$$\dot{\mathrm{P}}\mathrm{AH}_j + \mathrm{Soot} \xrightarrow{k_{ads_j}} \mathrm{Soot{-}PAH}_j \tag{R 2.13}$$

The rate constants of Reactions (R 2.10) and (R 2.11) are listed in Table 2.2. k_{inc_j} and k_{ads_j} are the rate constants of dimer production and PAH adsorption, respectively, which are calculated as:

$$k_{inc_j} = \beta_{jj,PAH} \cdot Av, \tag{2.138}$$

$$k^i_{ads_j} = \beta^i_{ads_j} \cdot Av, \tag{2.139}$$

where $\beta_{jj,PAH}$ and $\beta^i_{ads_j}$ are obtained using Equations (2.78) and (2.84), respectively. The rate of dimer formation and adsorption is calculated as:

$$\omega_{dim_j} = k_{inc_j}[\cdot \mathrm{PAH}_j]^2, \tag{2.140}$$

$$\omega^i_{ads_j} = k^i_{ads_j}[\mathrm{soot}^i][\dot{\mathrm{P}}\mathrm{AH}_j]. \tag{2.141}$$

The calculation of rate of inception and PAH adsorption from PAH_j requires the concentration of $\dot{\mathrm{P}}\mathrm{AH}_j$, which can be determined by applying the steady-state assumption for $\dot{\mathrm{P}}\mathrm{AH_j}$, i.e., $d[\dot{\mathrm{P}}\mathrm{AH}_j]/dt = 0$, resulting in a quadratic equation as:

$$a_{inc_j}[\dot{\mathrm{P}}\mathrm{AH}_j]^2 + b_{ads_j}[\dot{\mathrm{P}}\mathrm{AH}_j] + c_j = 0, \tag{2.142}$$

$$a_{inc_j} = k_{inc_j} \tag{2.143}$$

$$b_{ads_j} = k_{r,d_j}[\mathrm{H_2}] + k_{f,h_j}[\mathrm{H}] + \sum_{i=1}^{n_{sec}} k^i_{ads_j}[\mathrm{Soot}]^i \tag{2.144}$$

$$c_{inc_j} = k_{f,d_j}[\mathrm{PAH}_j][\mathrm{H}] \tag{2.145}$$

Solving the quadratic equation for each PAH gives the concentration of $\mathrm{P\dot{A}H}_j$ as:

$$[\mathrm{P\dot{A}H}_j] = \begin{cases} \dfrac{-b_{ads_j} + \sqrt{\Delta_j}}{2a_{inc_j}}, & \text{if } \Delta_j \geq 0 \\ 0 & \text{if } \Delta_j < 0 \end{cases} \tag{2.146}$$

$$\Delta_j = b^2_{ads_j} - 4a_{inc_j}c_j \tag{2.147}$$

The partial source terms of inception, $I_{\varphi,inc}$, are calculated in the E-Bridge Modified model as:

$$I_{N,inc} = \frac{1}{\rho}\sum_{j=1}^{n_{PAH}} 2\omega_{inc_j} n_{PAH_j,C}, \tag{2.148}$$

$$I_{C_{tot},inc} = \frac{1}{\rho}\sum_{j=1}^{n_{PAH}} 2\omega_{inc_j} n_{PAH_j,C}, \tag{2.149}$$

$$I_{H_{tot},inc} = \frac{1}{\rho}\sum_{j=1}^{n_{PAH}} \omega_{inc_j}\left(n_{PAH_j,H} - 2\right), \tag{2.150}$$

The partial source terms of PAH adsorption, $I^i_{\varphi,ads}$, are calculated for each section in this PAH growth model as:

$$I^i_{C_{tot},ads} = \frac{1}{\rho}\sum_{i=1}^{n_{PAH}} \omega^i_{ads_j} n_{C,PAH_j}, \tag{2.151}$$

$$I^i_{H_{tot},ads} = \frac{1}{\rho}\sum_{i=1}^{n_{PAH}} \omega^i_{ads_j}\left(n_{H,PAH_j} - 2\right). \tag{2.152}$$

The rate of removal of each PAH involved in soot inception and PAH adsorption, and release of H_2 to the gas mixture can be expressed as:

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{inc} = -2\sum_{k=1}^{n_{sec}} \omega_{inc_j}, \tag{2.153}$$

$$\left(\frac{\mathrm{d}\,[\mathrm{PAH}_j]}{\mathrm{d}t}\right)_{ads} = -\sum_{i=1}^{n_{sec}} \omega^i_{ads_j}, \tag{2.154}$$

$$\left(\frac{\mathrm{d}\,[\mathrm{H}_2]}{\mathrm{d}t}\right)_{inc} = \sum_{i=1}^{n_{sec}} \omega^i_{ads_j}. \tag{2.155}$$

Table 2.3 The summary of key features of PAH growth models.

Name	Reversibility	Temperature dependence	The sequence of steps in inception mechanism
Irrev.Dim. [39]	Irreversible	Weak, $\propto\sqrt{T}$	$[\mathrm{PAH}]\xrightarrow{\text{collision}}[\mathrm{Dim.}]$
Reac.Dim. [56]	Reversible	Strong, $\propto \mathrm{e}^{-E/(RT)}$	$[\mathrm{PAH}]\xleftrightarrow{\text{collision}}[\mathrm{Phy.Dim.}]\xrightarrow{\text{carbonization}}[\mathrm{Chem.Dim}]$
Dim.Coal. [85]	Irreversible	Weak, $\propto\sqrt{T}$	$[\mathrm{PAH}]\xrightarrow{\text{collision}}[\mathrm{Dim.}]\xrightarrow{\text{collision}}[\mathrm{Tetra.}]$
EBri.Mod. [24]	Reversible	Strong, $\propto \mathrm{e}^{-E/(RT)}$	$[\mathrm{PAH}]\xleftrightarrow{\text{dehydrogenation}}[\mathrm{Rad.}]\xrightarrow{\text{collision}}[\mathrm{Chem.Dim.}]$

Dim. denotes PAH dimers.

Phy.Dim. and Chem.Dim. represent physically- and chemically-bonded dimers, respectively.

Rad. and Tetra. refer to a PAH radical and a tetramer, respectively.

2.4.5 Summary of features

The comparison of key features of the implemented PAH growth models is provided in Table 2.3. As shown in Fig. 2.6 the inception flux predicted by the Reactive Dimerization and E-Bridge Modified models increases exponentially with temperature (that is, decreases with its inverse), whereas the Irreversible Dimerization and Dimer Coalescence models show only about a factor-of-two increase over the range 500-2500 K.

2.5 Surface reactions model

Heterogeneous surface reactions are described by HACA. The soot growth in HACA scheme is based on a sequential process similar to PAH growth. The hydrogenated armchair sites ($\mathrm{C_{soot}} - \mathrm{H}$) on the edge of aromatic rings are dehydrogenated by H abstraction forming $\mathrm{C_{soot}}{\circ}$ that bonds with C_2H_2 resulting in an additional aromatic ring with hydrogenated sites. These sites can also be attacked by O_2 or OH leading to removal of carbon from soot particles by oxidation. The elementary reactions that describe this sequential process are listed in Table 2.4. The rate of mass growth by HACA is obtained from the reaction of C_2H_2 with dehydrogenated sites as:

$$\omega_{gr}^{i} = \alpha^{i} k_{f4} [\mathrm{C_2H_2}][\mathrm{C}^{i}_{\mathrm{soot}}{\circ}], \tag{2.156}$$

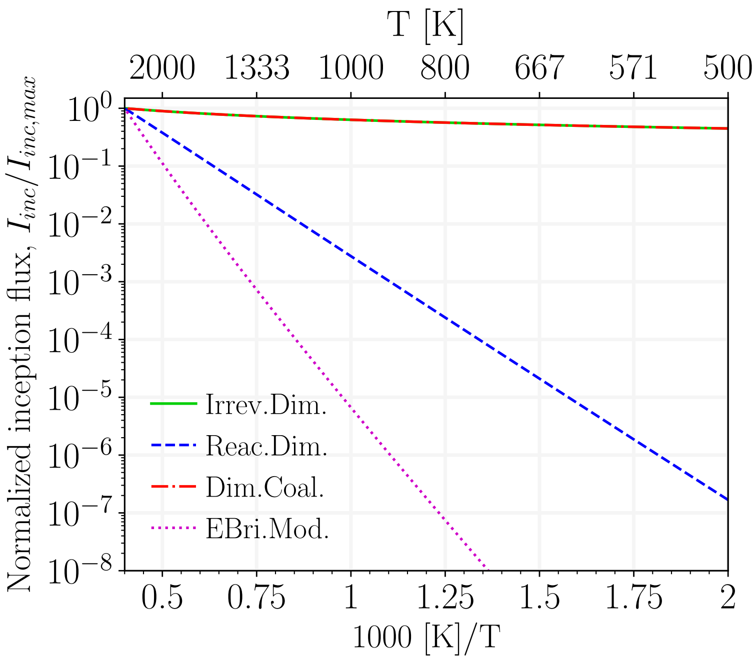


Fig. 2.6 The temperature dependence of inception flux, I_{inc}, predicted using various PAH growth models. I_{inc} of each model is normalized with its maximum to facilitate the comparison.

where k_{f4} denotes the forward rate of Reaction (R 2.17) in Table 2.4, and $[\mathrm{C}^{\mathrm{i}}_{\mathrm{soot}^\circ}]$ is obtained by multiplying the surface density of dehydrogenated sites, $\mathrm{C}^i_{\mathrm{soot}^\circ}$ by the total surface area of soot (per unit of mass of gas mixture) of each section as:

$$[\mathrm{C}^i_{\mathrm{soot}^\circ}] = \frac{\rho}{Av} A^i_{tot} \cdot \chi_{soot^\circ}, \tag{2.157}$$

where $\chi_{\mathrm{soot}^\circ}$ is the surface density of dehydrogenated sites, and is calculated by assuming the steady-state for $[\mathrm{C}_{\mathrm{soot}^\circ}]$ in the system of reactions in Table 2.4 as:

$$\chi_{soot^\circ} = \frac{k_{f1}[\mathrm{H}] + k_{f2}[\mathrm{OH}]}{k_{r1}[\mathrm{H_2}] + k_{r2}[\mathrm{H_2O}] + k_{f3}[\mathrm{H}] + k_{f4}[\mathrm{C_2H_2}] + k_{f5}[\mathrm{O_2}]} \chi_{soot-H}, \tag{2.158}$$

where χ_{soot-H} is the surface density of hydrogenated sites estimated based on the assumption that soot "surface is assumed to be composed of outwardlooking PAH edges with PAH molecular moieties assembled into turbostratic structures" [157]. Considering the layer spacing of 3.15Å and two C–H bonds per benzene ring length, χ_{soot-H} is calculated to be $0.23\,\mathrm{site/Å^2} = 2.3 \times 10^{19}$ site/m^2, which gives the maximum theoretical limit of the reaction sites.

In Equation (2.156), α^i is the surface reactivity factor between 0 and 1 that represents the decline of reaction sites from the theoretical limit due to PAH layer orientation, particle aging, growth and maturity [158, 159], and it has been observed to depend on temperature-time history [160, 59]. The value of α has been described using constant target-specific values as well as empirical equations based on particle size and flame temperature. A detailed review of these can be found in Chapter 4 of [161]. Here, the empirical equation proposed by Appel et al. [57] is used to calculate α^i as:

$$\alpha^i = \tanh\left(\frac{12.56 - 0.00563T}{\log_{10}\left(\frac{\rho_{soot}\cdot Av}{W_{carbon}}\frac{\pi}{6}{d_p^i}^3\right)} - 1.38 + 0.00068T\right). \tag{2.159}$$

Alternatively, α^i can be related to the H/C ratio of soot particles by assuming that all hydrogen atoms reside on the particle surface [84] as:

$$\alpha^i = \frac{H^i_{tot}}{C^i_{tot}}. \tag{2.160}$$

The contribution of HACA to growth source terms can be computed from HACA rates considering the number of carbon atoms in C_2H_2 and number of arm-chair and zig-zag hydrogenated sites on soot particle [85] using

$$I^i_{C_{tot},haca} = 2\omega^i_{gr}/\rho, \tag{2.161}$$

$$I^i_{H_{tot},haca} = 0.25\omega^i_{gr}/\rho. \tag{2.162}$$

The rates of concentration change of C_2H_2 and H radical due to HACA are written as:

$$\left(\frac{d\,[\mathrm{C_2H_2}]}{dt}\right)_{gr} = -\sum_{i=1}^{n_{sec}} \omega^i_{gr}, \tag{2.163}$$

$$\left(\frac{d\,[\mathrm{H}]}{dt}\right)_{gr} = 1.75\sum_{i=1}^{n_{sec}} \omega^i_{gr}. \tag{2.164}$$

The carbons on the surface of soot are oxidized via reaction with O_2 (Reaction (R 2.18)) and OH (Reaction (R 2.19)). As a result, oxidation decreases the total carbon of soot and releases CO and H_2 to the gas mixture. The O_2 and OH oxidation rates are calculated as:

$$\omega^i_{ox,O_2} = \alpha^i k_{f5}[\mathrm{O_2}][C^i_{soot^\circ}], \tag{2.165}$$

Table 2.4 Arrhenius rate coefficients of the surface reactions in HACA [57], $k = AT^n \cdot e^{-E/RT}$.

No.	Reaction		A [$m^3/mol \cdot s$]	n	E/R [K]
(R 2.14)	$C_{soot-H} + H \underset{k_{r,1}}{\overset{k_{f,1}}{\rightleftarrows}} C_{soot^\circ} + H_2$	f	4.17×10^7	0	6542.52
		r	3.9×10^6	0	5535.98
(R 2.15)	$C_{soot-H} + OH \underset{k_{r,2}}{\overset{k_{f,2}}{\rightleftarrows}} C_{soot^\circ} + H_2O$	f	10^4	0.734	719.68
		r	3.68×10^2	1.139	8605.94
(R 2.16)	$C_{soot^\circ} + H \xrightarrow{k_{f,3}} C_{soot-H}$	f	10^4	0.734	719.68
(R 2.17)	$C_{soot^\circ} + C_2H_2 \xrightarrow{k_{f,4}} C_{soot-H} + H$	f	80	1.56	1912.43
(R 2.18)	$C_{soot^\circ} + O_2 \xrightarrow{k_{f,5}} 2\,CO + \text{product}$	f	2.2×10^6	0	3774.53
(R 2.19)	$C_{soot}-H + OH \xrightarrow{k_{f,6}} CO + \text{product}$	f		$\gamma_{OH} = 0.13$	

$$\omega^i_{ox,OH} = \gamma_{OH}\beta^i_{OH}Av[\mathrm{OH}][\mathrm{soot}^i], \tag{2.166}$$

where $\gamma_{OH} = 0.13$ is the reaction probability of OH radicals with soot particles [57]. β_{OH} is the collision frequency of OH and soot particles calculated based on the kinetic theory of gases as:

$$\beta^i_{OH} = \sqrt{\frac{\pi k_B T}{2}\left(\frac{1}{m^i_{agg}} + \frac{1}{m_{OH}}\right)}\left(d^i_c + d_{OH}\right)^2, \tag{2.167}$$

where $m_{OH} = 2.824 \times 10^{-26}$ kg, and $d_{OH} = 0.3$ nm [162] are the mass and equivalent diameter of a OH radical, respectively. Then, the oxidation source term is calculated considering the number of carbon atoms removed from soot through each oxidation pathway as:

$$I^i_{C_{tot},ox} = (2\omega^i_{ox,O_2} + \omega^i_{ox,OH})/\rho. \tag{2.168}$$

The rate of change of concentration of CO, O_2 and OH by oxidation is calculates as:

$$\left(\frac{d\,[\mathrm{CO}]}{dt}\right)_{ox} = 2\sum_{i=1}^{n_{sec}}\omega^i_{ox,O_2}, \tag{2.169}$$

$$\left(\frac{d\,[\mathrm{O_2}]}{dt}\right)_{ox} = -\sum_{i=1}^{n_{sec}} \omega^i_{ox,O_2}, \tag{2.170}$$

$$\left(\frac{d\,[\mathrm{OH}]}{dt}\right)_{ox} = -\sum_{i=1}^{n_{sec}} \omega^i_{ox,OH}, \tag{2.171}$$

$$\left(\frac{d\,[\mathrm{H}]}{dt}\right)_{ox} = \sum_{i=1}^{n_{sec}} \omega^i_{ox,OH}. \tag{2.172}$$

2.6 Gas scrubbing rates

The rate of production/destruction of species involved in soot formation must be taken into account to preserve the mass and energy balance in reactive systems. In order to do that, the production rate of gaseous species calculated by Cantera must be corrected for the rate of release/consumption due to PAH growth and surface reaction models.

$$\left(\frac{d\,[\mathrm{PAH_j}]}{dt}\right)_{tot} = \left(\frac{d\,[\mathrm{PAH_j}]}{dt}\right)_{gas} + \left(\frac{d\,[\mathrm{PAH_j}]}{dt}\right)_{inc} + \left(\frac{d\,[\mathrm{PAH_j}]}{dt}\right)_{ads}. \tag{2.173}$$

H_2 is released to the gas mixture due to inception, PAH adsorption as well as oxidation.

$$\left(\frac{d\,[\mathrm{H_2}]}{dt}\right)_{tot} = \left(\frac{d\,[\mathrm{H_2}]}{dt}\right)_{gas} + \left(\frac{d\,[\mathrm{H_2}]}{dt}\right)_{inc} + \left(\frac{d\,[\mathrm{H_2}]}{dt}\right)_{ads} + \left(\frac{d\,[\mathrm{H_2}]}{dt}\right)_{ox}. \tag{2.174}$$

Surface growth consumes C_2H_2 and adds H_2 to the gas mixture.

$$\left(\frac{d\,[\mathrm{C_2H_2}]}{dt}\right)_{tot} = \left(\frac{d\,[\mathrm{C_2H_2}]}{dt}\right)_{gas} + \left(\frac{d\,[\mathrm{C_2H_2}]}{dt}\right)_{gr}. \tag{2.175}$$

$$\left(\frac{d\,[\mathrm{H}]}{dt}\right)_{tot} = \left(\frac{d\,[\mathrm{H}]}{dt}\right)_{gas} + \left(\frac{d\,[\mathrm{H}]}{dt}\right)_{gr}. \tag{2.176}$$

Oxidation uses O_2 and OH to remove carbon from soot particles and generates H_2 and CO.

$$\left(\frac{d\,[\mathrm{CO}]}{dt}\right)_{tot} = \left(\frac{d\,[\mathrm{CO}]}{dt}\right)_{gas} + \left(\frac{d\,[\mathrm{CO}]}{dt}\right)_{ox}. \tag{2.177}$$

$$\left(\frac{d\,[\mathrm{O2}]}{dt}\right)_{tot} = \left(\frac{d\,[\mathrm{O2}]}{dt}\right)_{gas} + \left(\frac{d\,[\mathrm{O2}]}{dt}\right)_{ox}. \tag{2.178}$$

$$\left(\frac{d\,[\mathrm{OH}]}{dt}\right)_{tot} = \left(\frac{d\,[\mathrm{OH}]}{dt}\right)_{gas} + \left(\frac{d\,[\mathrm{OH}]}{dt}\right)_{ox}. \tag{2.179}$$

Chapter 3

Code Validation

A set of simulations was performed to evaluate the accuracy and reliability of Omnisoot in predicting soot formation. The aerosol dynamics models were validated by comparing the results of the population balance models implemented in Omnisoot with DEM simulations reported in the literature. In addition, carbon and hydrogen mass and energy balances were rigorously assessed to ensure that residuals remain within the bounds of acceptable numerical error.

3.1 Validation of Collision Frequency

The collision frequency function determines the rate at which two particles collide, which results in the reduction of the total number of agglomerates and the increase in particle size. In the absence of strong flow shear or external forces, Brownian motion is the main driving force for particle coagulation. As explained in Sections 2.3.2 and 2.3.3, Omnisoot employs harmonic mean and Fuchs interpolations to calculate collision frequency of agglomerates from free-molecular (Kn $\geq$ 10) to continuum (Kn $\leq$ 0.1) regimes based on the gas mean free path and the particle size.

The test case for validation of collision frequency is based on the DEM simulation of 2000 monodisperse spherical particles with the density of 2200 kg/m^3 in a cubic cell with the constant temperature of 298 K and pressure of 1 atm [163]. Figure 3.1 depicts the collision frequency plotted against Knudsen number (Kn $= 2\lambda/d_m$) obtained by Omnisoot using harmonic mean (red solid line) and Fuchs interpolation (blue dashed line) and the DEM results of Goudeli et al. [163]. The Fuchs interpretation perfectly matches DEM data over the free-molecular regime to the continuum regime. Harmonic mean is also in good agreement with the DEM results in the free-molecular and continuum regimes, but slightly underpredicts the collision frequency in the transition regime with relative errors less than 16%.

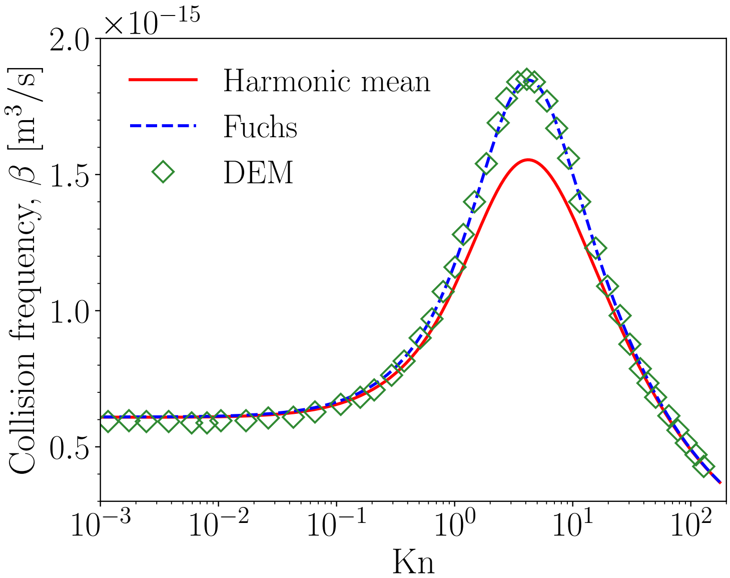


Fig. 3.1 The comparison of collision frequency, β, obtained by Omnisoot using harmonic mean (red solid line) and Fuchs interpolations (blue dashed line) with DEM results (symbols) [163].

3.2 Coagulation

Two test cases were set up to validate the coagulation sub-model of MPBM and SPBM in the free-molecular[1] and continuum[2] regimes. An adiabatic CVR with a volume of 1 m^3 was populated with 2.6261×10^{18} spherical particles with the initial diameters of 2 nm and 668 nm for the free-molecular and continuum cases, respectively. The initial temperature in the free-molecular and continuum simulations are 1800 K and 300 K, respectively.

Figure 3.2 illustrate the schematics of the designed test in which the particles collide and grow in size through coagulation in both simulations, without inception, surface growth, or oxidation. Figure 3.3 demonstrates the evolution of N_{agg}, N_{pri}, d_m, and d_g as predicted by MPBM and SPBM in the free-molecular regime, which are in good agreement with DEM results [150]. N_{pri} is conserved during coagulation, resulting in identical flat trends for both models. In contrast, N_{agg} decreases over time, with a

[1]https://github.com/mohammadadib-cu/omnisoot-cv/tree/main/validations/coagulation/free_molecular
[2]https://github.com/mohammadadib-cu/omnisoot-cv/tree/main/validations/coagulation/continuum

faster decay observed in SPBM due to its treatment of agglomerate polydispersity, which leads to a higher collision frequency compared to MPBM. Therefore, d_m and d_g predicted by SPBM, shown in Figure 3.3b, are slightly larger than those predicted by MPBM.

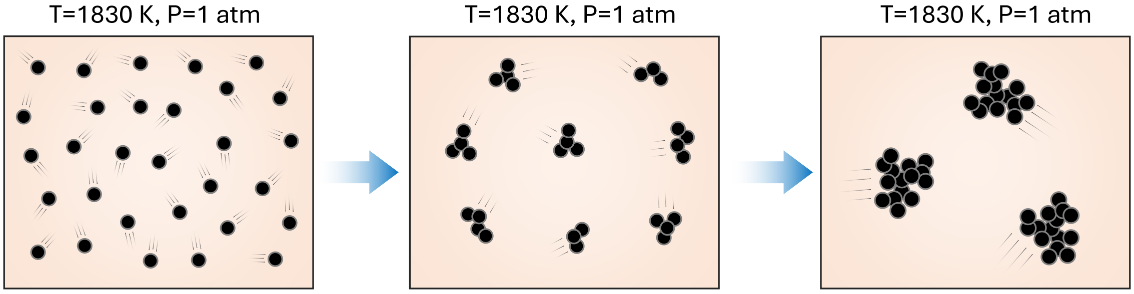


Fig. 3.2 The schematic of the agglomeration process in the coagulation test case where initially spherical particles collide and form agglomerates

The MPBM model cannot resolve the PSD due to the monodispersity assumption. In contrast, the SPBM tracks the number concentration of particles in discrete sections, which allows the construction of an evolving PSD, and the assessment of the size distribution spread. Figure 3.4 shows the evolution of the non-dimensional PSD from $t = 1$ ms to 500 ms for the free-molecular and continuum simulations. The PSD is plotted as a function of the normalized concentration, $\Psi = \bar{v} n_{agg}(v,t)/N_{agg,\infty}$ and dimensionless volume, $\eta = v/\bar{v}$, where $n_{agg}(v,t)$ is the size distribution function of agglomerate, v is the particle volume, $\bar{v}$ is the mean particle volume, $N_{agg,\infty}$ is the total number concentration of agglomerates. After the initial transient ($t > 22$ ms), the PSD rapidly evolves into a full bell curve and remains unchanged, indicating the attainment of a SPSD. This behavior is in good agreement with DEM results and confirms the capability of the SPBM implemented in Omnisoot to capture the SPSD for soot agglomerates in the free-molecular and continuum regimes, a characteristic outcome of Brownian-driven particle coagulation.

Figure 3.4a shows the standard deviation of the mobility diameter, σ_g, predicted by SPBM in the free-molecular regime, which is in close agreement with DEM results. Initially, $\sigma_g = 1$, indicating a monodisperse population, and it gradually increases, reaching a final value of 2.03, which corresponds to the characteristic standard deviation for the free-molecular regime [164].

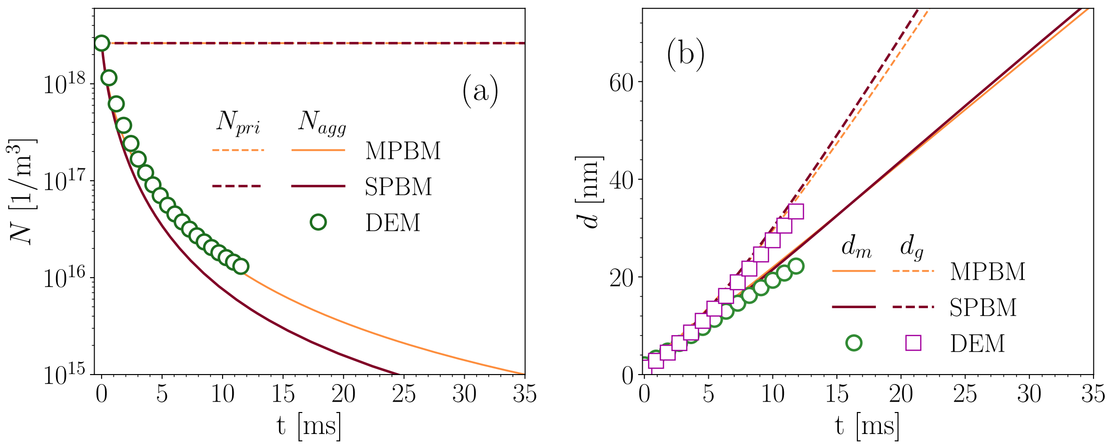


Fig. 3.3 The number concentration of agglomerates and primary particles (a), and mobility and gyration diameters (b) obtained by Omnisoot using MPBM and SPBM in the free-molecular regime, which are in close agreement with the DEM results [150] indicating the validity of the coagulation sub-models.

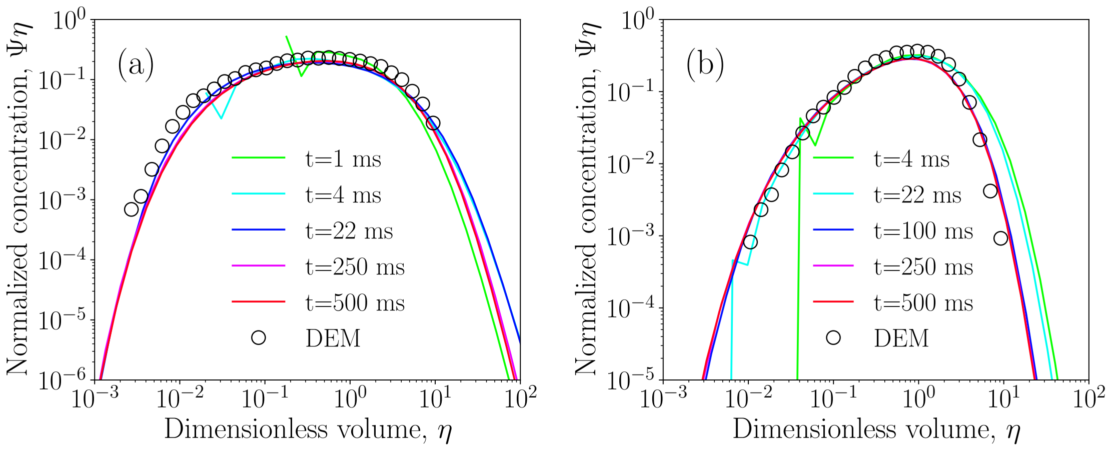


Fig. 3.4 The non-dimensional particle size distribution at different residence times in the free-molecular (a) and continuum regimes (b) overlaps after the initial transient phase, indicating the attainment of a self-preserving size distribution, which is also in good agreement with DEM results [163].

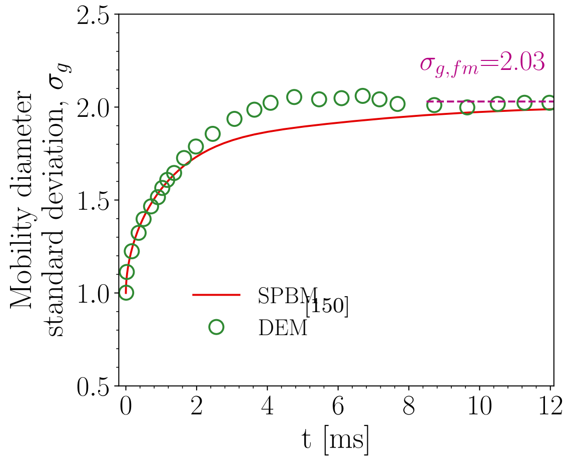


Fig. 3.5 Standard deviation of the mobility diameter, σ_g, in the free-molecular regime obtained by SPBM is in close agreement with DEM results [150].

3.3 Elemental and Energy Balance

To ensure the accuracy and reliability of the simulations, the conservation of elemental mass and energy was assessed across all reactor models implemented in Omnisoot. The elemental balances of carbon, hydrogen, and the total energy (internal or enthalpy depending on the reactor type) of the gas-particle system were evaluated during soot formation processes under various pyrolysis and combustion conditions. A set of simulations[3] was performed using different combinations of PAH growth and particle dynamics models, and the relative errors in elemental and energy balances were monitored over time or reactor length. The relative error of total carbon, hydrogen and energy are calculated for CVR, CPR, PSR, and PFR using different combinations of particle dynamics and inception models. The residuals for all reactor configurations remained within acceptable limits (typically below 10^{-10}), confirming that Omnisoot accurately preserves mass and energy during the coupled evolution of gas-phase and soot particles.

3.3.1 Constant Volume Reactor

The pyrolysis of 30% CH_4 diluted in N_2 with the initial temperature and pressure of 2455 K and 3.47 atm, respectively, was simulated using CVR with a residence time of 40 ms. The combination of available PAH growth and particle dynamics models

[3] https://github.com/mohammadadib-cu/omnisoot-cv/tree/main/validations/mass_energy

results in eight different cases, which were simulated to verify conservation of mass and energy. Here, we focus on the total elemental balance of carbon and hydrogen, as they are key elements involved in soot formation. Figure 3.6 shows the relative errors in total carbon, hydrogen, and energy for the different PAH growth and particle dynamics models. In all cases, the errors fall below 10^{-10}, confirming that CVR satisfies mass and energy conservation.

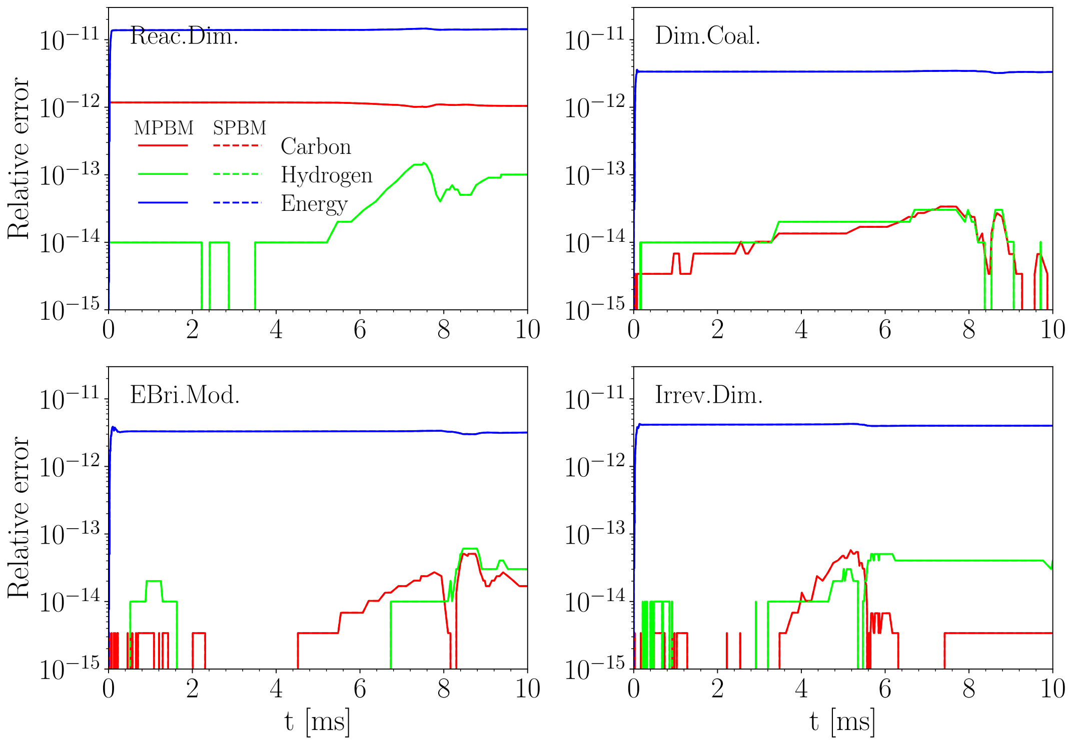


Fig. 3.6 The relative error of total carbon (red line) and hydrogen (green line) mass, and total internal energy residual of gas and soot (blue line) plotted against residence time during pyrolysis of 30% CH_4-N_2 at 2455 K and 3.47 atm in the constant volume reactor simulated using different PAH growth models along with MPBM (solid line) and SPBM (dashed line).

3.3.2 Imposed Pressure Reactor

The pyrolysis of 5%CH_4 in a shock-tube at post-reflected-shock temperature and pressure of T_5 =2355 K and P_5 = 4.64 atm was simulated using IPR model. The pressure was measured using an end-wall sensor by the Hanson Research Group at Stanford University. The measured pressure signal was smoothed using the Savitzky–Golay filter in SciPy[165] to reduce fluctuations in the raw signal and make it suitable for the numerical solver. The residual of carbon and hydrogen mass are less than 10^{-12}, while the energy residual reaches up to 10^{-6} for energy by the end of the simulation. The relatively higher energy residual is attributed to pressure work.

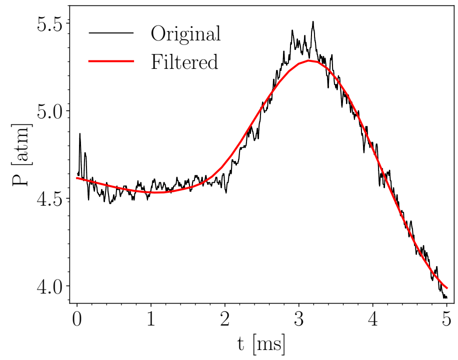


Fig. 3.7 The pressure time history (black line) measured by the Stanford group for pyrolysis of 5%CH_4 at T_5 =2355 K and P_5 = 4.64 atm, and filtered (red line) with reduced fluctuations used to specify the pressure of the reactor

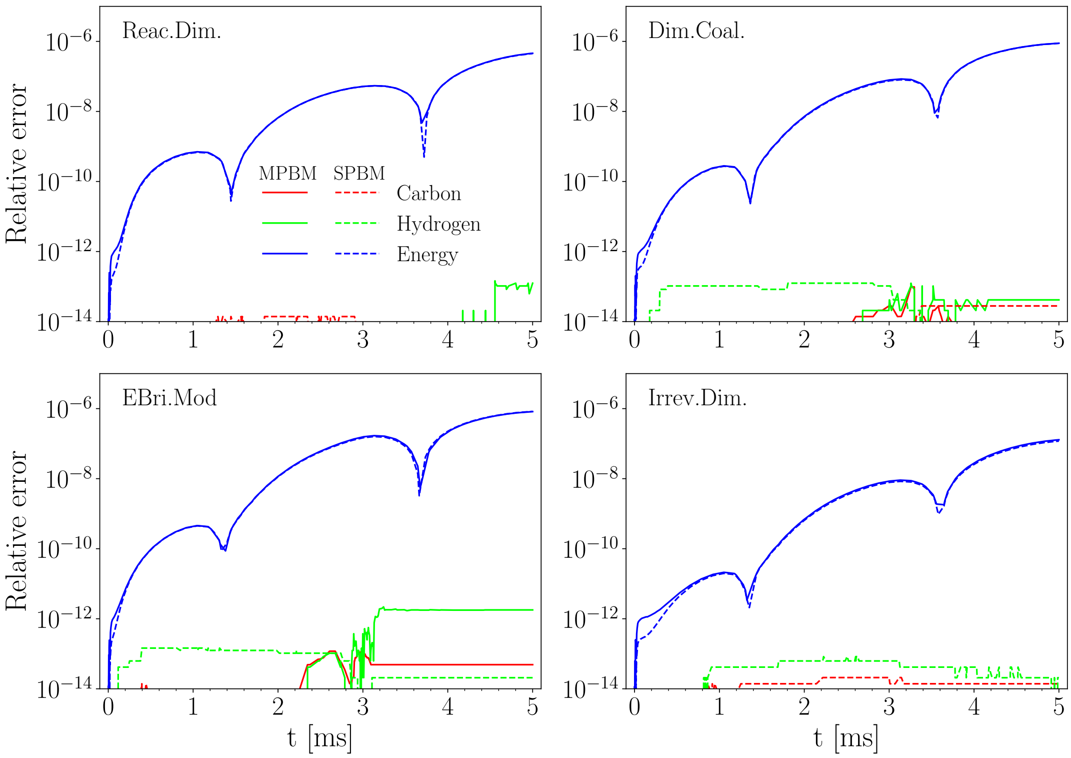


Fig. 3.8 The relative error of total carbon (red line) and hydrogen (green line) mass, and total internal energy residual of gas and soot (blue line) plotted against residence time during pyrolysis of 5% CH_4-Ar at post-reflected-shock temperature and pressure of T_5 =2355 K and P_5 = 4.64 atm with an imposed pressure profile simulated using different PAH growth models along with MPBM (solid line) and SPBM (dashed line)

3.3.3 Perfectly Stirred Reactor

The mass and energy balance are investigated for soot formation during ethylene-air oxidation at equivalence ratio of $\phi = 2$ in a perfectly stirred reactor. The simulation conditions were chosen based on the combustor implemented and utilized by Stouffer et al. [166]. The reactants enter the reactor with the volume of 250 ml at 300 K and atmospheric pressure. The simulation is initialized from a high temperature ($\approx$ 2000 K) to avoid trivial solution (cold reactant leaving the reactor with no chemical reactions) and to ensure the model captures a sustained combustion. The residence time of products in the reactor is 8.5 ms. Figure 3.9 shows the relative error of total elemental carbon and hydrogen mass and total enthalpy of gas and soot, which is less than 10^{-6} for all combinations of particle dynamics and PAH growth models.

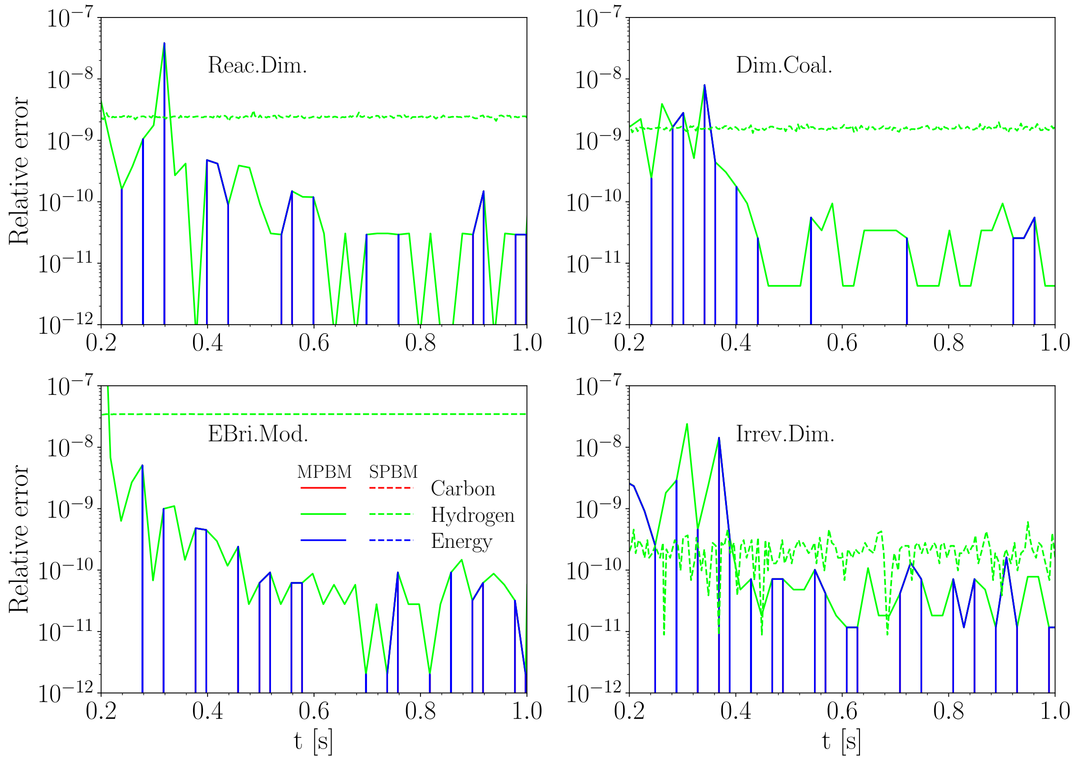


Fig. 3.9 The relative error of total carbon (red line) and hydrogen (green line) mass, and total internal energy residual of gas and soot (blue line) plotted in simulation time during adiabatic combustion of C_2H_4-air with $\phi = 2$ at 1 atm simulated using different combinations of PAH growth models and particle dynamics models: MPBM (solid line) and SPBM (dashed line).

3.3.4 Plug Flow Reactor

Methane pyrolysis in an adiabatic PFR was used to assess the conservation of elemental carbon, hydrogen, and energy. The inlet stream 30% CH_4 diluted in N_2 with an initial temperature of 2100 K and pressure of 1 atm. Figure 3.10 shows the residuals of total elemental carbon, hydrogen, and energy along the reactor length, up to 40 cm, for all combinations of PAH growth and particle dynamics models. The residuals remain below 10^{-11}, confirming that the PFR model in Omnisoot satisfies mass and energy conservation.

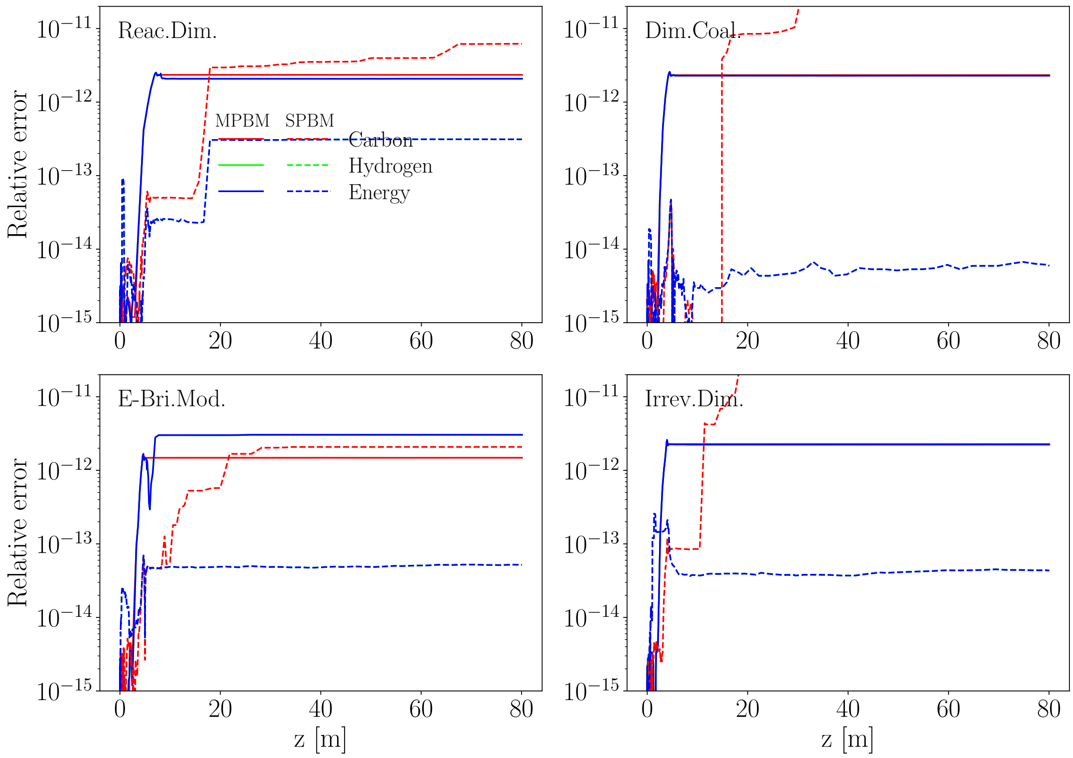


Fig. 3.10 The relative error of total carbon (red line) and hydrogen (green line) mass, and total internal energy residual of gas and soot (blue line) plotted against reactor length (cm) in the adiabatic flow reactor during pyrolysis of 30% CH_4-N_2 at 2100 K and 1 atm simulated using different PAH growth models and MPBM (solid line) and SPBM (dashed line)

Chapter 4

Results

This chapter shows the application of Omnisoot in various use-cases. Each section of the chapter represents a paper, in which Omnisoot was used for predicting soot yield, morphology and size distribution. The uncertainties from gas-phase chemistry and empirically-derived inception pathways were highlighted and discussed. The carbon flux from gas to solid via inception, HACA and PAH adsorption were investigated to understand their effect on soot properties. A special attention was paid to methane pyrolysis in shock-tubes with different compositions, and temperatures, and pressures because temperature and pressure ranges achieved in shock-tube experiments can reach 2300 K and 5 atm that are relevant to CB synthesis in industrial plasma reactors [167], which is a major motivations for developing Omnisoot. Additionally, a unique experimental data-set were compiled and processed in a joint program between Carleton-Monolith-Stanford. The shock-tube experimental campaign cover a wide range of high methane loadings (5%, 10%, and 30%) and temperature sweep of 1800 K-2500 K, and it provides simultaneous species and particulate measurements.

4.1 Fundamental analysis of PAH growth models

This section presents an extended version of the paper titled "*Omnisoot: a process design package for gas-phase synthesis of carbonaceous nanoparticles*" published in *Combustion and Flame*[1]. In the paper, we introduced Omnisoot as an integrated platform for modeling soot formation using 0D reactors. Theoretical foundation and governing equations were discussed in Chapter 2. The results and discussion in the paper focused on (i) understanding the uncertainties in predicted soot properties due to

[1]https://doi.org/10.1016/j.combustflame.2025.114758

gas-phase chemistry caused by the variability of reaction pathways and rate constants implemented in kinetic mechanisms (ii) the physio-chemical differences in PAH growth models when inception and PAH adsorption rates were adjusted by varying η_{inc} and η_{ads} to closely reproduce the experimental results.

4.1.1 Uncertainties in gas chemistry

Prior studies have shown that the predicted concentrations of large PAHs can vary by orders of magnitude depending on the reaction mechanism used [168]. This variation introduces substantial uncertainty in the predicted rates of soot inception and surface growth. As a result, the reliability of soot simulations depends more on the accuracy of upstream chemical kinetics than on the rate constants of the inception models.

To illustrate the impact of chemical mechanism selection on predict soot properties, a series of simulations[2] were performed using the CPR model of Omnisoot (with soot formation disabled) for the pyrolysis of 5% CH_4-Ar mixture at T_5 = 2203 K and P_5 = 4.9 atm [169]. Six widely used chemical mechanisms were evaluated: ABF [57], Caltech [170], KAUST [171], CRECK [172], ITV [173], and FFCM2 [174].

Fig. 4.1 shows the large variability in the predicted mole fraction of CH_4, C_2H_4, and C_2H_2 across different reaction mechanisms. The mole fraction of CH_4 serves as a useful indicator of carbon flux from the fuel to small and large intermediates. The CRECK and KAUST mechanisms significantly underpredict methane mole fraction compared to the measurements, but ABF and FFCM2 provide more accurate prediction considering the uncertainty of the experimental data. The carbon flux analysis by Wang et al. [168] highlighted the major role of C_2H_4 in the production of C_2H_2 via vinyl radical (C_2H_3). The overall trend of C_2H_4 mole fraction is captured by all mechanisms. After the rapid initial jump, C_2H_4 reaches its peak, which is underpredicted by ITV and overpredicted by KAUST. The subsequent decline was overestimated by all mechanisms that predict the final value of $\simeq 10^{-4}$ below the measurements. A similar variability is observed in the prediction of C_2H_2 mole fraction: it is best captured by FFCM2, overpredicted by CRECK, and slightly underpredicted by the ITV and Caltech mechanisms. Among all, FFCM2 shows the best overall agreement with the measurements; however, this mechanism only includes C_1-C_4 species and excludes benzene and larger PAHs, so it cannot be coupled with the soot models considered in this study.

Next, we examine the maximum possible soot yield predicted by the selected mechanisms (excluding FFCM2) during 5% CH_4 pyrolysis at T_5 = 2203 K and

[2]https://github.com/mohammadadib-cu/omnisoot-cv/tree/main/examples/pressure/mechanism_comparison

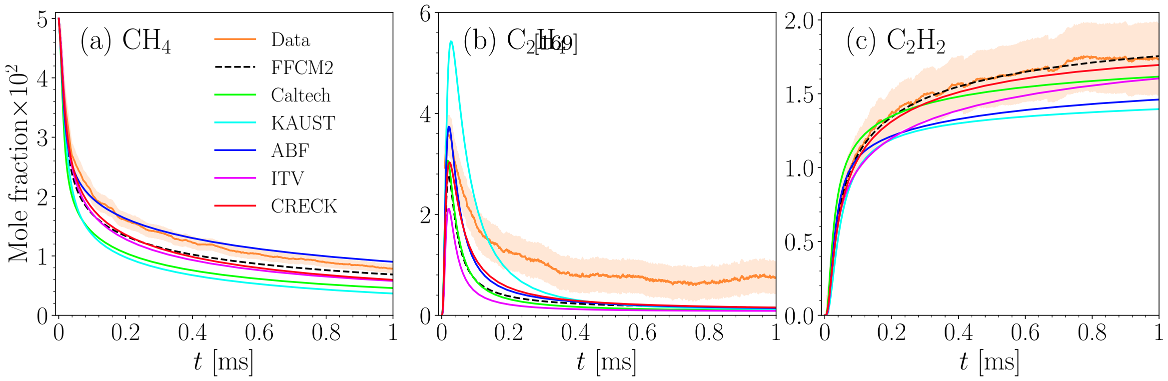


Fig. 4.1 The mole fraction of CH_4 (a), C_2H_4 and C_2H_2 (c) during pyrolysis of 5% CH_4-Ar predicted using different reaction mechanisms and compared with measurements [169].

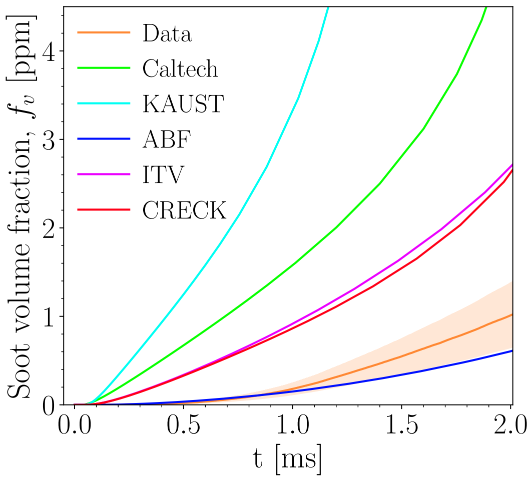


Fig. 4.2 The maximum soot volume fraction predicted using Irreversible Dimerization and different reaction mechanisms during pyrolysis of 5% CH_4-Ar was compared with measurements [169]. The shaded area represents the uncertainty in the reported experimental data.

$P_5 = 4.9$ atm. To do so, the Irreversible Dimerization model is employed with 100% efficiency for both inception and PAH adsorption (i.e., all collisions lead to inception or surface growth). As shown in Fig. 4.2, the maximum possible f_v predicted by the mechanisms shows greater relative variation than that observed for CH_4 and C_2H_2. Furthermore, the f_v predicted by the ABF mechanism is lower than the measured value, even when accounting for the reported uncertainty. The same analysis was performed for other data points with a different T_5 in [169], and similar results were observed indicating that ABF mechanism cannot provide enough precursors under pyrolysis conditions in these simulations. In contrast, the maximum possible f_v predicted by other mechanisms is significantly larger than the measurements, which presents the possibility of predicting f_v close to the measurements using more realistic assumptions (i.e., lower dimerization efficiencies). Caltech and KAUST mechanisms are used in the simulations of this study for description of gas chemistry as they provide the highest carbon flux to soot precursors, and they are still computationally more affordable than ITV and CRECK mechanisms for parametric studies.

In the rest of the results, the inception and PAH adsorption rates of PAH growth models are adjusted by calibrating η_{inc} and η_{ads} to match predict soot properties with available measurements. Note that, this approach does not resolve the mechanistic uncertainties of soot formation. Instead, it allows the model to calculate effective carbon fluxes from the gas phase into the particle phase that are consistent with measured soot properties. Our analysis showed that all PAH growth models require different sets of η_{inc} and η_{ads} to capture the measured soot properties in the studied conditions. The lack of universal factors highlights the uncertainties in concentration of gaseous species, potential missing inception and surface growth pathways, and unknown sensitivities to temperature and fuel composition.

4.1.2 Methane pyrolysis in shock tube

The pyrolysis of a 5% CH_4-Ar mixture was studied using the CPR model over a post-shock temperature range of T_5 = 1800-3000 K and a pressure range of P_5 = 4.7-7.1 bar. Since P_5 was not specified for individual experiments (each defined by a given T_5), a linear dependence of P_5 on T_5 was assumed within the stated range [175]. The simulations were performed using the KAUST and Caltech mechanisms, and the model predictions were compared with the CY measurements. The bell-shaped profile of CY was captured using both mechanisms; however, the T_5 corresponding to the peak CY was overpredicted by all inception models using KAUST by more than 100 K, in contrast to the Caltech mechanism, which shows closer agreement with the measurements. Therefore, the gas-phase chemistry was described using the Caltech mechanism for the rest of simulations in for the shock-tube pyrolysis of methane. Inception and PAH adsorption rates were adjusted using η_{inc} and η_{ads}, respectively, to match the predicted Carbon Yield (CY) at t = 1.5 ms with light extinction measurements at λ = 632 nm across the studied T_5 and P_5 ranges [175]. The experimental CY was extracted from the reported product of CY and the absorption function, E(m). However, this approach introduces uncertainty due to variability in E(m). It is common to assume the E(m) of mature soot, which may not be valid in shock tube experiments with short residence times ($\approx$ 2 ms) that involve initial stages of soot formation, as E(m) increases with particle size, composition, and maturity. For instance, during acetylene pyrolysis, E(m) can vary from 0.05 to 0.25 as d_p reaches 20 nm within 1.6 ms. We considered the reported variability of E(m) in the literature, ranging from 0.174 [176] to 0.37 [177].

A parametric study was conducted on η_{inc} and η_{ads} using a grid search approach. Each factor was varied across 11 logarithmically spaced values from 10^{-4} to 1, resulting in 121 simulation cases per data point and 847 simulations in total. Each η_{inc} and η_{ads} pair was ranked based on the average relative error of CY across all data points in the studied T_5 range, in order to identify the set of adjustment factors that minimized the CY prediction error.

To reduce the computational load of the parametric study, simulations were performed using MPBM for all inception models. The comparison of results obtained using SPBM and MPBM in Fig. A.1 shows that CY and d_p are not sensitive to the choice of particle dynamics model, indicating that MPBM results can be reliably used for CY error analysis. However, N_{agg} and n_p, shown in Fig. A.2, differ significantly between SPBM and MPBM because the central assumption of the monodisperse model is the rapid attainment of SPSD, which is not valid at short residence times near 1.5 ms. For example, the non-dimensional PSD at $T_5 = 2200$ K, shown in Fig. A.3, significantly changes from 1.0 ms to 2.5 ms for all inception models.

Fig. 4.3 shows a heat map of the mean relative error, normalized by the maximum value, for all inception models. The largest error occurs for the combination of maximum inception ($\eta_{inc} = 1$ or $\log_{10}(\eta_{inc}) = 0$) and minimum adsorption ($\eta_{ads} = 10^{-4}$ or $\log_{10}(\eta_{ads}) = -4$). The region of lowest normalized error (less than 5%), outlined in blue in Fig. 4.3, includes a broad set of adjustment factor combinations that enable the model to accurately predict CY. However, accurate prediction of CY does not necessarily imply accurate prediction of other properties, such as primary particle diameter or mobility diameter. The availability of data on other soot parameters or intermediate gaseous species, and including them as optimization targets, could help narrow down the range of optimal adjustment factors.

To highlight the variability in soot morphology across the low-error region of the heat map, we focus on three representative combinations of η_{inc} and η_{ads}: (i) minimum inception flux ($\eta_{inc} = 10^{-4}$), (ii) minimum PAH adsorption rate ($\eta_{ads} = 10^{-4}$), and (iii) equally scaled inception flux and PAH adsorption rate ($\eta_{inc} = \eta_{ads}$). It is important to note that soot CY and morphology are closely coupled to gas-phase chemistry under the short residence times studied. Therefore, a set of simulations was performed without soot to isolate gas-phase chemistry and examine the temperature dependence of soot precursors and C_2H_2. Subsequently, inception models were optimized using equal

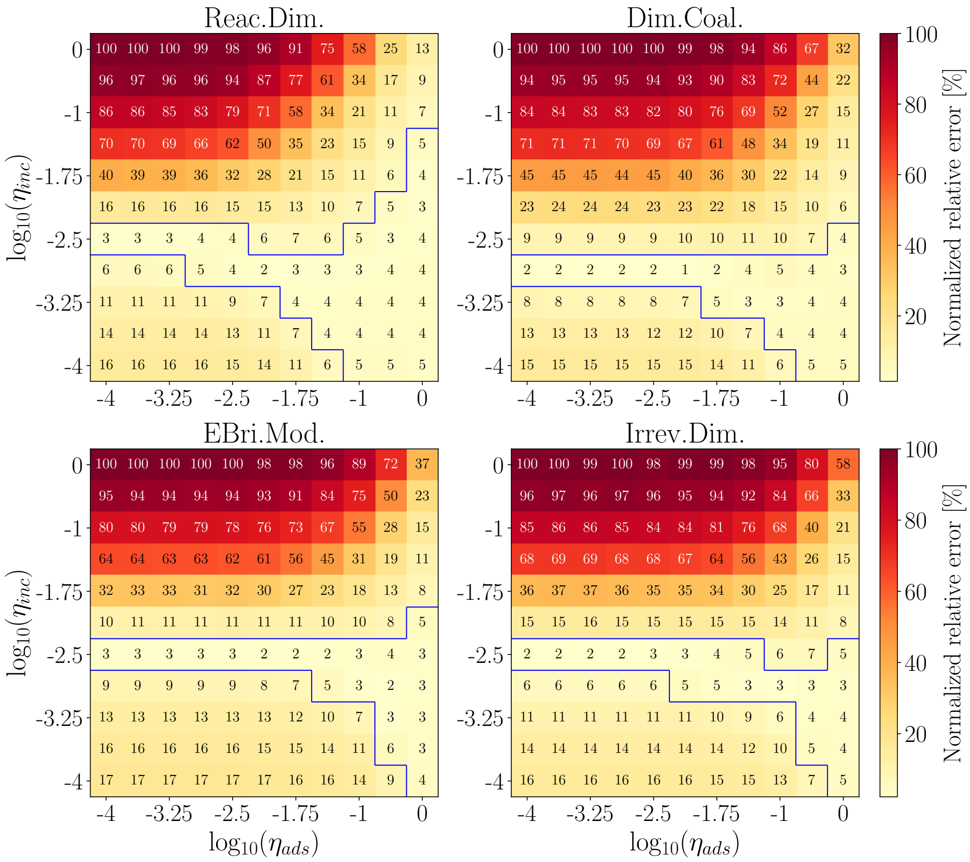


Fig. 4.3 The relative error of CY normalized by the maximum value at $t = 1.5$ ms for 5% CH_4-Ar obtained using Caltech mechanism and all PAH growth models. Each cell is annotated with its corresponding normalized relative error expressed in percent.

adjustment factors[3] ($\eta_{inc} = \eta_{ads}$), and separate optimizations were carried out for the minimum adsorption case ($\eta_{ads} = 10^{-4}$) and the minimum inception case ($\eta_{inc} = 10^{-4}$), denoted as *Min-Ads* and *Min-Inc*, respectively.

Fig. 4.4a shows the CMF of soot precursors (individual PAHs and total) and C_2H_2 at 1.5 ms across the studied temperature range, with the soot model deactivated. The CMF is defined as the ratio of the carbon mass contained in a given species to the total carbon mass in the system. While PAH precursors exhibit a bell-shaped profile, the CMF of C_2H_2 increases linearly with temperature and plateaus near 85% at $T_5 \approx 2400$ K. Among the considered PAHs, A2, A2R5, and A4R5 account for most of the carbon and are likely major contributors to soot inception and surface growth. The strong variation in PAH CMF highlights the impact of precursor selection on inception flux, surface growth rates, and their temperature dependence. Fig. 4.5 illustrates the

[3] https://github.com/mohammadadib-cu/omnisoot-cv/tree/main/examples/pressure/methane_pyrolysis

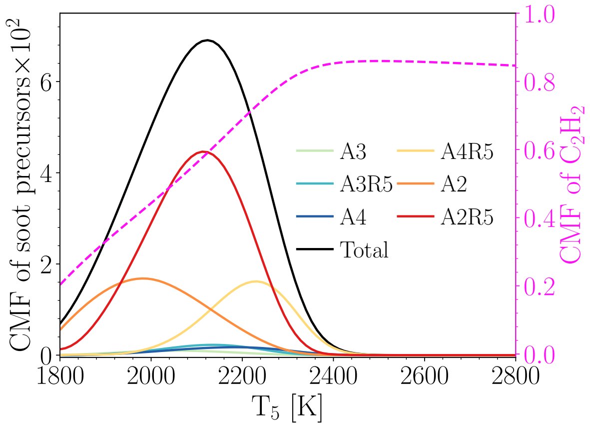


Fig. 4.4 The bell-shaped temperature profile of carbon mass fraction of soot precursors (A2 and larger) combined and C_2H_2 at t = 1.5 ms during pyrolysis of 5% CH_4-Ar obtained using Caltech mechanism without considering soot.

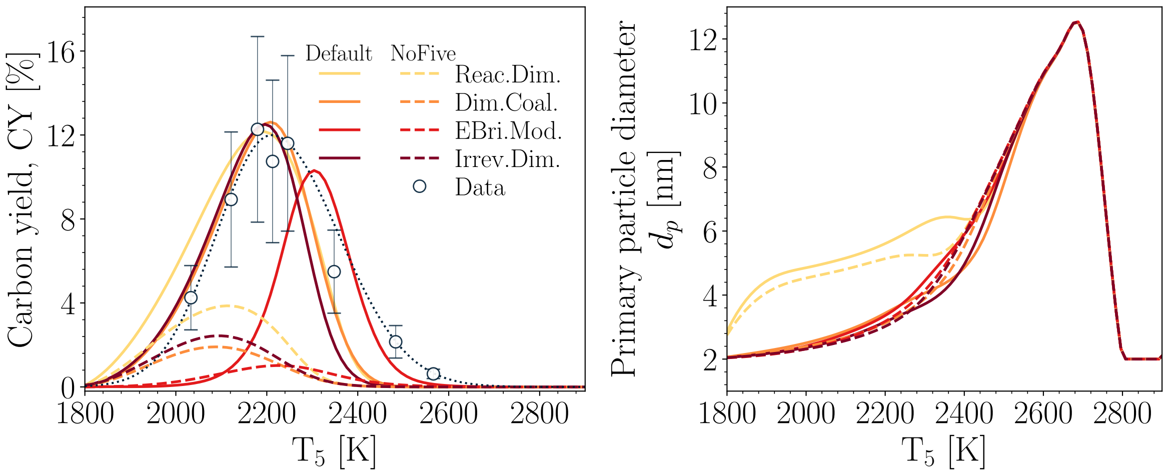


Fig. 4.5 The soot carbon yield, CY, at t =1.5 ms (a) and primary particle diameter, d_p (b) obtained using the default soot precursors (denoted by solid line and labeled as “Default") compared with the same results when five-membered ring PAHs excluded from soot precursors (denoted by dashed line and labeled as “NoFive"). Both cases were obtained using Caltech mechanism and the same equal adjustment factors. The black dashed line was added to show the trend in the measurements [175].

effect of excluding five-membered ring PAHs (A2R5, A3R5, and A4R5) from the soot precursors on the CY and d_p. As expected, excluding these species reduces CY for all inception models, potentially highlighting the significant contribution of five-membered rings PAHs.

Next set of simulations were conducted by using equal adjustment factors ($\eta_{inc} = \eta_{ads}$) to minimize the prediction error of CY. Table 4.1 lists the adjustment factors used for each inception model. Note that the efficiencies of inception (γ_{inc} in Equation (2.87))

Table 4.1 The value of adjustment factors used in the simulation of 5% CH_4-Ar pyrolysis.

	Reac.Dim.	Dim.Coal.	EBri.Mod.	Irrev.Dim.
η_{inc}	0.17	0.21	0.36	2.3
η_{ads}	0.17	0.21	0.36	2.3

and PAH adsorption (γ_{ads} in Equation (2.88)) for Irreversible Dimerization were set to 10^{-4} and 10^{-3}, respectively [48], so η_{inc} and η_{ads} values greater than unity do not result in collision rates exceeding the theoretical limit of kinetic theory. Fig. 4.6a compares CY from various inception models with experimental data [175]. A skew exponential curve (black dotted line) was fitted to the data to highlight the CY trend and its peak near 12%. Soot CY follows a bell-shaped profile similar to that of soot precursors, as inception flux and PAH adsorption rate directly depend on precursor concentrations. Reactive Dimerization model predicts a higher CY at lower temperatures due to stronger PAH adsorption rates, while E-Bridge Modified model shifts the peak to higher temperatures compared to the other models because in this model, inception is initiated and controlled by the dehydrogenation rate of PAH molecules, which is strongly depends on temperature and the concentration of H radicals. In contrast, the other models begin with physical PAH collision, which is primarily a function of PAH concentration.

As shown in Fig. 4.6b, the predicted d_p increases with T_5, reaching a maximum of 11 nm at 2700 K, where yield is very low ($\approx 10^{-7}$), and then drops to the model's minimum allowed value of 2 nm. The differences between inception models in predicting d_p is overall negligible except for Reactive Dimerization model, which predicts a larger d_p in $T_5 < 2500$ K. The d_p trends can be better understood by examining Equation (4.7) indicating that d_p is proportional to the third-root of C_{tot}/N_{pri}. C_{tot} describes total carbon mass converted to soot through inception and surface growth while N_{pri} is only determined by inception flux. As a result, d_p is controlled by the ratio of total surface growth rate (via HACA and PAH adsorption) to inception flux. Here, we define the parameter θ as the ratio of carbon mass gained via each pathway to the total soot carbon mass. As shown in Fig. 4.7, PAH adsorption is the dominant soot mass growth pathway for Reactive Dimerization model in $T_5 < 2000$ K (Fig. 4.7a), which results in larger C_{tot}/N_{pri} and d_p values, but inception is dominant for the other inception models. The contribution of inception decreases with temperature for all inception models while the contribution of HACA becomes dominant up to 2700 K leading to higher d_p (Fig. 4.6b). This is due to the decrease of precursor concentration and the

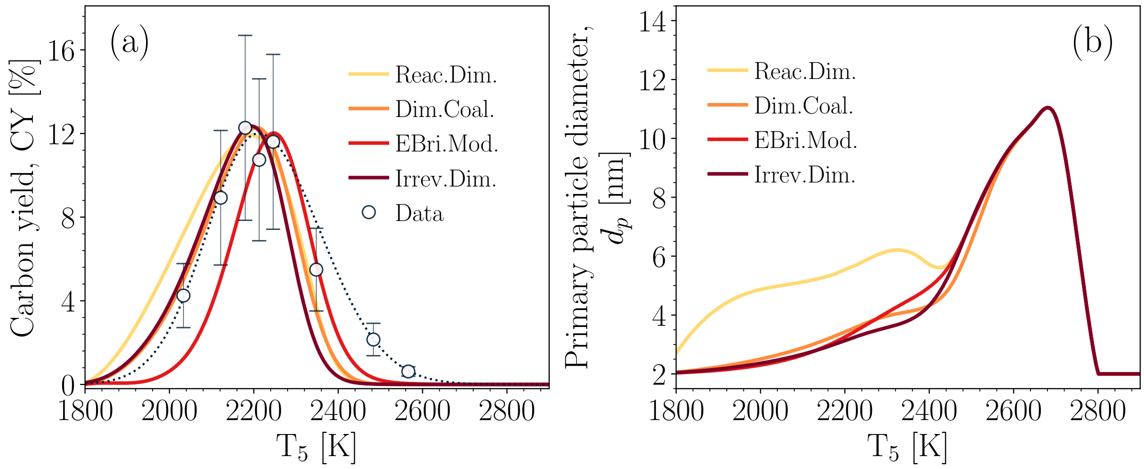


Fig. 4.6 The CY (a) and primary particle diameter, d_p, (b) at t = 1.5 ms during the pyrolysis of 5% CH_4-Ar obtained using equal adjustment factors to minimize the prediction error with extinction measurements [175]. The dashed line was added to show the trend in the measurements.

increase of C_2H_2 CMF. Beyond 2700 K, the contribution of inception rapidly increases with temperature leading to the decrease of d_p (Fig. 4.6b).

Fig. 4.8a shows that N_{pri} peaks near 2200 K, aligning with the peak in CY. This is expected because N_{pri} depends solely on the inception flux, which is the highest when PAH concentrations peak. Increased coagulation rates reduce N_{agg} in the 2000-2500 K range. Reactive Dimerization model yields the fewest particles (N_{pri} and N_{agg}) because more precursors are directed toward PAH adsorption. Consequently, n_p is lowest for Reactive Dimerization model. Differences between inception models are more evident in n_p than in d_p, influencing predicted particle morphology.

Fig. 4.9a and Fig. 4.9b shows that the optimized models with minimum η_{inc} and η_{ads} yield similar CY across inception models, except E-Bridge Modified model, which shifts slightly toward higher temperatures. The Min-Ads case has a higher peak and narrower profile compared with the Min-Inc case. As shown in Fig. 4.9c, d_p predicted in Min-Ads mode is smaller and less sensitive to the selection of inception model compared with Min-Inc mode. Two peaks can be seen in Min-Inc mode for all inception models: The first peak occurs due to high PAH adsorption rates at the temperature range close to the peak of CMF of soot precursors (Fig. 4.4a); the second peak which occurs around T_5 = 2700 K and d_p = 12 nm, which is very similar to d_p trends predicted using equal adjustment factors (Fig. 4.6b). Reactive Dimerization model has the largest variation at the peak from 2 to 19 nm between Min-Ads and Min-Inc modes. The existence of such a variation highlights the need for characterizing primary particle size

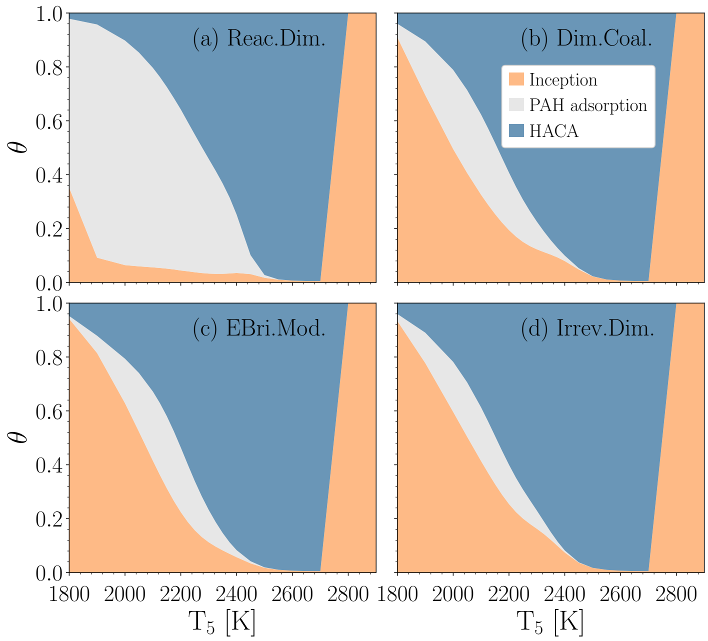


Fig. 4.7 The variation of θ, defined as soot carbon mass from inception, PAH adsorption and HACA normalized by total soot carbon mass at $t = 1.5$ ms during the pyrolysis of 5% CH_4-Ar.

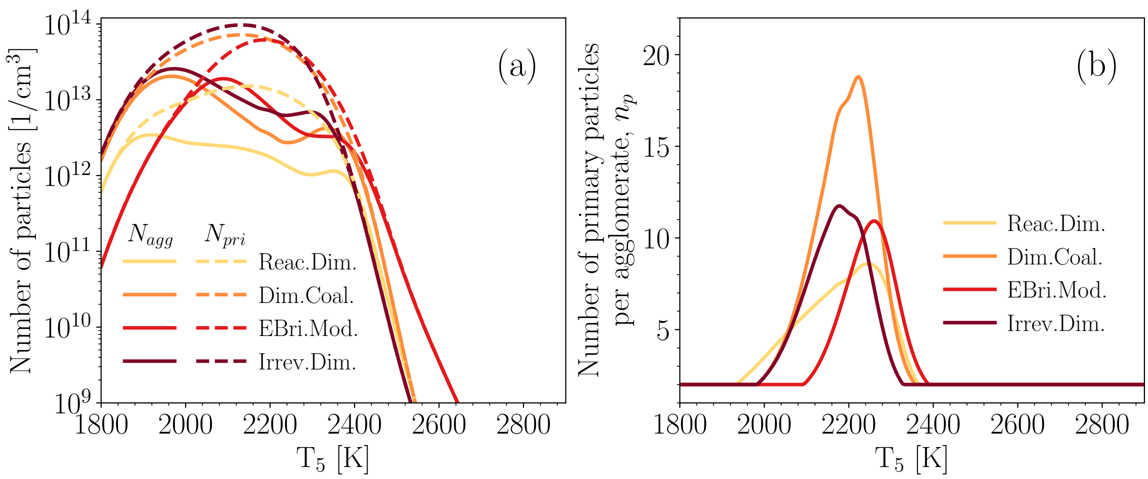


Fig. 4.8 The temperature dependence of total number of agglomerates, N_{agg} (a), number of primary particles per agglomerate, n_p (b) at $t = 1.5$ ms during the pyrolysis of 5% CH_4-Ar.

in the experiment to reduce the uncertainty in the soot model and restrict the range of expected inception and surface growth rates required to accurately predict soot yield and morphology.

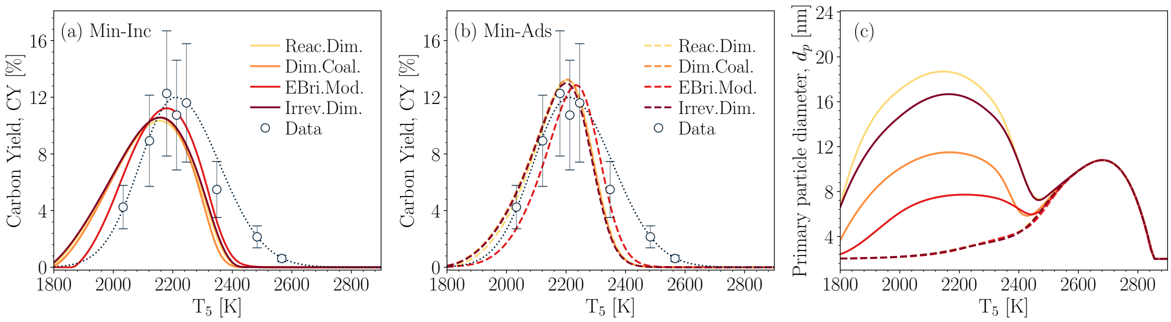


Fig. 4.9 The comparison of CY and d_p at $t = 1.5$ ms for 5% CH_4-Ar when the minimum inception (solid lines) and the minimum PAH adsorption adjustment factors were applied to minimize the prediction error. The dashed line was added to show the trend in the measurements [175].

4.1.3 Ethylene pyrolysis in a flow reactor

The PFR model was used to simulate[4] the pyrolysis of 0.6% C_2H_4-N_2 in a 1.4 m long, and 16 mm diameter flow reactor. Fig. 4.10 shows the axial temperature profile used in the simulations. The maximum temperature, $T_{max} = 1673$ K, was imposed on the model based on thermocouple measurements reported in [178] for various flow rates along the reactor centerline. The temperature increases from 300 K at the reactor inlet to the hot zone, where it reaches a value within 10% of T_{max}. The length of the hot zone varies with flow rate due to advection, ranging from 0.71 m at Q = 8 L/min to 0.76 m at Q = 12 L/min. Near the reactor outlet, the temperature drops to approximately 650 K.

The inception and PAH adsorption adjustment factors were varied to match the predicted PSD with Scanning Mobility Particle Sizer (SMPS) measurements conducted on centerline at the reactor exit for the flow rates of 8, 11, and 12 L/min [178], as shown in Fig. 4.11. The parametric analysis revealed that each inception model requires a unique set of adjustment factors to minimize the PSD prediction error across different flow rates. The KAUST mechanism [171] was used to describe gas-phase chemistry because it provided better agreement with the measured PSDs than the Caltech mechanism, shown in Fig. A.4. Nevertheless, the choice of reaction mechanism does not change the findings about the differences between the inception models because they are primarily due to their reversibility and sensitivity to temperature, which is not affected by the base kinetic mechanism. Table 4.2 indicates the adjustment factors used with the KAUST mechanism for simulations of ethylene pyrolysis in the studied flow reactor.

[4] https://github.com/mohammadadib-cu/omnisoot-cv/tree/main/examples/plug_flow/sectional

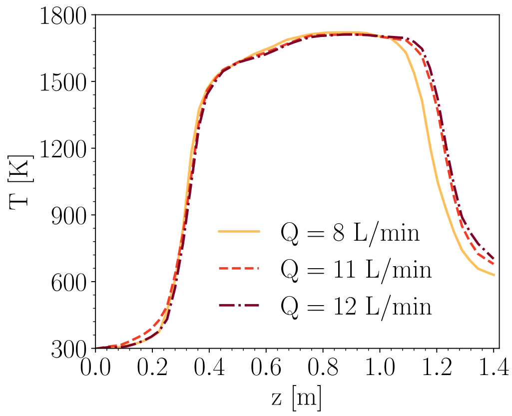


Fig. 4.10 The centerline temperature along the reactor for Q = 8, 11, and 12 L/min interpolated from the thermocouple measurements [178].

Table 4.2 The value of adjustment factors used in the simulation of ethylene pyrolysis in the flow reactor.

	Reac.Dim.	Dim.Coal.	EBri.Mod.	Irrev.Dim.
η_{inc}	0.01	0.004	0.01	0.08
η_{ads}	0.001	0.01	0.1	1

A bimodal size distribution can be observed for Q = 8 L/min (Fig. 4.11a) that was attributed to the continuous inception [26]. The comparison of simulations with measurements reveals different temperature dependence of inception models. Irreversible Dimerization and Dimer Coalescence models capture the bimodality of PSD in good agreement with the measurements, but the other two models predict a nearly unimodal PSD. Specifically, the number concentration of the first section is lower by more than three orders of magnitude for the Reactive Dimerization and E-Bridge Modified models indicating the lack of active inception at the sampling location due to the temperature drop near the end of reactor that suppresses soot inception.

All inception models capture the disappearance of the PSD shoulder at flow rates of 11 and 12 L/min. In these cases, the shorter particle residence time limits coagulation, preventing formation of a second peak. However, Reactive Dimerization and E-Bridge Modified models still underpredict the number concentration of nascent particles by approximately one order of magnitude compared to the irreversible models (Irreversible Dimerization and Dimer Coalescence).

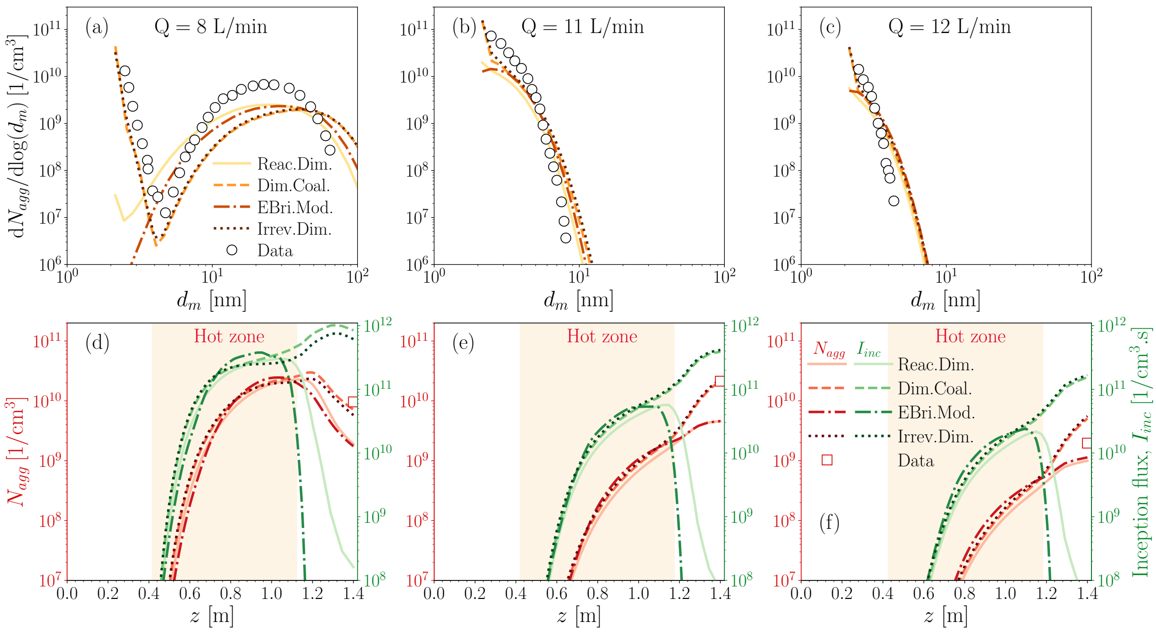


Fig. 4.11 The particle size distribution at the end of PFR, and the total number of agglomerates, N_{agg}, and inception flux, I_{inc} along the PFR for Q = 8 (a, d), 11 (b, e), and 12 L/min (c, f) obtained using KAUST mechanism, SPBM and different inception models calibrated to match the predictions with measurements [178].

As shown in Fig. 4.11d-f, all inception models predict nearly the same N_{agg} up to $z \approx 1.1$ m, corresponding to the end of the hot zone. Beyond this point, N_{agg} continues to increase for the Irreversible Dimerization and Dimer Coalescence models, which allows soot inception to proceed in cooler downstream regions. These models agree well with experimental data at 8 and 11 L/min but tend to overpredict N_{agg} at 12 L/min.

Soot inception flux rises sharply as the gas enters the hot zone. The axial location where the flux exceeds 10^7 cm^{-3} s^{-1} is similar across all inception models, as it is primarily determined by PAH chemistry. However, this location shifts downstream with increasing flow rate due to reduced residence time. Within the hot zone, differences among the inception models are minimal. Outside the hot zone, Irreversible Dimerization and Dimer Coalescence models maintain high inception flux, while Reactive Dimerization and E-Bridge Modified models predict a rapid decrease of more than three orders of magnitude due to cooling. The difference between the models is primarily due to their temperature dependence as indicated in Table. 2.3 and illustrated in Fig. 2.6. The Arrhenius rates embedded in the carbonization step in Reactive Dimerization and the dehydrogenation in E-Bridge cause a exponential decay in these models with the temperature drops for $z > 1.1$ m. Fig. 4.12 demonstrates the contribution of each

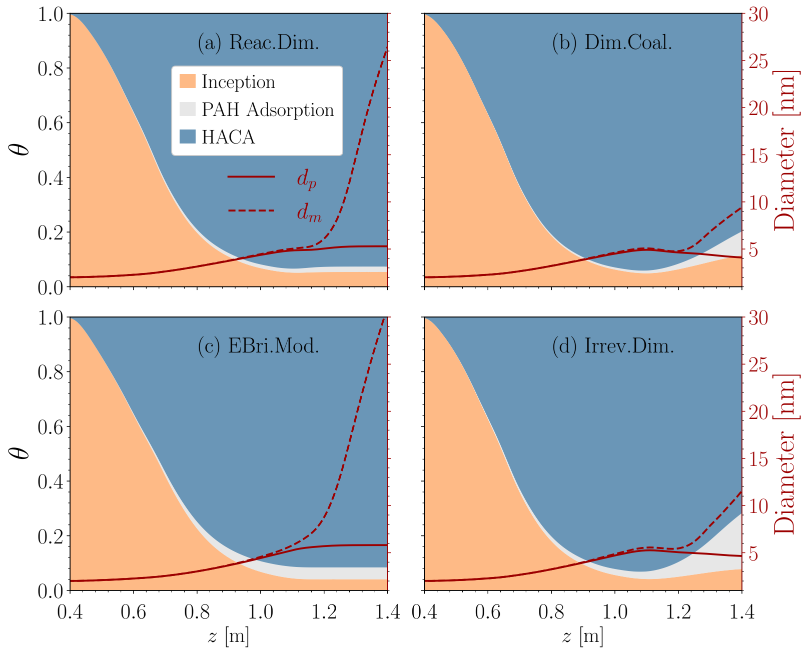


Fig. 4.12 The variation of θ, defined as soot carbon mass from inception, PAH adsorption and HACA normalized by total soot carbon mass for Q = 8 L/min along with d_p and d_m obtained using KAUST mechanism, SPBM and different inception models.

pathway to total carbon mass of soot along the reactor, which elucidates the link between particle morphology and the balance of inception and surface growth. The relative contribution of inception decreases from 100% to less than 5% by z = 1.1 m for all inception models leading to a gradual increase in d_p and d_m. Irreversible Dimerization and Dimer Coalescence models predict an increase in the contribution of inception beyond $z > 1.1$ m resulting in the decline of d_p while Reactive Dimerization and E-Bridge Modified models predict a nearly constant d_p close to 5 nm. The final increase in d_m beyond the end of the hot zone near z = 1.1 m is more pronounced for the Reactive Dimerization and E-Bridge Modified models. This is because the inception rate and surface growth rate (Fig. A.5) rapidly decrease in these models, and soot morphology becomes primarily governed by coagulation, which conserves d_p and increases d_m.

4.1.4 Ethylene combustion in a perfectly stirred reactor

The PSR model of Omnisoot[5] was used to simulate ethylene combustion at different equivalence ratios, ϕ = 1.9, 2.0, and 2.1 in a perfectly stirred reactor connected

[5]https://github.com/mohammadadib-cu/omnisoot-cv/tree/main/examples/psr_pfr

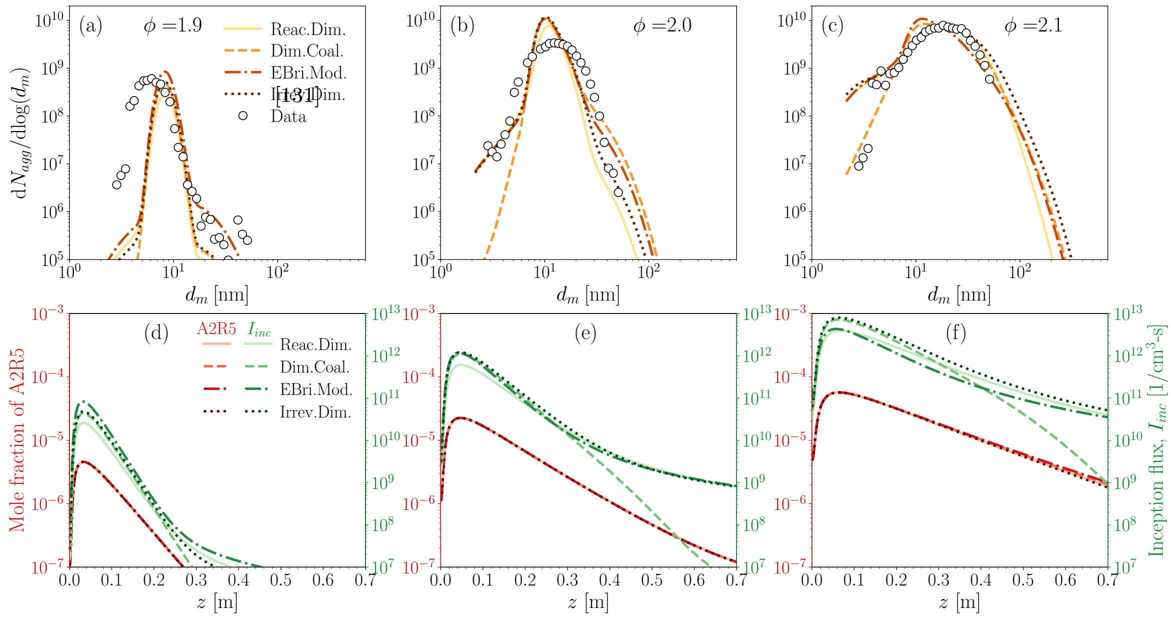


Fig. 4.13 The particle size distribution at the end of PFR, and the mole fraction of A2R5, and inception flux, I_{inc} along the PFR for $\phi = 1.9$ (a, d), 2.0 (b, e), and 2.1 (c, f) obtained using KAUST mechanism, SPBM and different inception models calibrated to match the predictions with the PSD measurements [131].

to a flow reactor [131]. PSD measurements were conducted at the end of the flow reactor using a Nano-Differential Mobility Analyzer (Nano-DMA). The volume (V) and nominal residence time (τ_{psr}) of the PSR are 250 ml and 11 ms, respectively [131]. The flow reactor is 70 cm long with an inner diameter of 5.1 cm. The reactants enter the adiabatic PSR at 300 K with an inlet mass flow rate of $\dot{m}_{in} = \rho V/\tau_{psr}$, where ρ was calculated at the reactor temperature of 1723 K [179]. The PSR simulation was continued until a steady-state condition was reached for all the solution variables. The final gas composition, temperature, and soot properties from the PSR were used as inlet conditions for the PFR simulations, yielding an average axial velocity of 14.5 m/s in the PFR, which is consistent with the values reported in [131].

The KAUST mechanism was used to describe the gas-phase chemistry. The inception and PAH adsorption adjustment factors were varied for each PAH growth model to match the predicted PSD with the measurements as closely as possible for all three equivalence ratios. A similar match could not be reached using the Caltech mechanism because the number concentration of particles was underestimated by at least four orders of magnitude even with $\eta_{inc} = \eta_{ads} = 1$. Table 4.3 lists the adjustment factors used with the KAUST mechanism for simulations of ethylene oxidation in the studied perfectly stirred reactor.

Table 4.3 The value of adjustment factors used in the simulation of ethylene oxidation in the perfectly stirred reactor.

	Reac.Dim.	Dim.Coal.	EBri.Mod.	Irrev.Dim.
η_{inc}	0.0005	0.001	0.0002	0.01
η_{ads}	0.01	0.01	0.01	1

Table 4.4 The geometric mean mobility diameter, $d_{m,g}$, and the geometric mobility standard deviation, $\sigma_{m,g}$, obtained using different inception models compared with the value calculated from the measured PSD [131].

	$\phi = 1.9$		$\phi = 2.0$		$\phi = 2.1$	
	$d_{m,g}$ [nm]	$\sigma_{m,g}$	$d_{m,g}$ [nm]	$\sigma_{m,g}$	$d_{m,g}$ [nm]	$\sigma_{m,g}$
Data [131]	**6.04**	**1.25**	**12.40**	**1.49**	**17.66**	**1.64**
Reactive Dimerization	8.27	1.14	11.10	1.21	16.88	1.58
Dimer Coalescence	8.18	1.14	10.76	1.24	16.99	1.68
E-Bridge Modified	8.27	1.15	11.02	1.27	14.56	1.58
Irreversible Dimerization	8.19	1.14	10.96	1.25	18.44	1.78

As shown in Fig. 4.13, all inception models capture the peak number concentration and the unimodal shape of the PSD, indicating that particle inception has ceased and coagulation is the dominant mechanism for agglomerate size growth. As reported in Table 4.4, the geometric mean mobility diameter, $d_{m,g}$, and the geometric mobility standard deviation, $\sigma_{m,g}$, obtained using all inception models are in good agreement with the values calculated from the measured PSD [131]. As shown in Fig. 4.13a-b, the spread of the predicted PSD is narrower than that of the measurements for $\phi = 1.9$ and 2.0, corresponding to an underprediction of $\sigma_{m,g}$ for these equivalence ratios.

All inception models underpredict the number concentration of small particles ($d_m < 6$ nm) at $\phi = 1.9$. The discrepancy decreases at $\phi = 2.0$, and the model predictions align well with the measurements except for Dimer Coalescence model that underpredicts the number concentration of small particles ($d_m < 5$ nm). In the simulation results, the number concentration of the smallest particles with a diameter of 2 nm is strongly affected by the equivalence ratio, as the precursor production rate increases with ϕ. This behavior can be better understood by comparing the acenaphthylene (A2R5) mole fraction at different ϕ values, as shown in Fig. 4.13d-f.

The inlet mole fraction of A2R5 at $z = 0$ m corresponds to the value obtained from the PSR, which increases by factors of 10 and 4 when ϕ is increased incrementally from 1.9 to 2.0, and from 2.0 to 2.1, respectively. This relative increase is nearly the same across all inception models. The jump in A2R5 mole fraction leads to a substantial increase in the inception flux, which explains the sensitivity of the smallest particle concentrations to the equivalence ratio. The strong similarity between the A2R5 mole fraction profiles and the soot inception profiles along the reactor highlights the dominant role of A2R5 in soot inception. Another important observation is that the A2R5 mole fraction is not affected by the choice of inception model.

As shown in Fig. 4.14a-c, d_p along the PFR is nearly identical for all inception models at $\phi = 1.9$, but the differences become more noticeable at higher ϕ values. The Reactive Dimerization model predicts the largest final d_p because it channels more PAH mass into PAH adsorption rather than inception, which is consistent with the lower peak number concentration observed in Fig. 4.13. The value of d_p decreases rapidly at the beginning of the reactor due to a surge in the production of incipient particles (spherical particles with a diameter of 2 nm), which lowers the mean d_p. It then increases toward the end of the reactor as surface growth becomes dominant. Although the final d_p increases slightly with ϕ, the overall range remains similar across ϕ values, indicating a comparable balance between inception and surface growth.

As shown in Fig. 4.14, the surface growth rate (HACA and PAH adsorption combined) and the soot volume fraction, however, are highly sensitive to ϕ, which can be attributed to the strong influence of ϕ on precursor production, inception flux, and the number of particles. An analysis of surface growth rates at the end of the PFR revealed that more than 95% of the soot mass is acquired through the HACA mechanism. The apparent difference in f_v between the inception models originates from the different PAH adsorption rate predicted by the inception models. For example, at $\phi = 1.9$, the E-Bridge Modified model predicts the highest surface growth rate and soot volume fraction due to a higher number of agglomerates (as shown in Fig. 4.13a), which leads to a larger total surface area, which is the key factor in calculating the HACA growth rate.

4.1.5 Conclusions

Omnisoot was employed to elucidate differences among inception models in shock tubes, flow reactors, and perfectly stirred reactors. Global adjustment factors were applied to inception and surface growth rates to accurately reproduce experimental data for each target scenario across a range of parameters, including temperature, pressure,

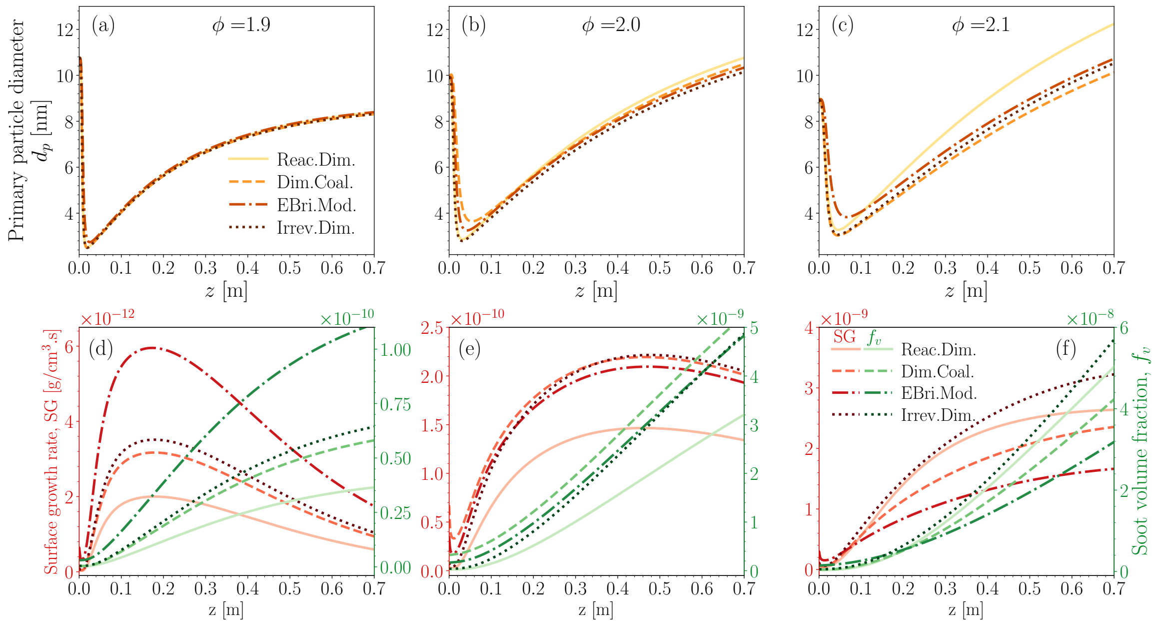


Fig. 4.14 The primary particle diameter, d_p, the total surface growth rate by HACA and PAH adsorption, and the soot volume fraction, f_v along the PFR for $\phi = 1.9$ (a, d), 2.0 (b, e), and 2.1 (c, f) obtained using KAUST mechanism, SPBM and different inception models calibrated to match the predictions with the PSD measurements [131].

composition, and flow rates. The performance of inception models was evaluated based on soot yield, morphology, size distribution, inception and surface growth rates, and carbon mass source analysis. Although different inception models predicted similar carbon yields that closely matched experimental observations, they produced varying morphological characteristics. This highlights the importance of characterizing soot morphology to better constrain inception flux and surface growth rate predictions.

Simulations of ethylene pyrolysis in a flow reactor demonstrated the temperature sensitivity of inception models. Irreversible models (Irreversible Dimerization and Dimer Coalescence) effectively captured the bimodal PSD resulting from high inception flux in the lower-temperature region near the reactor exit due to their weak dependence. However, the Reactive Dimerization and E-Bridge models generated nearly unimodal distributions due to their exponential temperature dependence, causing a significant drop in inception flux in the same region.

The PSD sampled at the outlet of a flow reactor downstream of a perfectly stirred reactor showed good agreement with experimental measurements across three equivalence ratios. Simulations indicated that active inception and surface growth primarily occur at high temperatures within the perfectly stirred reactor and at the entrance region of the flow reactor, far upstream of the sampling location. Under these conditions, the particle size distribution, morphology, and inception flux predictions were largely consistent across all inception models, although notable differences were observed in soot volume fraction due to variations in predicted HACA growth rate.

4.2 Soot formation at high fuel loadings

This section presents the paper titled "*Gas-phase chemistry and soot formation during shock-tube pyrolysis of methane at high fuel loadings*" and submitted to ****, where Omnisoot was utilized to extend the application of kinetic models from dilute to high-fuel-loading regime typical in cogeneration of carbon black and hydrogen. The paper provides the first simultaneous investigation of methane pyrolysis chemistry and particulate formation under high fuel concentrations accompanied with numerical simulations that accounts for the evolution of gas chemistry and formation of particulate. Detailed measurement-model comparisons enabled the assessment of several kinetic models to elucidate the combined effect of temperature and fuel loading on methane carbon flux to intermediates that determine the yield of carbonaceous particles.

Direct decomposition of methane offers a low-emission route for large-scale co-production of carbon black (CB) and hydrogen [180]. Achieving CB with desired specific surface area, morphology, and internal structure requires precise control of gas-phase chemistry as well as CB inception and surface growth [181]. This remains challenging due to the complex reaction pathways [23] and short time scales [34] involved in methane-to-CB conversion. Although kinetic models coupled with particle dynamics have been developed to describe polycyclic aromatic hydrocarbon (PAH) formation and particle inception [24], most were developed for dilute conditions to facilitate the inference of rate constants [182], and to minimize spectral interference in laser diagnostics [183]. However, high fuel loadings (local mole fractions $\sim$1) are typical in industrial CB synthesis [167], highlighting the need for data to benchmark kinetic models across low- to high-fuel regimes.

Prior studies on soot formation from methane at high fuel loadings in flames [184] and flow reactors [185] lack time- or space-resolved species data due to diagnostic challenges such as strong radiation, interference from intermediates, and limited optical access [186]. Time-resolved CH_4 measurements during 10% CH_4 pyrolysis were initially conducted to refine decomposition reaction rates [187]. Later, simultaneous measurements of CH_4, C_2H_4, and C_2H_2 were performed for 10–40% CH_4 pyrolysis with a focus on maximizing H_2 yield without soot characterization [188].

In this work, we investigate the shock-tube pyrolysis of 5%, 10%, and 30% CH_4-Ar mixtures at P_5=4±0.9 atm and 1800<T_5<2500 K, providing simultaneous measurements of key intermediates and soot volume fraction via optical diagnostics. Simulations were conducted using detailed kinetic models coupled with a sectional population balance model that accounts for inception, surface growth, and agglomeration. Comparison between measurements and model predictions enables (i) evaluation of kinetic model performance for CH_4 conversion and the resulting carbon flux governing C_2H_4 and C_2H_2 formation and (ii) assessment of soot volume fraction evolution over extended times to examine their dependence on temperature and fuel loading.

4.2.1 Experimental method & numerical model

Methane pyrolysis experiments were conducted using the Stanford Flexible Applications Shock Tube (FAST) facility with a 3.35 m driver, 9.70 m driven section, and a 14.12 cm inner diameter. Fig. 4.15 depicts a schematic of the diagnostic suite viewed from the end wall. More details about the FAST facility can be found in [189]. The post-reflected shock pressure (P_5) was measured using an endwall pressure transducer (Kistler 603B1). Time histories of key intermediates were quantified using laser absorption spectroscopy (LAS). C_2H_4 was measured at 10532 nm using an updated diagnostic technique originally developed by Ren et al. [190] and later refined by Pinkowski et al. [191]. To reduce C_2H_2 interference with CH_4 absorption near 2998 nm [192], a NanoPlus distributed feedback laser was employed to detect C-H stretch vibrations of CH_4 at 3366 nm, with temperature-dependent absorption coefficients (σ_v) determined over 1500–2450 K and 1–5 atm. C_2H_2 was detected at 2998 nm using absorption cross-sections reported by Stranic et al. [193]. To address CH_4 interference at 2998 nm, the absorption coefficient of CH_4 at this wavelength was also characterized for the range of experimental temperatures and pressures, allowing correction of the C_2H_2 measurements.

Light extinction at λ=633 nm (Coherent Inc. 70 mW OBIS 633 nm LX laser) was employed to measure time-resolved volume fraction of soot particles with optical properties in the Rayleigh limit [194] as:

$$f_v = \ln\left(\frac{I_0}{I_t}\right)\frac{\lambda}{6\pi \cdot L \cdot E(m)}, \tag{4.1}$$

where L=14.12 cm is the optical path length, and $E(m)$=[0.17, 0.26] [176, 102] is the soot absorption function at 633 nm.

The measured pressure trace was filtered to reduce noise and fluctuations, then applied to the pressure-reactor model in Omnisoot [195] to simulate methane pyrolysis in the shock tube. Four kinetic models were used to describe the gas-phase chemistry: ABF [57], KAUST [171], CRECK [196], and FFCM2 [174]. All models except FFCM2 include PAH chemistry required for soot modeling and can therefore be coupled with a sectional population balance model. The *BIN* species representing solid particles were removed from CRECK to avoid double-counting soot mass. Soot inception

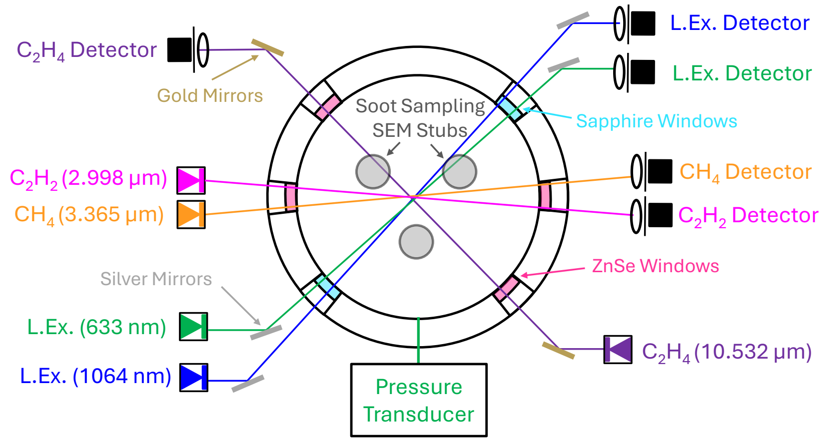


Fig. 4.15 Schematics of the laser diagnostics and shock-tube setup. Spatial and spectral filtering is employed to ensure high-quality absorbance measurements to infer species mole fractions and soot volume fraction. All lasers are aligned in a plane 1 cm from the endwall

was modeled using a modified E-bridge formation based on the inception model proposed by Frenklach and Mebel [24] with some modifications detailed in [195]. Surface growth was described by H-abstraction–C_2H_2/carbon-addition (HACA) [57] and PAH adsorption. Soot precursors include naphthalene, phenanthrene, pyrene, acenaphthylene, acephenanthrylene, and cyclopentapyrene. An Arrhenius equation is used to account for the loss of hydrogen by the carbonization of soot particles as [197]:

$$\left(\frac{\mathrm{d}H_{tot}}{\mathrm{d}t}\right)_{\mathrm{carb}} = -A_{carb} \cdot \exp\left(\frac{-E_{carb}}{RT}\right), \tag{4.2}$$

where A_{carb}=6×10^6 [1/s] and E_{carb}=2×10^5 [J/mol] are the exponential prefactor and activation energy for carbonization, respectively.

4.2.2 Gaseous species

Fig. 4.16a-c presents the time-histories of CH_4 mole fraction at two representative conditions, $T_5\approx$ 1900 K and 2200 K, for 5%, 10%, and 30% CH_4. Absorption cross-sections (σ_v) were calculated using simulated temperatures with FFCM2 and shown in Fig. A.14. Overall, model predictions agree more closely with measurements at 5% and 10% CH_4 than at 30% CH_4.

Methane conversion, defined as the fractional reduction in mole fraction, is $\approx$29% for both 5% and 10% CH_4 at $T_5\approx$1900 K by 1 ms, but decreases from 78% to 66%

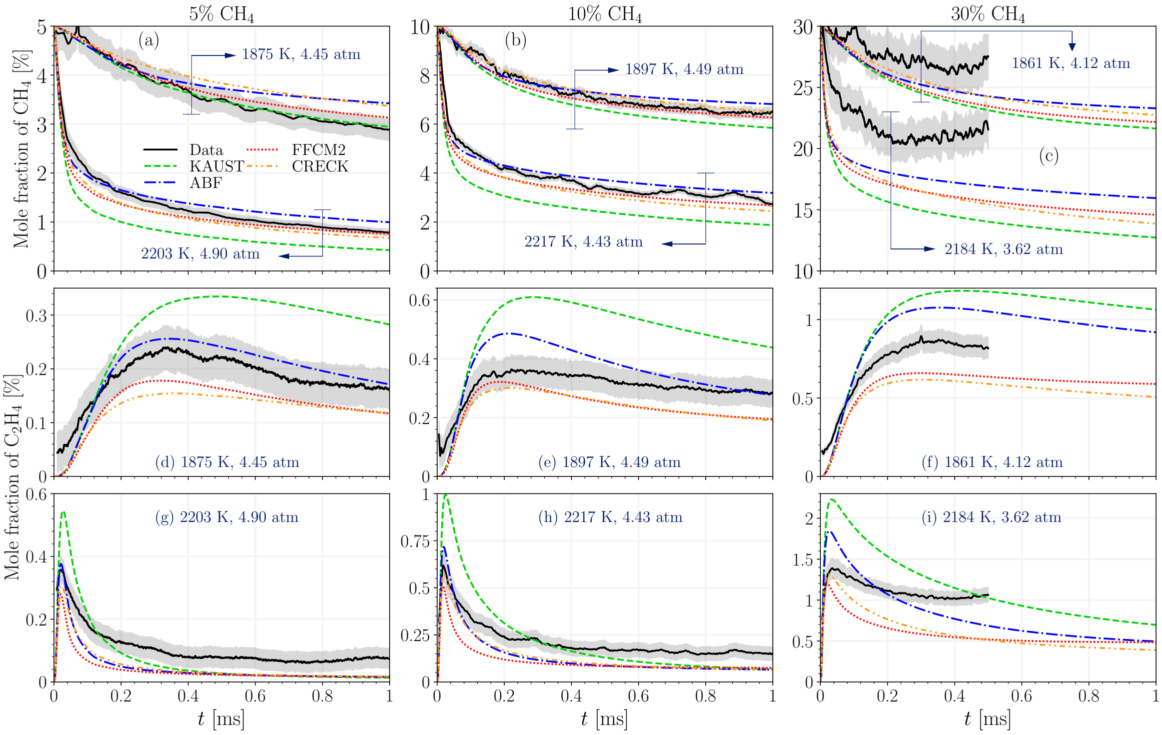


Fig. 4.16 The mole fraction time-history of (a-c) CH_4 and (d-i) C_2H_4 for 5%, 10%, and 30% CH_4 pyrolysis for T_5≈1900 K and 2200 K.

at T_5≈2200 K as fuel loading increases from 5% to 10%. These changes arise from variations in T_5, P_5, and coupled chemical–thermal effects. To isolate these effects, two additional sets of simulations were performed for 5%, 10%, and 30% fuel loadings at T_5=1900 K and 2200 K and P_5=4.5 atm. These cases are referred to as “Const-TP” and “Const-P”. In the Const-P simulations, a constant pressure of 4.5 atm was imposed instead of the measured pressure trace. In the Const-TP simulations, the temperature was also held constant at T_5 and the energy equation was not solved. Fig. 4.17 shows representative Const-TP and Const-P results using FFCM2, while the complete set of results is provided in Fig. A.16.

At T_5=1900 K, Const-TP simulations indicate that CH_4 conversion by 1 ms increases by approximately 2% and 7% at 10% and 30% loadings, respectively, relative to 5% loading. Similar relative trends were observed for the other kinetic models. At T_5=2200 K, FFCM2 predicts full conversion (>99%) by 1 ms, with the time to reach full conversion decreasing as fuel loading increases. In contrast, Const-P simulations show a much stronger reduction in CH_4 conversion with increasing fuel loading. At T_5=1900 K, the conversion at 1 ms decreases by nearly 20% and 38% for 10% and 30% loadings, respectively, compared to 5% loading. These results indicate that the

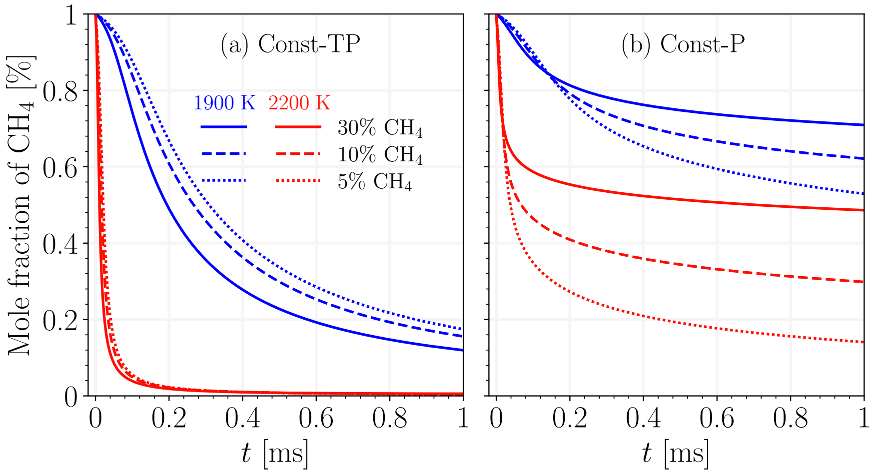


Fig. 4.17 The CH_4 conversion predicted using FFCM2 in Const-TP and Const-P simulations.

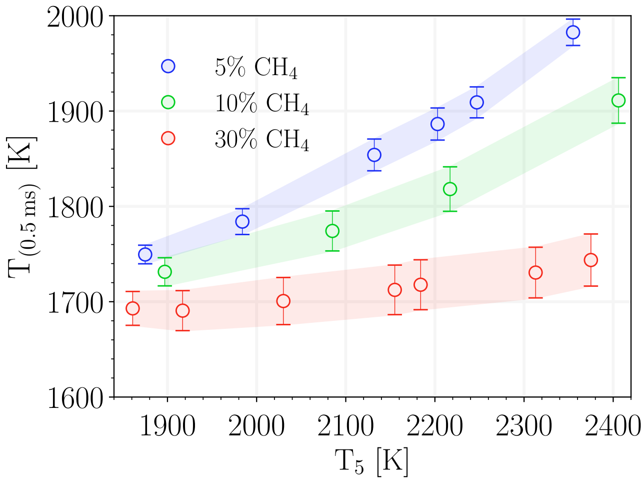
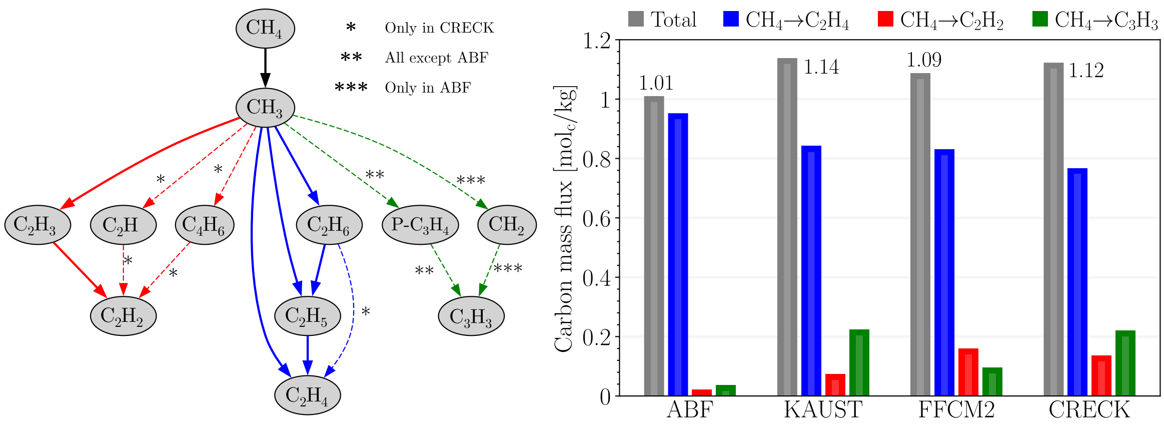


Fig. 4.18 Variation in predicted temperature at 0.5 ms for 5%, 10% and 30% loadings over the studied T_5 range. Error bars and shaded regions represent the spread in predictions from four employed kinetic models.

observed reduction in CH_4 conversion at higher fuel loadings is primarily driven by the endothermicity of pyrolysis, which causes faster and deeper temperature drops and consequently slows the decomposition reactions. The range of predicted temperature at 0.5 ms, $T_{(0.5\,ms)}$, is compared for various loadings in Fig. 4.18, which demonstrates that $T_{(0.5\,ms)}$ decreases with fuel loading. Moreover, the spread in the predicted temperature from kinetic models is ≈60 K for 30% loading, larger than ≈47 K and ≈30 K for 10% and 5%, respectively.

As discussed previously, methane conversion is defined here as the fractional reduction in CH_4 mole fraction, normalized by its initial value, $1 - X_{CH_4}/X_{CH_4,0}$. This



(a) (b)

Fig. 4.19 (a) The common pathways for CH_4 conversion predicted using various kinetic models, and (b) the comparison of total flux from CH_4 to C_2H_4, C_2H_2, and C_3H_3 through exclusive pathways for 5% loading at ≈2200 K.

definition is widely used in shock-tube studies [198, 188, 199] to compute conversion from experimental measurements. However, it can deviate from the actual conversion defined based on the reduction in the carbon mass fraction of CH_4, $M_{c,CH_4}/M_{c,CH_4,0}$, because the total number of moles is not conserved during pyrolysis, whereas total carbon mass is conserved. Evaluation of all kinetic models shows that the mole-based definition systematically overpredicts methane conversion. The difference between mole- and mass-based conversion at 1 ms increases from an average of 1.1% at 5% loading to 5.8% at 30% loading. A comparison of conversion calculated using both definitions for all kinetic models is shown in Fig. A.17.

The comparison of predicted and measured CH_4 mole fraction (Fig. 4.16a, b) indicates that among the kinetic models, FFCM2 provides the best agreement with the 5% and 10% data based on the root mean square error over 1 ms. At 30% CH_4 (Fig. 4.16c), all models overpredict conversion, with discrepancies increasing at higher T_5. Additionally, the measured CH_4 mole fraction rises at 0.2 ms for T_5=1861 K and at 0.4 ms for T_5=2184 K, which is not observed in simulations. The apparent increase is likely caused by interference from overlapping C–H absorption bands of intermediate species formed more abundantly at high fuel loadings [188].

The ABF and KAUST models form the upper and lower bounds of the predicted CH_4 mole fraction across both conditions and the full T_5 range. Methane conversion predicted by CRECK is comparable to ABF at T_5≈1900 K but increases with temperature, approaching FFCM2 predictions at T_5≈2200 K. Differences among models in the prediction of CH_4 conversion are not primarily due to variations in the rate

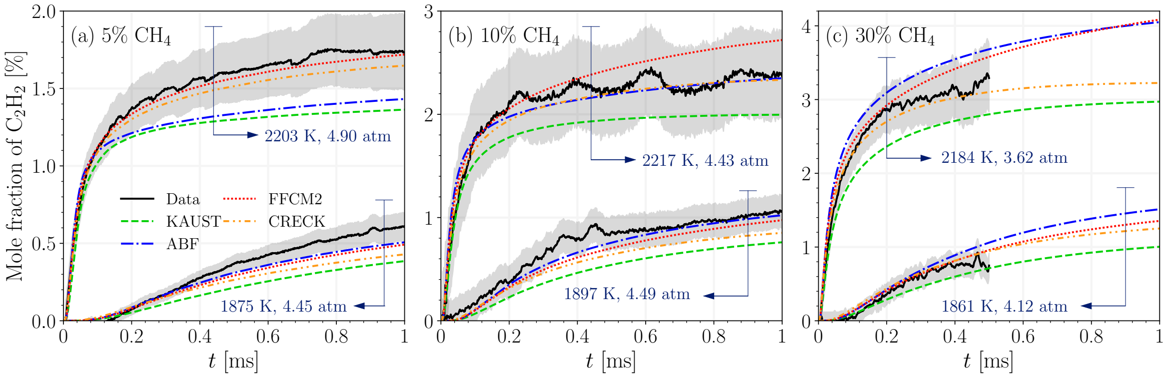


Fig. 4.20 The time history of C_2H_2 mole fraction for (a) 5%, (b) 10% and (c) 30% CH_4 pyrolysis for T_5≈1900 K and 2200 K

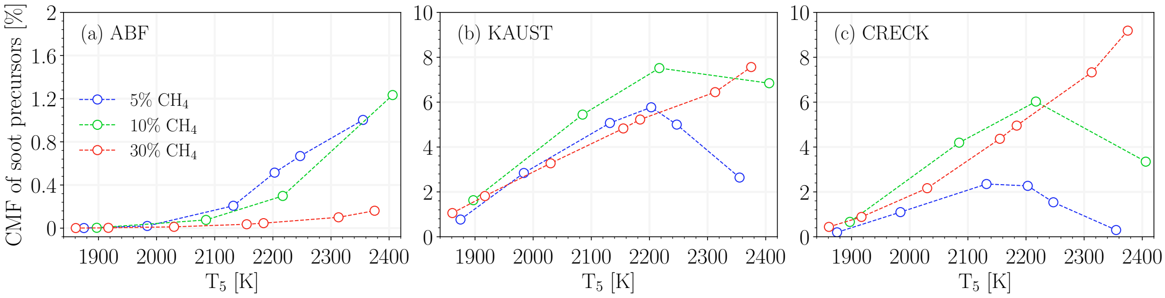


Fig. 4.21 The CMF of soot precursors combined at 1 ms over the studied T_5 range predicted using (a) ABF, (b) KAUST, and (c) CRECK for 5%, 10%, and 30% loadings.

constants of the dominant methane decomposition reactions, $CH_4 + (M) \leftrightarrow CH_3 + H$ and $CH_4 + H \leftrightarrow CH_3 + H_2$. A series of simulations were conducted using modified ABF, KAUST, and CRECK models, where the forward and reverse rate constants of both reactions were taken from FFCM2. As shown in Fig. A.19, the modification of dominant CH_4 decomposition reactions only alters temperature by <1% and CH_4 mole fraction by <7% over 1 ms of simulation.

Thus, modeling suggests the variation between the four kinetic models does not arise from the variability in the rate constants of methane decomposition reactions. Instead, we focus on the carbon flux through the reaction network and the carbon mass fraction (CMF) of species. The net progress rates of reactions were integrated to construct a directed graph [200], where species are represented as nodes, and an edge connects species involved in a reaction. The direction of each edge indicates the net transfer of carbon from reactant to product, and its weight corresponds to the moles of carbon transferred along that reaction pathway over a given time interval, normalized by the total gas mass.

The exclusive carbon pathways from CH_4 to C_2H_2, C_2H_4, and C_3H_3 were identified, and the corresponding carbon fluxes were compared for the kinetic models. Fig. 4.19a illustrates major identified pathways of CH_4 consumption. Fig.4.19b shows the representative results of carbon flux analysis for 5% loading at 2200 K, suggesting that the persistent underprediction of CH_4 conversion by ABF is primarily due to lower carbon flux from CH_4 to C_2H_2 and C_3H_3 compared with the other models. While in ABF, C_3H_3 is predominantly produced via $C_2H_2 + CH_2 \leftrightarrow C_3H_3 + H$, in the other models propyne (P-C_3H_4) dehydrogenation (P-$C_3H_4 \leftrightarrow C_3H_3 + H$) is responsible for C_3H_3 production. As shown in Fig. A.20, CMF of P-C_3H_4 predicted by ABF remains below 0.03% by 1 ms for all loadings and temperatures, which is nearly 60 times lower than the predictions of other kinetic models. Analysis of the reaction fluxes indicates that propyne (P–C_3H_4) is produced primarily via $C_2H_2 + CH_3 \leftrightarrow$ P-$C_3H_4 + H$. The forward rate constant of the reaction in ABF is nearly two orders of magnitude lower than the other models (Fig. A.18), leading to underprediction of P-C_3H_4. FFCM2 predicts a larger CMF of P-C_3H_4 compared to KAUST and CRECK due to the lower subsequent consumption of C_3H_3 leading to carbon accumulation in P–C_3H_4.

The lower consumption of C_3H_3 in FFCM2 is due to the lack of propargyl recombination reaction, $2C_3H_3 \leftrightarrow$ A1, which is dominant in ABF and KAUST, and a major consumption reaction in CRECK. On other hand, C_3H_3+$CH_3$$\leftrightarrow$$C_4H_6$ is the dominant C_3H_3 consumption reaction in CRECK and FFCM2. The peak CMF of C_3H_3 consistently decreases in all kinetic models is primarily due to lower CH_4 conversion at higher loadings. KAUST exhibits the highest carbon flux from C_3H_3 to A1, which rapidly propagates to larger PAHs. At 5% and 10% loadings, nearly 23% of the total carbon mass resides in C_6 and larger species ($C_{\geq 6}$) in KAUST, whereas the CMF of $C_{\geq 6}$ remains below 12% for CRECK and ABF.

More than 55% of the carbon flux from CH_4 is directed to C_2H_4 across all models and conditions through two dominant pathways: $CH_4 \rightarrow CH_3 \rightarrow C_2H_5 \rightarrow C_2H_4$ and $CH_4 \rightarrow CH_3 \rightarrow C_2H_6 \rightarrow C_2H_5 \rightarrow C_2H_4$. As shown in Fig. 4.16d-i, ABF and KAUST overpredict the C_2H_4 peak at 10% and 30% loadings. FFCM2 and CRECK yield similar C_2H_4 profiles and reproduce the peak more accurately. The subsequent decline in C_2H_4 results from carbon transfer to C_2H_2 through the dominant pathway $C_2H_4 \rightarrow C_2H_3 \rightarrow C_2H_2$ except for CRECK, which includes an additional direct pathway from C_2H_4 to C_2H_2 via the third-body reaction $C_2H_4 + M \leftrightarrow C_2H_2 + H_2 + M$.

Time-resolved mole fraction of C_2H_2 is shown in Fig. 4.20. At 30% loading, C_2H_2 measurements are better captured by the kinetic models compared to CH_4 because the selected transitions near 2998 nm targeting the asymmetric C-H stretch of acetylene are spectrally isolated from other hydrocarbons [201]. At 5% loading, a noticeable drop is observed in ABF's prediction of C_2H_2 relative to the other models from 1875 K to 2203 K due to the rapid accumulation of carbon in A1. The CMF of $C_{\geq 6}$ changes from 0.7% to 10.7% for ABF and from 3.3% to 6% for CRECK. FFCM2 best captures C_2H_2 yield at 5% over the entire T_5 range. At 10% loading and 2217 K, all model predictions fall within the measurement uncertainty, but their trends differ slightly. FFCM2 predicts a faster rise after 0.4 ms in C_2H_2 compared to ABF and CRECK, which better reproduce the trends in the measurements. At 30% loading, ABF and KAUST closely predict the upper and lower uncertainty range of C_2H_2 mole fraction at both temperatures, respectively. The high C_2H_2 predictions by ABF, despite its low CH_4 conversion values, are the compound effect of having the highest carbon flux from C_2H_4 to C_2H_2 and the lowest carbon flux from C_2H_2 to C_3H_3.

Fig. 4.21 shows the dependence of the carbon mass fraction (CMF) of soot precursors on T_5 and fuel loading at 1 ms. Results for FFCM2 are not included because it does not contain PAH chemistry. ABF predicts the lowest CMF ($\leq$1.2%) among the kinetic models, with a monotonic increase with T_5 and a decrease with fuel loading. In contrast, KAUST and CRECK exhibit a bell-shaped temperature dependence at 5% and 10% loadings, while the CMF increases monotonically with T_5 at 30% loading.

These trends arise from the combined effects of endothermic cooling during pyrolysis and the intrinsic temperature dependence of precursor formation in each kinetic model. The Const-TP results (Fig. A.21) show that all models predict a bell-shaped CMF profile with temperature. The peak CMF occurs near 2075 K for ABF, but around 1875 K for KAUST and CRECK. As shown in Fig. 4.18, the predicted gas temperature at 0.5 ms never exceeds 2075 K under the studied conditions, so all ABF cases remain on the rising side of the bell-shaped curve, yielding a persistently increasing trend with T_5. For KAUST and CRECK, the predicted temperature exceeds 1875 K at 5% and 10% loadings when $T_5 >$ 2200 K, causing the CMF of soot precursors to decrease beyond this point. At 30% loading, stronger endothermic cooling limits the temperature rise, preventing the system from reaching the descending branch of the bell-shaped curve and resulting in a monotonic increase in CMF with T_5.

4.2.3 Soot particles

As previously discussed, the time histories of species mole fraction predicted using CRECK are in overall closer agreement with the measurements, so it can provide a more accurate description of carbon flux from $C_{1,2}$ species to $C_{6\geq}$ chemistry including PAHs, necessary for description of soot inception and surface growth. Soot model

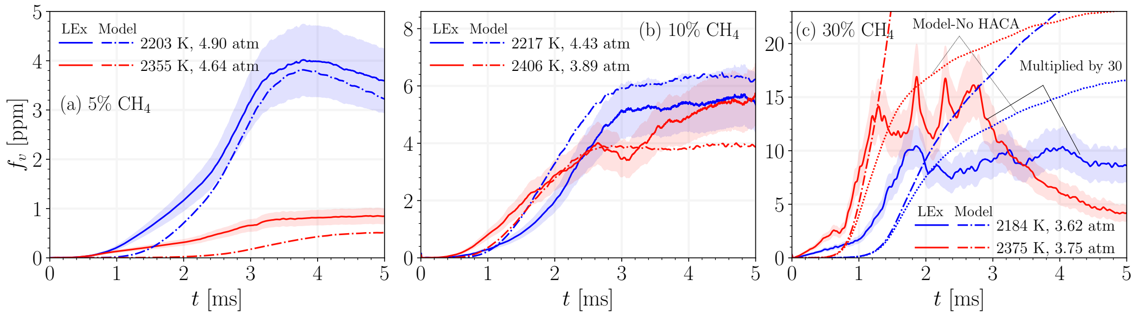


Fig. 4.22 The time history of soot volume fraction, f_v, for (a) 5%, (b) 10%, and (c) 30% CH_4 pyrolysis from laser extinction (LEx) and model prediction using CRECK at $T_5 \approx 2200$ K and 2400 K.

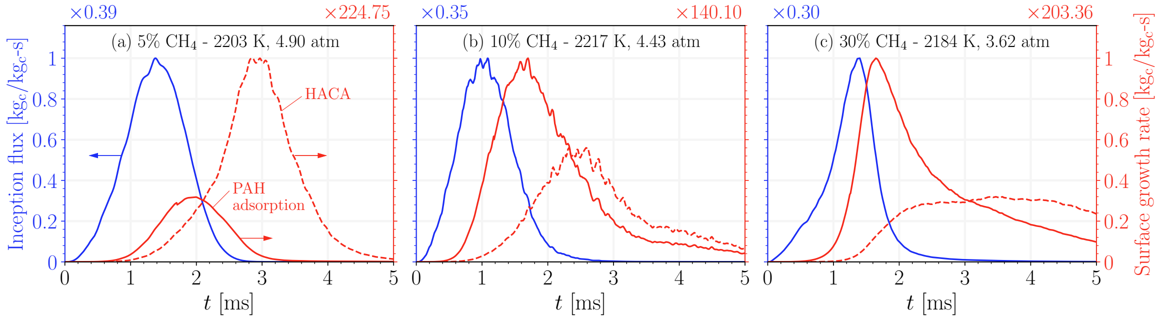


Fig. 4.23 The soot inception flux and surface growth rates via PAH adsorption and HACA (a) 5%, (b) 10%, and (c) 30% CH_4 pyrolysis predicted using CRECK at $T_5 \approx 2200$ K.

is coupled to CRECK, and it affects the gas chemistry mainly by consumption of precursors (Section 4.2.1) and C_2H_2. Additional simulations were performed without considering soot, and the analysis of results suggests that soot formation has negligible effect on the mole fraction of CH_4, C_2H_4, and C_2H_2 in 1 ms of simulation.

Fig. 4.22 compares the predicted soot volume fraction, f_v, using CRECK with the measurements of 633 nm laser extinction recorded for 5 ms. The shaded area represents the uncertainty in f_v due to the variability of the soot absorption function, $E(m)$, discussed in Section 4.2.1. The f_v at $T_5 \approx 1900$ K were below the signal noise level, thus not shown here.

The model successfully captures the time trend of f_v at 5% and 10% loadings, but a noticeable underprediction at 5% is observed. The predicted primary particle and mobility diameters are below 20 nm and 100 nm, respectively, supporting the assumption of negligible contribution of scattering (less than 7%) to total extinction [**?**]. The late-time ($t \geq 3.5$ ms) decrease in the predicted and measured f_v at 5% loading (Fig. 4.22a) is due to the pressure drop by nearly 2 atm between 3-5 ms that expands the gas, fictitiously reducing f_v=$V_{\mathrm{soot}}/V_{\mathrm{gas}}$. In fact, as shown in Fig. A.22a, the soot yield ($M_{\mathrm{c,soot}}/M_{\mathrm{c,CH_4,0}}$) from predictions at 2203 K and 2322 K asymptotically approaches 39% and 6.2%, respectively. The model captures the reduction of f_v with temperature from ≈2200 K to ≈2400 K observed at 5% and 10% loadings, consistent with the bell-shaped temperature dependence of f_v [175], which closely follows that of soot precursors shown in Fig. 4.21.

At 30% loading (Fig. 4.22c), there is a large discrepancy between the model and the measurements that were multiplied by 30 to better illustrate the trends. The measured f_v remains below 0.6 ppm until 5 ms, which is even lower than the f_v of 5% and 10% loadings. Despite overprediction, model captures the overall early-time behavior of the measurements that begins with a slow rise (<1 ms for T_5=2184 K) followed by a faster increase (1-1.8 ms). However, the subsequent decrease in the measurements at nearly 1.8 ms for T_5=2184 K and 1.2 ms for T_5=2375 K differs from the model predictions that continue to grow. Interestingly, the time of decrease of measured f_v at both T_5 is close to the inflection point of the predicted f_v, which represents the transition of the dominant surface growth mechanism from PAH adsorption to HACA. The simulation results without HACA (dotted line in Fig. 4.22c) highlights the significant contribution of HACA to f_v growth beyond the infection point, and reveals remarkable similarity in their time history and the measurement. Such similarity could indicate potential missing pathways or mechanisms that limit HACA surface growth. Nevertheless, experimental uncertainties inherent at high fuel loadings, especially non-idealities such as bifurcation [202] and shock-boundary layer interaction [203] can influence the optical path, potentially introducing artifacts that the 0D model does not capture.

Figure 4.23 presents the inception flux and surface growth rates for all fuel loadings at ≈2200 K, expressed as the rate at which each mechanism converts a fraction of the total carbon mass into soot. Inception and PAH adsorption reach their maxima before 2 ms and subsequently decline due to the depletion of soot precursors. The transition in the dominant surface growth mechanism is observed for all fuel loadings at ≈2200 K, with the peak of HACA occurring later than that of PAH adsorption. In addition, the relative contribution of PAH adsorption increases with fuel loading compared to HACA. At 30% loading (Fig. 4.23c), HACA does not exhibit a pronounced peak, but instead maintains a relatively uniform rate between 2 and 5 ms. This trend in the relative strength of HACA and PAH adsorption is not observed at ≈2400 K (Fig. A.23), highlighting the strong dependence of inception and surface growth rates on the temperature history, which is governed by the post-reflected-shock pressure trace and the endothermicity of methane pyrolysis.

4.2.4 Conclusions

This study presents the first simultaneous time-resolved measurements of key gas-phase species and soot volume fraction during shock-tube pyrolysis of 5%, 10%, and 30% methane over a temperature range of 1800–2500 K and pressures near 4 atm. Using multi-wavelength laser diagnostics, we quantified the evolution of CH_4, C_2H_4, and C_2H_2. The results show that methane conversion and intermediate yields strongly depend on temperature and fuel concentration, with significant model–measurement deviations for CH_4 mole fraction emerging at 30% loading due to increased absorption interference. Comparisons with detailed kinetic models revealed that the FFCM2 and CRECK models capture gas-phase species more accurately despite significant differences in pathways. Carbon flux analysis highlighted the dominant role of C_3H_3 in channeling carbon toward C_2H_2 and PAHs, and the role of P-C_3H_4 as a key precursor of C_3H_3. The model captured the time and temperature trends of f_v at 5% and 10% loadings, but a substantial overprediction with similarities in time trends is observed at 30% loading, indicating potential missing chemistry or physical effects that could inhibit surface growth at higher loadings.

4.3 Identifying stages of carbon black evolution

This section presents the paper titled "*Yield, morphology and maturity of carbon black made by methane pyrolysis in a shock tube*" and submitted to Carbon journal, and it is focused on using Omnisoot to identify different stages of CB formation during methane pyrolysis in a shock-tube.

Despite extensive studies on the chemical kinetics of methane pyrolysis (Refs. in [204]), the mechanisms governing CB formation remain poorly understood. This limitation arises from incomplete knowledge of the pathways leading to CB inception and from uncertainties in existing kinetic mechanisms [175]. These uncertainties originate with methane dissociation reactions [205] and propagate through key intermediates such as methyl (CH_3) [182], ethylene (C_2H_4), and acetylene (C_2H_2) [206]. The uncertainty is further amplified for polycyclic aromatic hydrocarbons (PAHs) [187], which are widely accepted as the primary precursors to soot and CB [18]. Previous studies have shown that predicted concentrations of large PAHs can vary by orders of magnitude depending on the kinetic mechanism employed [168]. This variability underscores the need for time-resolved measurements of gas-phase species and particulate properties under well-controlled conditions to improve quantitative understanding of methane conversion, intermediate formation, and carbon flux to CB through inception and surface growth.

Shock tubes are powerful tools for this purpose because a wide range of temperatures (1700-3000 K [175]) and pressures (1-100 atm [207]) can be achieved by adjusting the pre-shock pressure ratio; the instant heating of the mixture behind the reflected shock creates a nearly isothermal and isobaric condition over a short test time (1-3 ms) with minimal diffusion and mixing [194], which is an ideal setting for kinetic studies of pyrolysis pathways and CB inception and surface growth.

Capturing the rapid decomposition of methane and the subsequent formation of intermediates can be achieved with the high temporal resolution offered by continuous-wave (CW) laser absorption spectroscopy [191]. In this technique, the attenuation of light in an absorbing medium is related to the gas density through the Beer–Lambert law, provided that the absorption cross-section is known. Accurate application, however, is complicated by the strong dependence of the absorption cross-section on temperature, pressure, and gas composition. CB yield and volume fraction can also be quantified with good accuracy from light extinction measurements based on the same principle, as long as the particles remain in the Rayleigh limit and the particle volume fraction is below 10 ppm [194]. This approach requires prior knowledge of the optical properties of CB, represented by the absorption function $E(m)$.

While a constant $E(m)$ value of mature CB is commonly used to retrieve volume fractions from extinction data [208, 175, 209], the optical properties are noticeably different for incipient and mature CB [134]. During maturation, CB particles undergo structural and chemical changes that result in the reduction of hydrogen-to-carbon ratio (H/C) due to carbonization [44] that increases the density and specific heat [134]. The maturity level of CB is linked to the spectral dependence of absorption [210] quantified by the ratio of $E(m)$ at two different wavelengths [211] that approaches unity for mature CB; therefore, performing extinction measurements at multiple wavelengths can reduce the uncertainties of $E(m)$ and provide a good measure of CB maturity.

This section deals with simultaneous measurements of gaseous species and carbon black (CB) during the pyrolysis of 5% CH_4 at post-reflected-shock pressures near 5 atm and temperatures between 1800 and 2500 K. In that work, a computational framework coupling detailed gas-phase chemistry with CB inception, surface growth, and coagulation is developed to describe methane-to-CB conversion, successfully reproducing the measured time histories of key gas-phase species and CB volume fraction, as well as the final primary particle diameter. A detailed analysis of the simulation results, complemented by experimental measurements, is conducted to delineate distinct stages of CB formation based on the time evolution of f_v and carbon yield. Within each stage, the evolution of inception flux and surface growth mechanisms is examined to elucidate their influence on f_v, yield, and morphology, as well as their coupling with gas-phase chemistry. The deviation of model predictions from f_v, d_p, and induction time measurements are discussed in detail. Particular emphasis is placed on the role of carbonization in describing CB maturity, its effect in limiting surface growth rates, and its connection to the spectral dependence of CB particles characterized through multi-wavelength light extinction.

4.3.1 Experimental methods

A brief overview of the employed experimental techniques is presented here. Mole fraction time histories of methane (CH_4), acetylene (C_2H_2), and ethylene (C_2H_4) were captured using CW laser absorption at 3.365 μm, 2.998 μm, and 10.532 μm, respectively [193, 191, 192]. CB volume fraction was measured using laser light extinction at λ=633 nm and 1064 nm:

$$f_v = \frac{1}{L}\frac{\ln(I_0/I_t)\lambda}{6\pi E(m)}, \tag{4.3}$$

Table 4.5 The post-reflected-shock pressure (P_5) and temperature (T_5) of the executed shock-tube tests.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
T_5 [K]	1875	1984	2132	2203	2247	2322	2355
P_5 [atm]	4.45	4.30	4.17	4.90	4.51	4.55	4.64

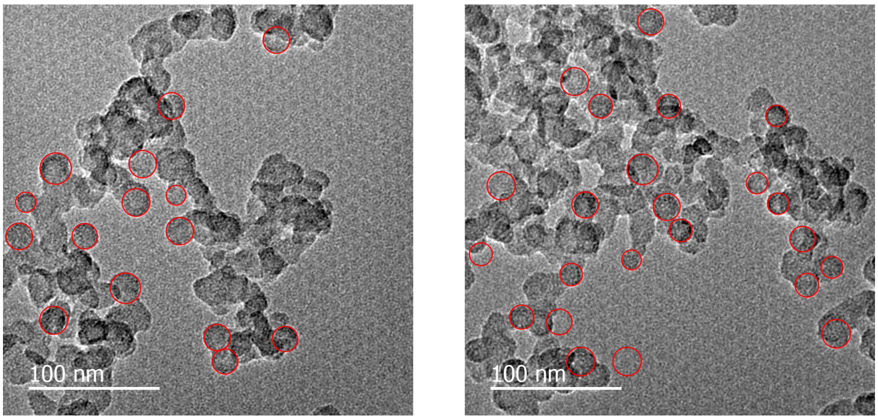


Fig. 4.24 Samples of TEM images collected for T_5=2203 K. Red circles represent the identified primary particles used to measure d_p.

where L=14.12 cm is the absorption path length, equal to the inner diameter of the shock tube. λ is the absorption wavelength, and I_0 and I_t are the incident and transmitted intensities, respectively. Table 4.5 indicates the conditions of the shock-tube tests with 1800 K<T_5<2400 K and P_5=4.5±0.5 atm. f_v was determined from the measured extinction at 633 nm and 1064 nm, considering the reported variability of $E(m)$ in the literature, ranging from 0.174 [176] to 0.26 [102] for 633 nm and from 0.2 to 0.3 for 1064 nm [212].

Additionally, CB samples were collected onto imaging stubs mounted in the shock-tube endwall. Samples were extracted from the interior surface of the shock tube endwall after each experiment to allow for imaging and analysis of the particulates. TEM images were recorded with a FEI Tecnai G2 F20 X-TWIN microscope and Gatan SC200 camera for all test cases. Primary particle diameter, d_p, was obtained by visually identifying the smallest circumscribed circle enclosing each primary particle in the TEM images and calculating the arithmetic and geometric means. A minimum of 200 primary particles were measured for each condition. Fig. 4.24 depicts representative TEM images collected for the test at T_5=2203 K, with identified primary particles shown as superimposed red circles.

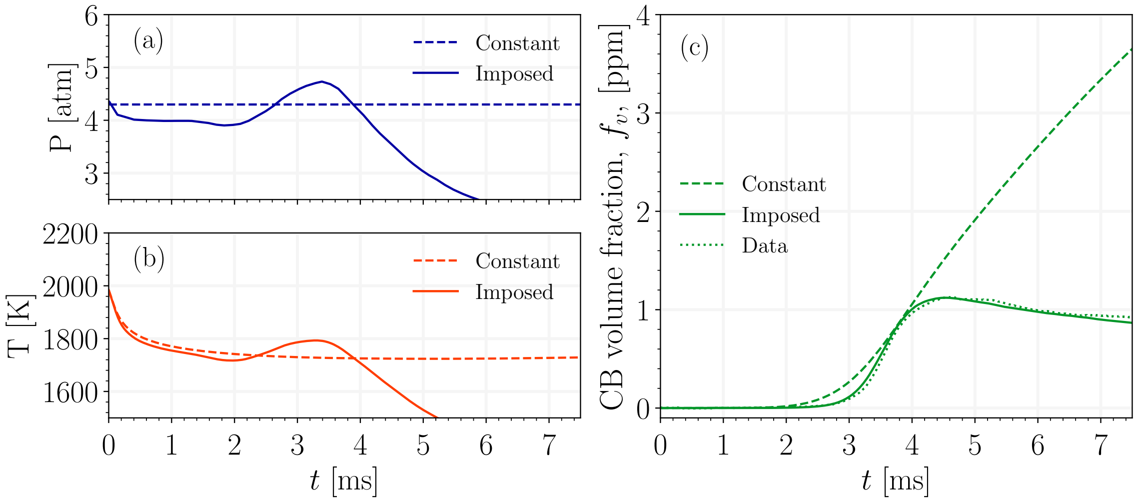


Fig. 4.25 The comparison of (a) pressure (b) temperature (c) CB volume fraction for simulations with constant and imposed pressure trace at T_5=1984 K.

4.3.2 Numerical model

The 0-D imposed-pressure reactor (IPR) model of Omnisoot [195] was used to simulate CB formation coupled with chemical kinetics during the pyrolysis process under post-reflected-shock conditions in the shock tube. IPR incorporates a sectional population balance model (SPBM) to describe the evolving size distribution of CB agglomerates by tracking the number of agglomerates, N_{agg}, primary particles, N_{pri}, total carbon, C_{tot}, and hydrogen content, H_{tot}.

The measured pressure trace was filtered to reduce noise and fluctuations, and then imposed on the model. The measured pressure remains within 10% of the initial value until 2 ms for all tests. Fig. 4.25a highlights the deviation of the (filtered) pressure time history from the initial pressure of P_5=4.30 atm at T_5=1984 K. The pressure starts to rise at ≈2.2 ms, reaches its peak at ≈3.1 ms, and drops towards the end of the test time due to the arrival of the expansion wave. Fig. 4.25b shows a similar deviation in the predicted temperature. The analysis of species mole fractions showed that the imposed pressure results in lower CH_4 conversion and C_2H_2 yield, thereby reducing PAH formation and CB inception flux. As shown in Fig. 4.25c, f_v values predicted using constant and imposed pressure are close until 3.9 ms, but the discrepancy grows after that, where only the simulation with the imposed pressure captures the experimental time trends.

Soot precursors considered in this study include naphthalene (A2), phenanthrene (A3), pyrene (A4), acenaphthylene (A2R5), acephenanthrylene (A3R5), and cyclopentapyrene (A4R5), due to extensive support in the literature for their role in the initial steps of CB formation [213, 214]. The E-Bridge Modified model is used to describe the dimerization and adsorption of precursors that contribute to CB inception and surface growth, respectively. The general formulation of the model is based on the inception model proposed by Frenklach and Mebel [24], with some modifications detailed in [195]. Two scaling factors, η_{inc} and η_{ads}, between 0 and 1 are used to adjust the inception and PAH adsorption rates, respectively [195].

The surface growth is described by the H-abstraction-C_2H_2/Carbon-addition (HACA) scheme [57], but the surface density of hydrogenated sites, $\chi_{soot\text{-}H}$, is dynamically calculated as the ratio of hydrogen content to the surface area of agglomerates, i.e., $\chi_{soot\text{-}H} = H_{tot}/A_{tot}$ [85]. An Arrhenius equation is used to account for the loss of hydrogen by the carbonization of CB particles as [197]:

$$k_{carb} = A_{carb} \cdot \exp(\frac{-E_{carb}}{RT}), \tag{4.4}$$

$$\left(\frac{\mathrm{d}H_{tot}}{\mathrm{d}t}\right)_{\mathrm{carb}} = -k_{carb} \cdot H_{tot}, \tag{4.5}$$

where A_{carb} and E_{carb} are the exponential prefactor and activation energy for carbonization, respectively. H_{tot}/A_{tot} depends on the relative contributions of PAH adsorption and HACA, and the carbonization rate. The comparison of $\chi_{soot\text{-}H}$ with its nominal value of 2.3×10^{19} sites/m^2 used in the original HACA formulation [39] shows that the dynamic method strongly depends on the temperature time history, and it predicts an overall larger number of reaction sites for T_5<2300 K (Fig. A.7).

4.3.3 Precursor chemistry

A detailed discussion of the time histories of measured species was presented in our previous work [169]. The performance analysis of six kinetic mechanisms indicated that FFCM2 [174] provides the best agreement across the entire temperature range, especially for CH_4 conversion and C_2H_2 yield, but it does not include PAH chemistry necessary for inception and surface growth, so it cannot be used for soot modeling. As shown in Fig. 4.26a and b, the CRECK mechanism predicts CH_4 and C_2H_2 mole fractions in close agreement with the measurements, considering their uncertainty range. CRECK includes the PAH chemistry necessary for CB inception and surface growth. Fig. 4.26c and d depict the mole fractions of A2 and A2R5, which similarly increase with time. For T_5>2200 K, the mole fraction of PAHs reaches its peak and

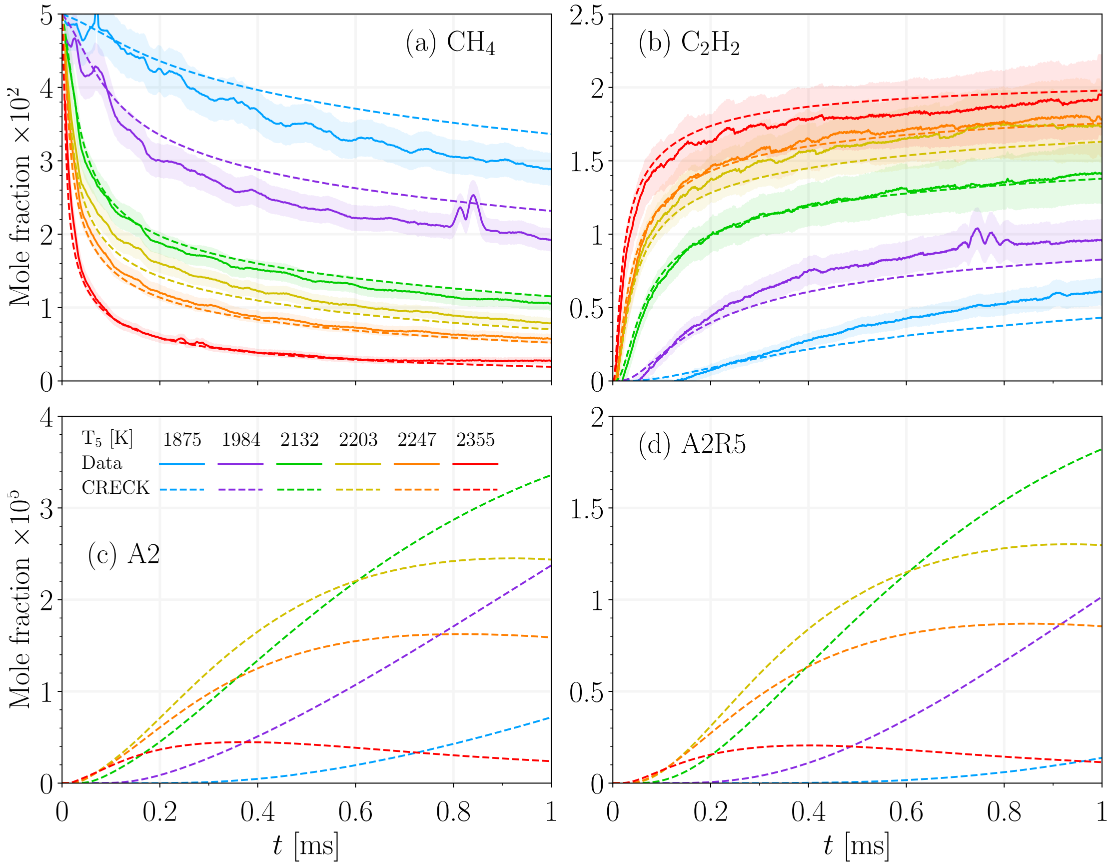


Fig. 4.26 The time history of (a) CH_4 and (b) C_2H_2 predicted using CRECK mechanisms in close agreement with the laser absorption measurements. The shaded area represents the uncertainty in the measurements.

gradually decreases within 1 ms, primarily due to conversion to C_2H_2 rather than CB inception. The mole fraction of A2 and A2R5 sampled at 1 ms follows the well-known bell-shaped behavior with the peak near 2200 K. The analysis of the carbon mass fraction (CMF) of precursors over the studied T_5 range indicated that A2 and A2R5 are the major contributors to CB inception due to their larger CMFs compared to the other precursors. The initial composition of CB is dictated by the H/C ratio of constituent precursors, which changes within a narrow range, slightly decreasing from 0.8 at 1875 K to 0.67 at 2355 K.

4.3.4 Stages of CB formation

As shown in Fig. 4.27a, the optimized model reproduces the time evolution of f_v measured using 1064 nm light extinction at 1984 K. The reported f_v measurements are the average of values calculated using the min and max $E(m)$ of each wavelength. The predicted f_v profile exhibits four distinct stages: (I) an initial period with negligible CB mass due to low inception and surface growth, known as the CB induction time, which is slightly overestimated by the model (≈2.3 ms) compared to the measurements (≈1.9 ms). A detailed discussion of induction time over the entire T_5 range is provided in Section 4.3.5. In stage (II), rapid growth occurs in f_v, driven by increasing PAH adsorption and HACA. The end of stage (II) signifies the inflection point in f_v, where both surface growth mechanisms reach their peak, as shown in Fig. 4.27b. At 1984 K, PAH adsorption is the dominant surface growth, accounting for nearly 79% of total CB mass by 7.5 ms, and the rest (≈21%) originates from HACA. Inception is mainly active in stages (I) and (II), and its contribution to total CB mass remains less than 0.3%. In stage (III), the increase of f_v slows down due to the decline in both surface growth mechanisms. The time histories of f_v and carbon yield, CY, are similar in the first two stages, but f_v deviates from CY in stage (III). While f_v gradually decreases at t≈4.6 ms, CY gradually increases to reach 14.3%. The decline of $f_v = V_{CB}/V_{gas}$ at ≈4.56 ms is attributed to gas-phase expansion caused by the arrival of the expansion wave at the endwall, reducing pressure (Fig. 4.25a) and increasing V_{gas} while V_{CB} remains constant.

In stage (IV), surface growth rates become negligible, and CB agglomerates are chemically frozen, growing primarily through Brownian coagulation. As shown in Fig. 4.27c, CB morphology undergoes a corresponding transition. Here, d_p and d_m denote the average primary particle diameter and mobility diameter, respectively. These diameters remain identical until approximately 3 ms, near the peak of inception in stage (II), when most CB particles are spherical. Beyond 3 ms, the growth of d_p slows and reaches 99% of its terminal value of 14.4 nm before the end of stage (II) at approximately 5.89 ms, while d_m continues to increase due to ongoing coagulation.

The rapid decline in inception rates (Fig. 4.27b) is primarily driven by the consumption of precursors through PAH adsorption, as evidenced by a factor of 2.3 decrease in their CMF between approximately 3 ms and 4.5 ms (Fig. 4.27c). Beyond 4.5 ms, PAH adsorption decreases as the temperature falls below 1600 K, which suppresses PAH dehydrogenation in the E-Bridge Modified model [195] and reduces the availability of PAH radicals for adsorption onto CB particles. Consequently, the CMF of precursors continues to decline after 4.5 ms and approaches 0.172 by the end of stage (III).

The reduction in HACA rates during stage (III) is mainly caused by a temperature-driven decrease in H radical concentration, which lowers the number of dehydrogenated surface sites, χ_{soot°, in the HACA formulation [57]. In addition, depletion of hydrogenated surface sites due to CB carbonization suppresses HACA rates throughout the

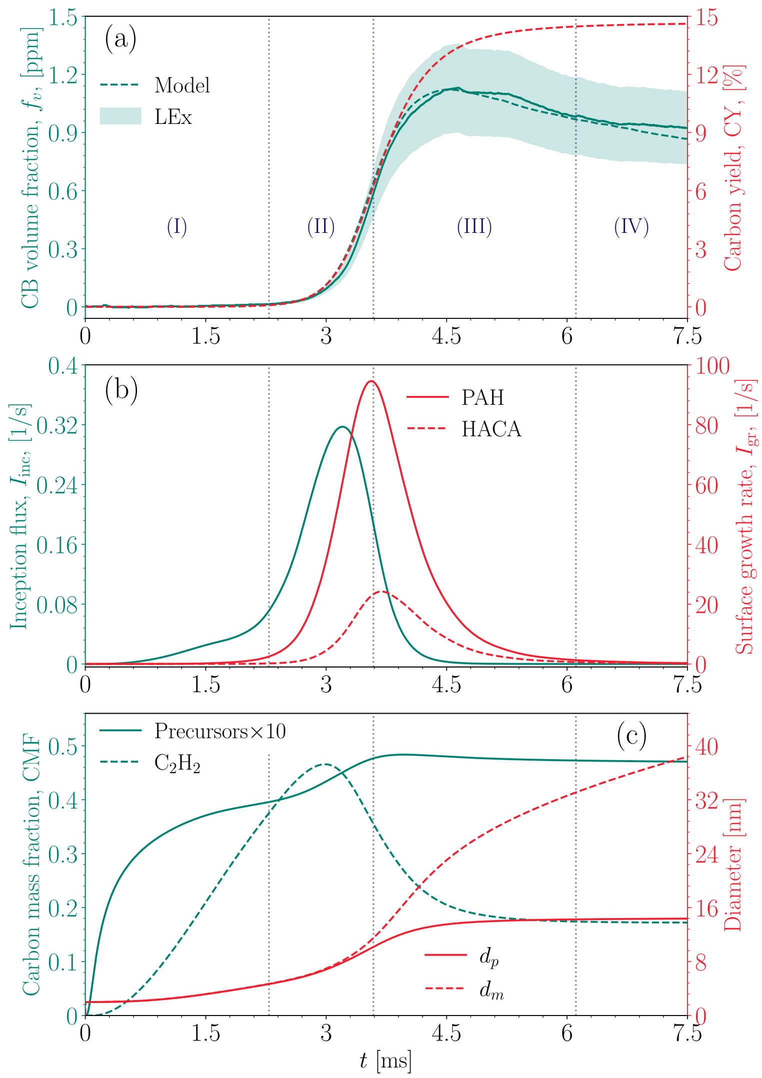


Fig. 4.27 The four stages of CB formation demonstrated at T_5=1984 K for (a) CB volume fraction, f_v, and carbon yield, CY (b) surface growth rate, I_{gr}, via HACA and PAH adsorption, and inception flux; (c) the carbon mass fraction, CMF, of C_2H_2 and CB precursors combined, and primary particle diameter, d_p, and mobility diameter d_m, of CB agglomerates. Panel (a) depicts the light extinction measurements at $\lambda = 1064$ nm with the shaded area representing the uncertainty from the variability in E(m).

simulation. Simulations performed without carbonization indicate an overprediction of the peak HACA rate by a factor of 2.4 and an overestimation of the final carbon yield by 46%. While carbonization reduces the magnitude of the HACA rate, it does not affect the timing of its peak. Accordingly, the four stages identified in Fig. 4.27 remain unchanged by carbonization.

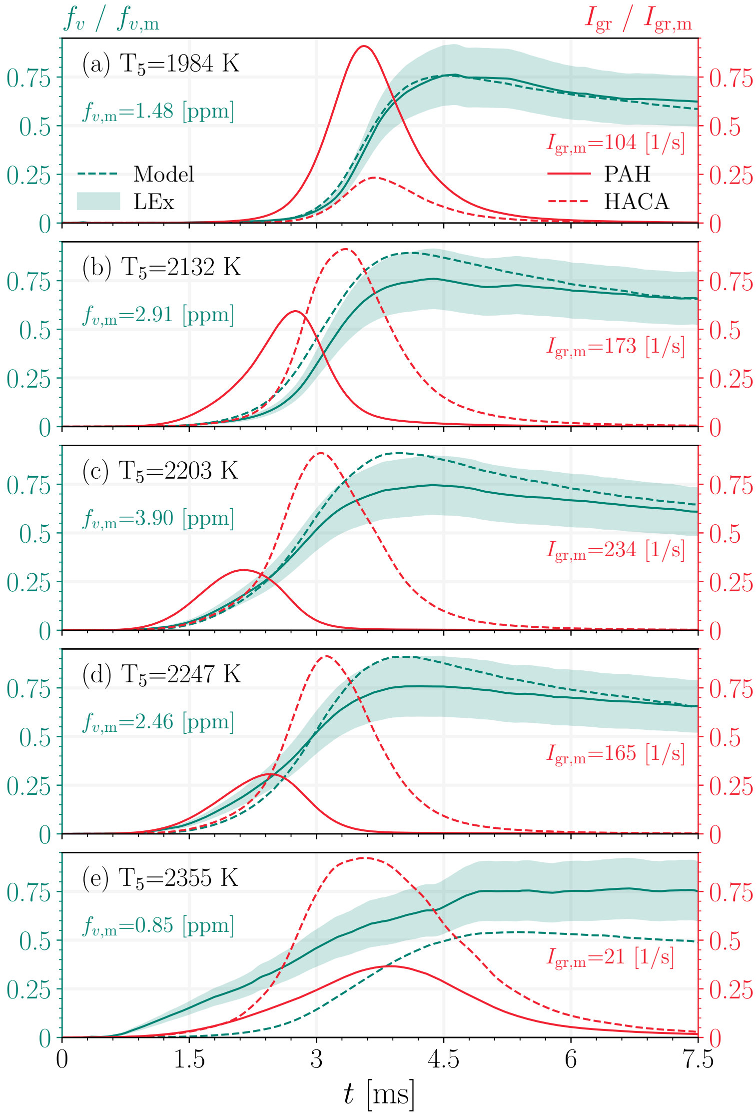


Fig. 4.28 The time histories of normalized CB volume fraction, $f_v/f_{v,m}$ and normalized surface growth, $I_{gr}/I_{gr,m}$ at various temperatures. The measurements are from light extinction at $\lambda = 1064$ nm with the shaded area representing the uncertainty from the variability in E(m).

Fig. 4.28 depicts the normalized CB volume fraction, $f_v/f_{v,m}$, and surface growth rate $I_{\mathrm{gr}}/I_{\mathrm{gr,m}}$ for the entire studied T_5 range. The time history of f_v is predicted by the model for the entire T_5 range in good agreement with light extinction measurements, considering the sensitivity of f_v to the pressure trace. The measured f_v demonstrates that the onset of CB formation occurs earlier for higher T_5. In other words, stage (I) becomes shorter with increasing T_5. The model captures this shift from 1984 K to 2203 K, but noticeably predicts longer CB induction times at 2247 K and 2355 K compared to the measurements. The behavior of the model is dictated by precursor chemistry. PAH concentration has a bell-shaped temperature dependence with the peak at T_5=2203 K. Therefore, the inception and PAH adsorption rates decrease when T_5 increases to 2247 K and 2355 K.

HACA becomes the dominant surface growth mechanism for T_5 >1984 K, as indicated by its higher peak compared to PAH adsorption. The analysis of integrated surface growth rates indicates that the contribution of HACA to the total carbon mass of CB increases from 21% at 1984 K to 64% at 2132 K and 77% at 2203 K due to the increase in the inception flux producing more particles (active growth sites) and higher C_2H_2 concentration (Fig. 4.26b). However, the contribution of HACA slightly decreases to 75% and 69% for 2247 K and 2355 K, respectively, because the suppression of HACA by carbonization increases with temperature. Without carbonization, carbon yield reaches ≈100% for T_5 ≥2132 K with HACA accounting for more than 90% of carbon mass.

As shown in Fig. 4.29, d_p starts from 2 nm, which is the diameter of incipient CB defined in the model, increases through surface growth, and asymptotically approaches its final value, $d_{p,\infty}$. The characteristic time $\tau_{d_{p,\infty}}$, defined as the time at which d_p reaches 99% of $d_{p,\infty}$, remains below 6 ms for all T_5. These results indicate that no significant evolution of the primary particle diameter occurs beyond $\tau d_{p,\infty}$. Therefore, d_p measured from collected TEM samples provides a representative measure of CB morphology within the studied time window. The predicted $d_{p,\infty}$ is in close agreement with the measurement, except for a noticeable underprediction at T_5=1984 K. dp, ∞ varies in a narrow range of 14-21 nm across various conditions. Primary particles have a relatively narrow size distribution; the standard deviation of d_p increases due to persistent inception and surface growth, reaches a maximum of 3.8 nm, and finally converges to ≈1.5 nm. Table 4.6 lists arithmetic and geometric means and their corresponding standard deviations for the primary particle diameter computed from TEM measurements and model predictions. The difference between arithmetic (d_p) and geometric ($d_{p,g}$) remains less than 0.8% across various conditions.

4.3.5 The effect of temperature

The CB induction time (τ_{ind}) is defined as the time extinction level exceeds the noise floor, and it provides a measure of the earliest detectable CB mass and avoid effects of pressure gradients at later times. The noise floor is $f_v E(m) = 2 \times 10^{-9}$, or f_v≈8 ppb (set by beam steering effects). Fig. 4.30 demonstrates the predicted τ_{ind} along with the measurements at 1064 nm and 633 nm that closely follow an Arrhenius-like behavior,

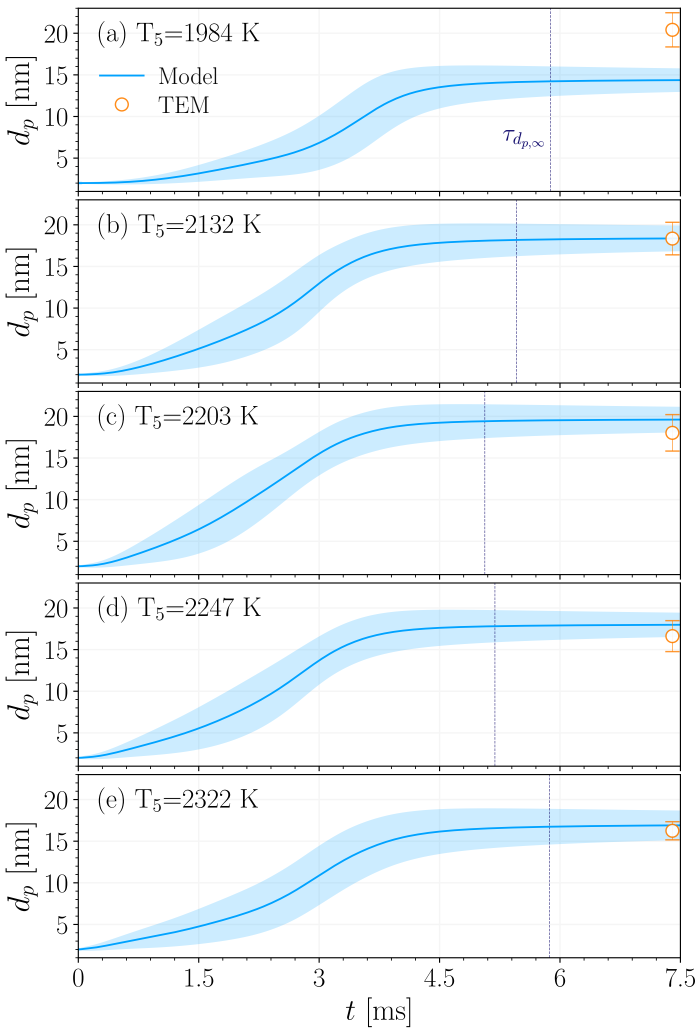


Fig. 4.29 The time histories of average primary particle diameter, d_p compared with TEM measurement. The shaded area, and error bar represents the d_p arithmetic standard deviation from the simulations and measurements, respectively. The dashed line specifies the time when d_p reaches 99% of its final value.

expressed as $\tau_{ind} = A_{ind} \exp\left(E_{ind}/RT\right)$ [215], where E_{ind} is defined as the activation energy of CB inception. The value of E_{ind} is ≈130 kJ for both wavelengths, consistent with the range suggested in the literature [194], but 1064 nm measurements result in longer induction times. τ_{ind} obtained from 633 nm measurements are in agreement with the measurements by Agafonov et al. [215] for 5% CH_4 using the same wavelength. The observed difference in τ_{ind} between two wavelengths is due to higher absorbance at 633 nm, which has been attributed to the non-negligible contribution of large PAHs to absorption at 633 nm [103]. In fact, Migliorini et al. [216] showed that the absorption

Table 4.6 The primary particle diameter measurements from TEM images. d_p is arithmetic mean, $d_{p,g}$ geometric mean, σ_p arithmetic standard deviation, and $\sigma_{p,g}$ geometic standard deviation.

	d_p [nm]		$d_{p,g}$ [nm]		σ_p [nm]		$\sigma_{p,g}$	
T_5 [K]	TEM	Model	TEM	Model	TEM	Model	TEM	Model
1984	**20.4**	14.4	**20.3**	14.3	**2.1**	1.3	**1.10**	1.10
2132	**18.3**	18.4	**18.3**	18.3	**2.0**	1.5	**1.11**	1.08
2203	**18.0**	19.6	**17.9**	19.6	**2.2**	1.5	**1.13**	1.08
2247	**16.6**	18.0	**16.5**	17.9	**1.9**	1.4	**1.12**	1.08
2322	**16.3**	16.9	**16.2**	16.8	**1.1**	1.7	**1.07**	1.11

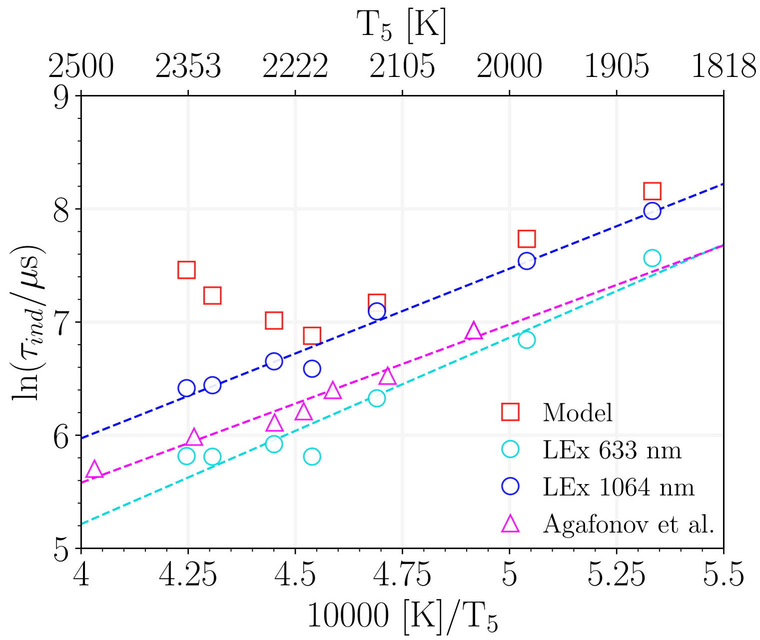


Fig. 4.30 The predicted CB induction time, τ_{ind}, compared with light extinction measurements at 633 nm and 1064 nm, and with the data from Agafonov et al. [215]. The dashed line is the Arrhenius fit to the measurements.

of gaseous species extends up to 700 nm. However, PAHs smaller than circumcoronene ($C_{54}H_{10}$) are not likely to absorb in wavelengths above 550 nm [217].

The model closely reproduces the experimental trends in the low-to-mid temperature range (1800 K-2200 K), but deviates from the measurements for T_5>2200 K. In fact, the model predicts an inverse temperature dependence for τ_{ind} relative to PAH concentration, reaching a minimum near the temperature of maximum PAH abundance (≈2200 K) and increasing at higher temperatures where PAH concentrations decline.

The carbon yield at τ_{ind} is close to 0.1% for $T_5 \geq 1984$ K. The analysis of simulations results at τ_{ind} suggests that inception accounts for 4.5%-6.7% of total CB mass, while the contribution of PAH adsorption decreases from 88% at T_5=1875 K and 1984 K to 50% at 2355 K. Therefore, the majority of carbon mass at τ_{ind} originates from precursors that exhibit a bell-shaped temperature dependence. Further analysis suggests that carbonization prolongs the induction time by suppressing HACA rates, but does not alter the temperature dependence of τ_{ind} (Fig. A.8). Additional simulations were conducted to assess the sensitivity of τ_{inc} to precursor selection, in which individual PAHs in the CRECK mechanism were treated separately as the sole precursor. All PAHs produced a similar inverted bell-shaped temperature dependence of τ_{inc}, as shown in Fig. 4.30.

As shown in Fig. 4.31, a bell-shaped temperature dependence of f_v is observed in the light extinction measurements at 633 nm and 1064 nm at 2, 3, 4, and 5 ms. As discussed earlier, larger f_v values are inferred at 633 nm across the studied T_5 range due to stronger absorption at this wavelength compared to 1064 nm. The temperature dependence of f_v is primarily governed by that of CB precursors (Fig. A.9) and evolves with time without significant changes in its overall shape, with the peak remaining consistently at T_5=2203 K. The model reproduces this temperature dependence in close agreement with the measurements. However, at 2 ms, the model noticeably underpredicts f_v for T_5>2203 K, consistent with the overestimation of induction time in this temperature range. This discrepancy may be attributed to missing chemical pathways for PAH formation at high temperatures, which would otherwise accelerate inception and PAH adsorption [218].

Fig. 4.32 reveals a significant discrepancy between the predicted and measured temperature dependence of d_p. While the TEM measurements exhibit a decreasing trend with increasing T_5, the model predicts a bell-shaped profile similar to that of f_v, with a peak at 2203 K. No data point was obtained for TEM at T_5=1875 K because the collected particles appear as compact carbonaceous lumps with poorly defined boundaries between primary particles rather than chain-like aggregates (Fig. A.13), which makes the identification of individual primary particles by visual inspection highly uncertain. Nevertheless, some identifiable structures are close to 30 nm, supporting the consistently decreasing trend in d_p inferred from the TEM measurements.

The significant underprediction of d_p at 1984 K, despite the close agreement in the predicted and measured f_v time histories (Fig. 4.27a), can be attributed to the coalescence of primary particles, which increases d_p but is not accounted for in the model. Higher temperatures promote the formation of turbostratic graphite-like order,

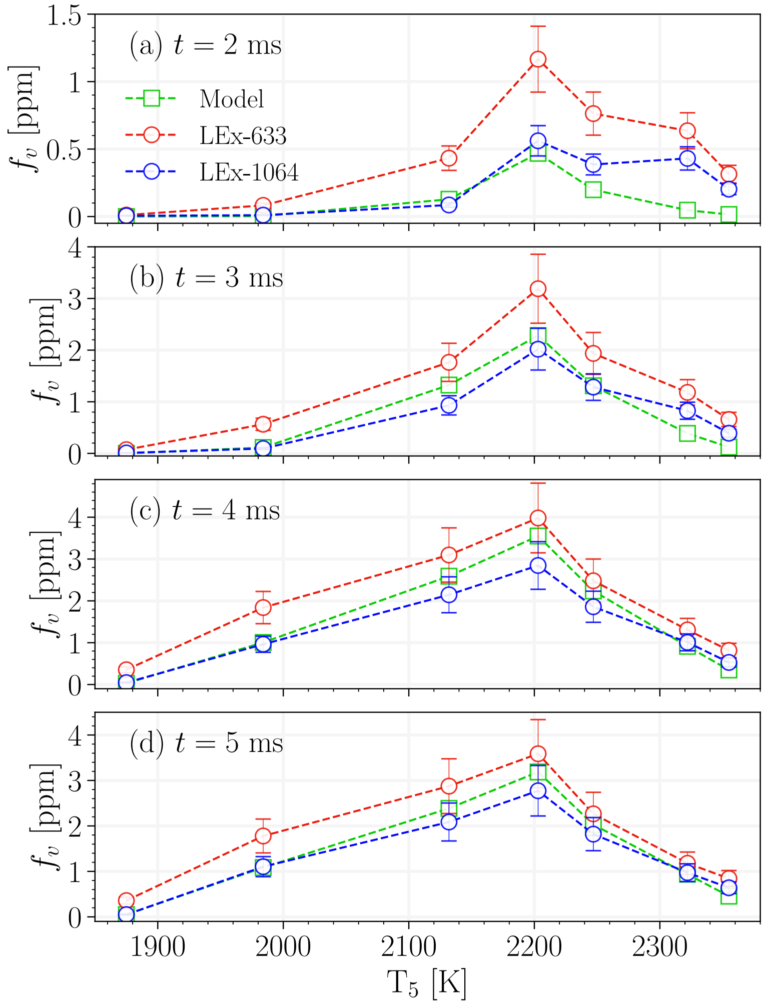


Fig. 4.31 The CB volume fraction, f_v, from light extinction at $\lambda = 633$ nm sampled at 2 ms (a), 3 ms (b), 4 ms (c), and 5 ms and compared with the simulation results for T_5 = 1875, 1984, 2132, 2203, 2247, 2355 K. Errors bars indicate experimental uncertainty due to variability in $E(m)$, and the dashed line is added to show the trends.

which inhibits complete merging of particles [219]. At the same time, coalescence alters particle surface area and therefore affects surface growth via HACA, which in turn influences f_v predictions.

In addition, condensation of residual aromatic species beyond 7.5 ms may further contribute to larger primary particle sizes. An upper limit for the increase in d_p was estimated for all conditions by assuming that (i) all residual species containing six or more carbon atoms, equivalent to benzene and larger species, are converted to CB and (ii) the total number of primary particles (N_{pri}) remains constant. The revised d_p was calculated as [195]:

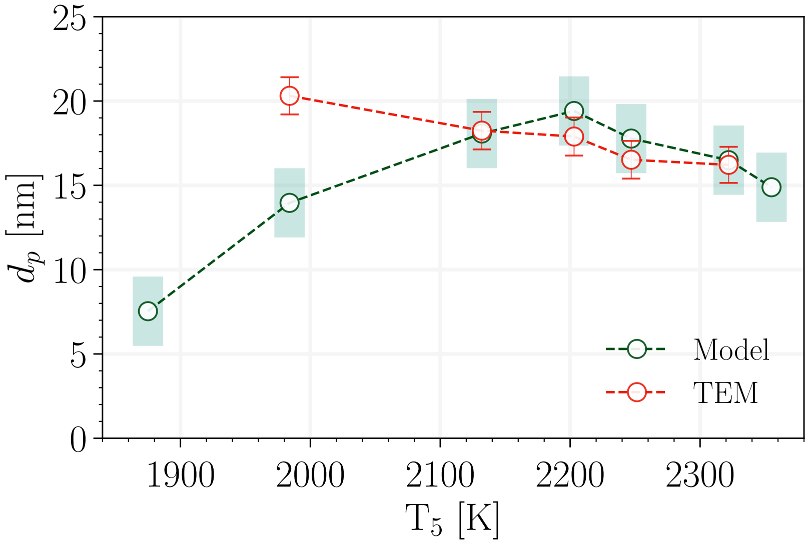


Fig. 4.32 The final primary particle diameter, d_p, from TEM measurements compared with the model predictions.

$$d_p = \left(\frac{6W_{carbon}}{\pi \rho_{soot}} \frac{C_{tot} + C_{6\geq}}{N_{pri} \cdot Av} \right)^{1/3}, \tag{4.6}$$

where C_{tot} is the total carbon mass of CB gained through surface growth mechanisms, and $C_{6\geq}$ is the total carbon mass in benzene and larger species. Under these assumptions, d_p increases to 25.6 nm at 1875 K from 7.5 nm and to 16.6 nm at 1984 K from 14 nm, whereas the increase at higher T_5 remains below 1 nm (Fig. A.10) resulting in a decreasing temperature dependence similar to the profile obtained from TEM measurements.

Fig. 4.33a depicts the maturity index, β, inferred from the extinction ratio at 633 nm and 1064 nm [211] as

$$\beta = \ln \left(\frac{E(m, \lambda_{633})}{E(m, \lambda_{1064})} \right) / \ln \left(\frac{\lambda_{1064}}{\lambda_{633}} \right). \tag{4.7}$$

The progression from high to low β with increasing time reflects the transformation of CB from disordered PAH clusters to particles with concentric graphitic layers. This structural evolution is accompanied by chemical carbonization, manifested as a decrease in the Hydrogen-to-Carbon (H/C) ratio due to dehydrogenation [220]. As shown in Fig. 4.33b, β exhibits strong qualitative agreement with the predicted H/C ratio of CB particles in both time history and temperature dependence. In all cases, both quantities show an initial rapid decline whose rate increases with T_5, followed by a slower decrease

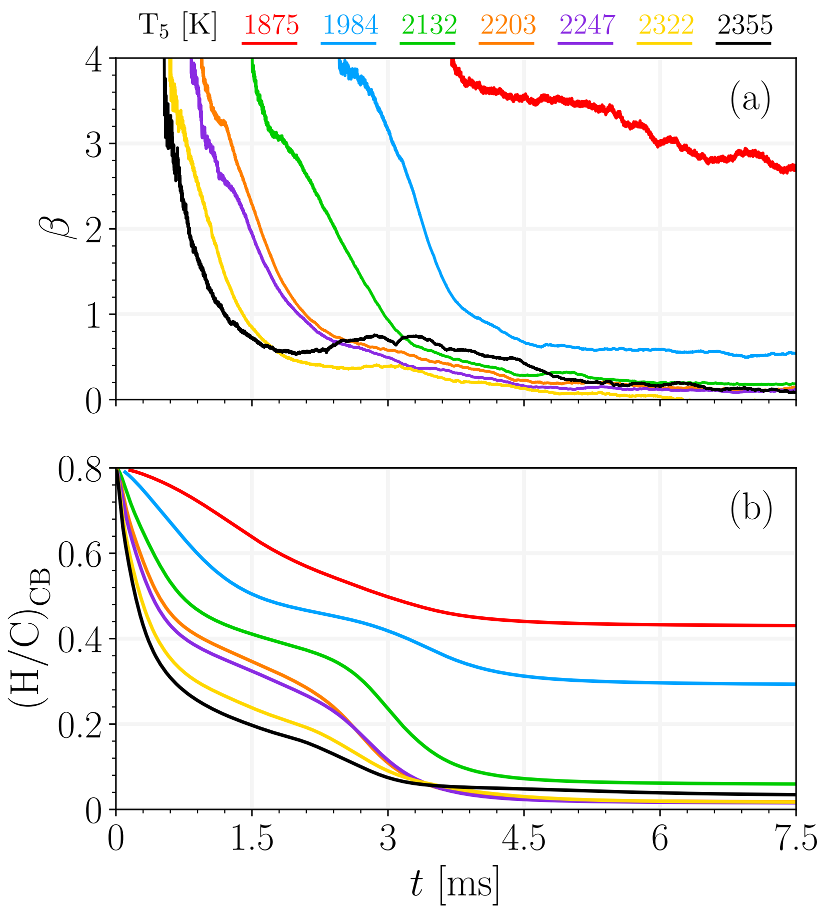


Fig. 4.33 The evolution of (a) maturity index, β, obtained from extinction measurements at 633 nm and 1064 nm, and (b) the predicted H/C ratio of CB particles at various T_5.

toward asymptotic values $\approx$0, indicating increasing CB maturity and weak spectral dependence. Notably, the β and H/C profiles at 1875 K and 1984 K converge to higher final values than those at $T_5 \geq 2132$ K, indicating incomplete carbonization and lower structural ordering under these conditions. In contrast, the higher-T_5 cases approach similar late-time values, suggesting that once a threshold temperature is exceeded, CB maturity becomes relatively insensitive to further increases in T_5 within the studied range.

4.3.6 Conclusions

The present work provides a combined numerical and experimental investigation of CB formation during shock-tube pyrolysis of 5% CH_4 in argon at a post-reflected-shock pressure of P_5=4.5±0.5 atm and temperatures spanning 1800 K<T_5<2500 K. Multi-wavelength laser diagnostics were employed to measure the time-resolved mole fractions of CH_4, C_2H_4, and C_2H_2. The CB volume fraction, f_v, was determined using light extinction at 633 nm and 1064 nm, while collected TEM images were analyzed to extract the primary particle diameter, d_p, as a quantitative measure of CB morphology. Simulations using the CRECK mechanism accurately reproduced methane conversion and acetylene yields. The gas-phase chemistry solver was coupled with a sectional population balance model that accounts for CB inception, surface growth, and the reduction in H/C due to carbonization.

Simultaneous calibration of the model using both f_v and d_p provided strong constraints on inception, PAH adsorption, HACA surface growth, and carbonization rates, highlighting the importance of morphology data in avoiding non-unique calibrations based solely on f_v. The model reproduced the time history of f_v and the final d_p in good agreement with the measurements. Four stages of CB evolution were identified: (I) an induction period dominated by precursor formation, (II) rapid increase in f_v driven by rising surface growth rates, (III) transition toward final yield, composition, and d_p as surface growth wanes, and (IV) a coagulation-dominated stage in which primary particle diameters are chemically frozen while the agglomerate mobility diameter continues to increase.

The analysis showed that PAH adsorption dominates CB mass growth at lower temperatures and early times, whereas HACA becomes the primary contributor with increasing T_5. The model reproduced the bell-shaped temperature dependence of f_v but overpredicted induction times at $T_5>2200$ K, suggesting the presence of missing high-temperature PAH formation pathways that would accelerate inception and PAH adsorption. TEM measurements revealed a decreasing trend in primary particle diameter with increasing T_5, while the model predicted a bell-shaped dependence. The discrepancy at 1984 K was attributed to possible late-time condensation of residual aromatics.

The maturity index, β, inferred from the extinction ratio at 633 nm and 1064 nm, exhibited strong similarities to the predicted H/C ratio of CB particles, which is used in the model as a measure of surface reactivity. Both β and H/C decrease rapidly at early times before approaching near-asymptotic values, with the rate of the initial decline increasing with T_5.

Chapter 5

Conclusions and Future Work

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Appendix A

Supplementary results

This chapter includes the supplementary results mentioned in the thesis but are the central part of discussions.

A.1 Fundamental analysis of PAH growth models

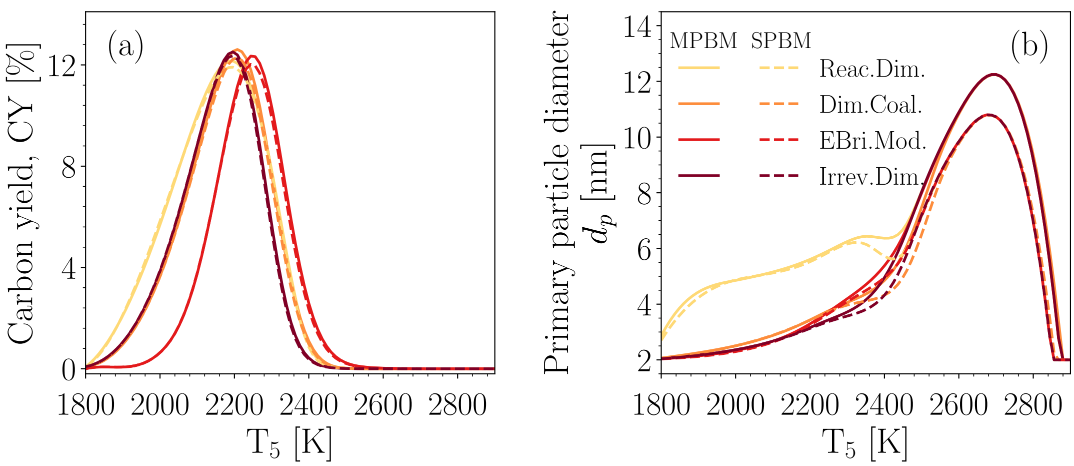

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Fig. A.1 The comparison of CY (a) and primary particle diameter, d_p, at t =1.5 ms obtained using MPBM and SPBM models for the case optimized using equal adjustment factors to minimize the prediction error.

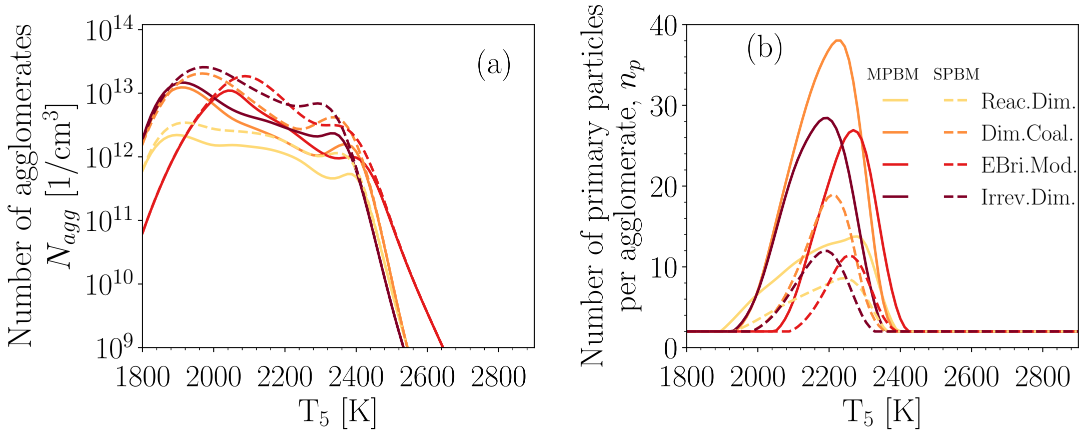

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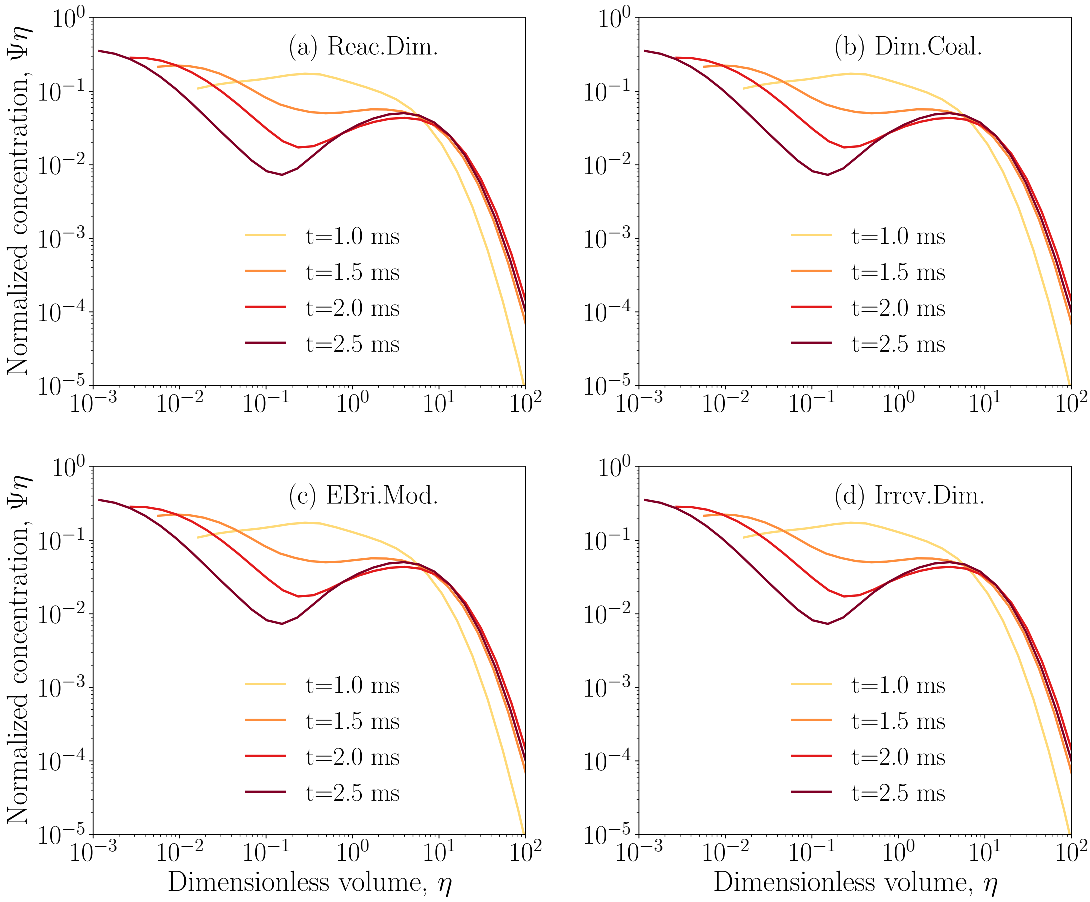
Fig. A.2 The temperature dependence of total number of agglomerates, N_{agg} (a), number of primary particles per agglomerate, n_p (b), at t =1.5 ms obtained using MPBM and SPBM models for the case optimized using equal adjustment factors to minimize the prediction error.



;

Fig. A.3 The non-dimensional particle size distribution during 5%CH_4-Ar at T_5 = 2200 K that evolves from 1 ms to 2.5 ms indicating SPSD is not attained yet.

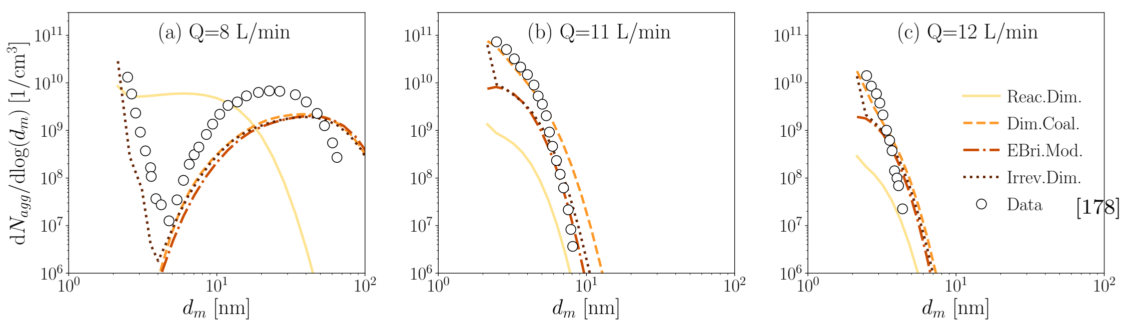


Fig. A.4 The particle size distribution at the end of PFR for Q = 8 (a), 11 (b), and 12 L/min (c) obtained using Caltech mechanism, SPBM and different inception models calibrated to match the predictions with measurement [178].

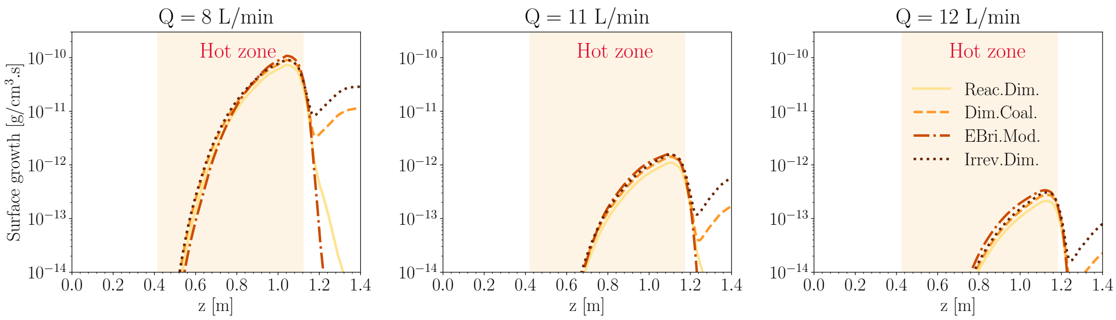


Fig. A.5 The surface growth by HACA and PAH adsorption combined along the PFR for Q = 8 (a), 11 (b), and 12 L/min (c) obtained using KAUST mechanism, SPBM and different inception models. The yellow area represents the hot zone ($T > 0.9T_{max}$).

A.2 Identifying stages of carbon black formation

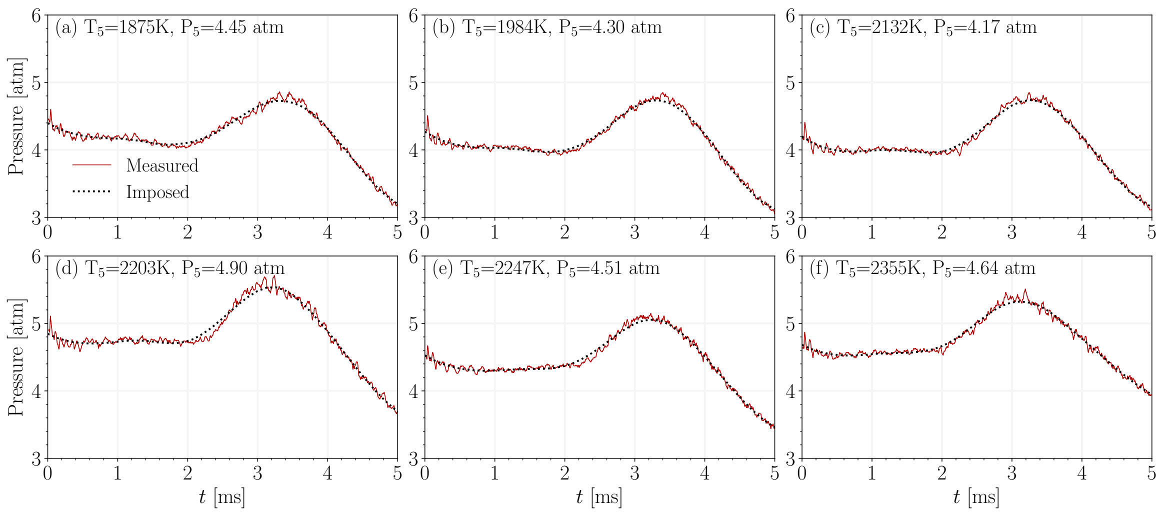


Fig. A.6 The measured pressure trace and the filtered profile imposed in the model for the entire temperature range

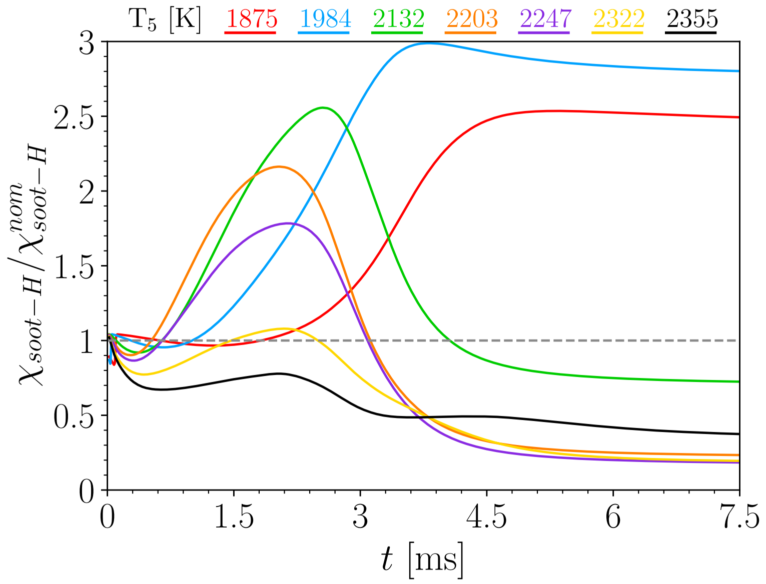


Fig. A.7 The number of hydrogenated sites, $\chi_{soot\text{-}H}=H_{tot}/A_{tot}$ normalized by its nominal value of $\chi^{nom}_{soot\text{-}H} = 2.3\times10^{19}$ sites$/m^2$.

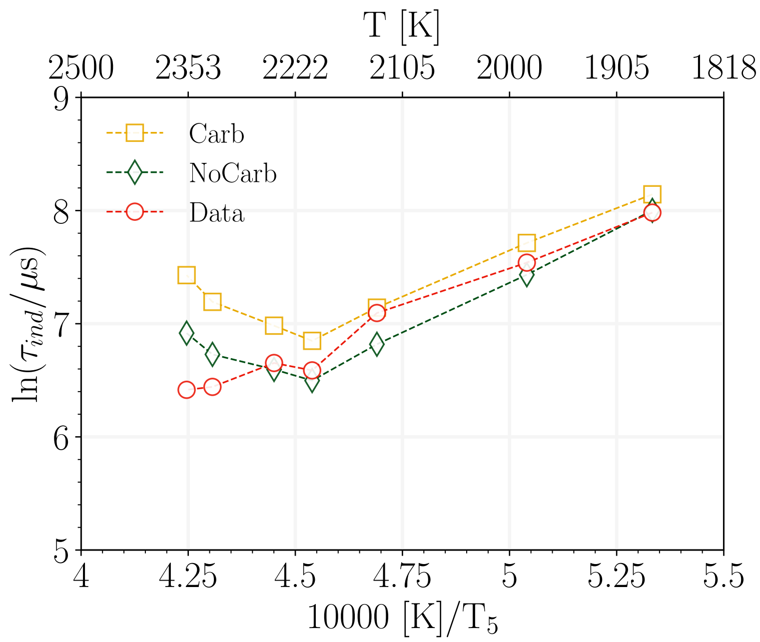


Fig. A.8 The effect on carbonization on CB induction time.

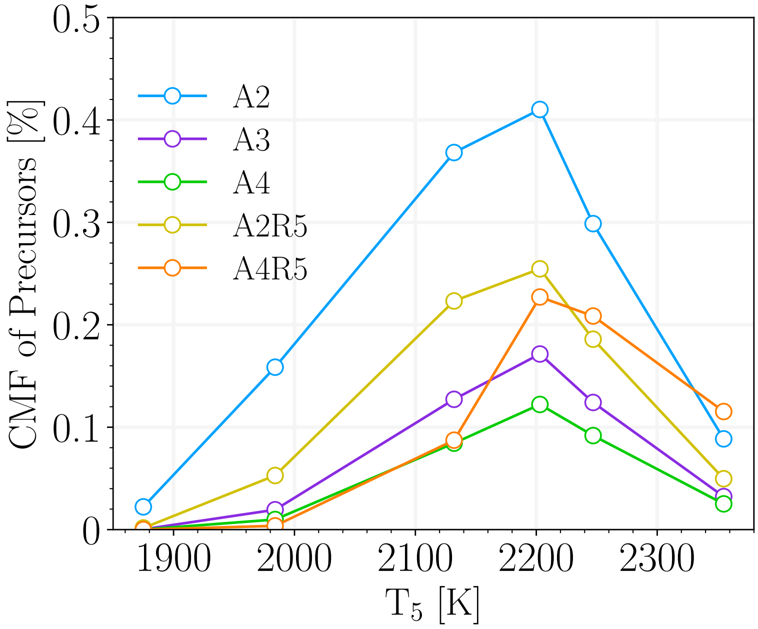


Fig. A.9 The carbon mass fraction of CB precursors at 0.5 ms

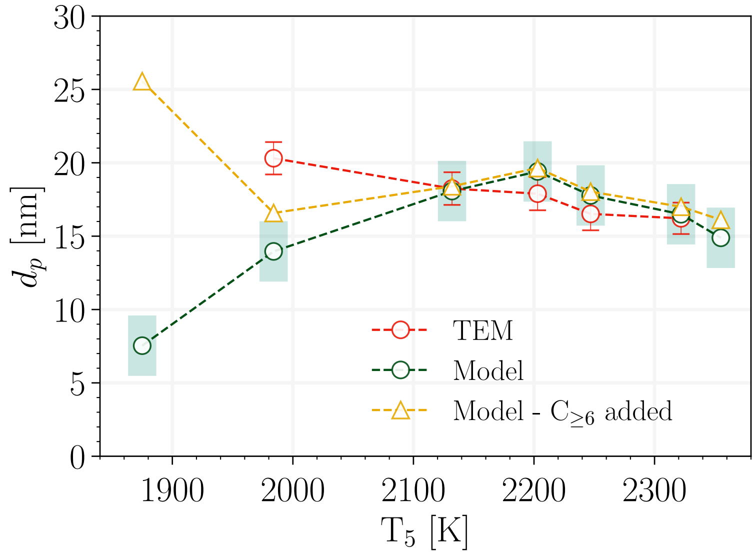


Fig. A.10 The final primary particle diameter, d_p, considering the additional carbon mass from condensation of benzene and larger species on CB compared with d_p from TEM measurements and the model predictions at 7.5 ms.

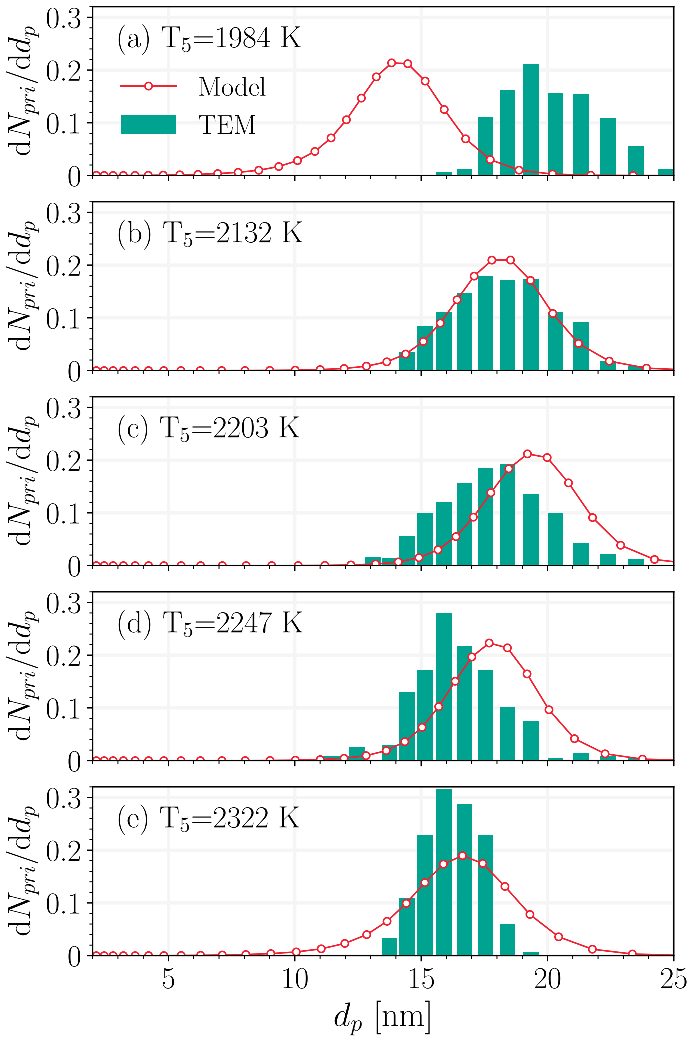


Fig. A.11 The size distribution of primary particles from TEM measurements compared with the model predictions.

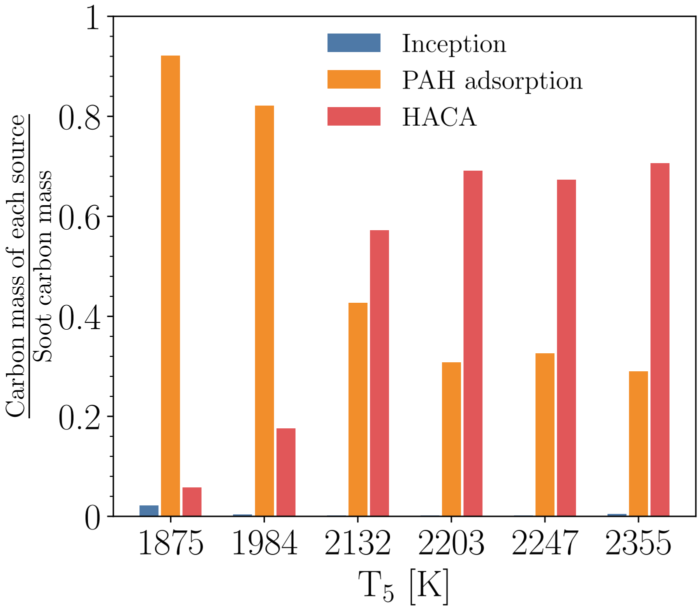


Fig. A.12 The CB mass from inception, PAH adsorption and HACA normalized by total carbon mass at t=5 ms

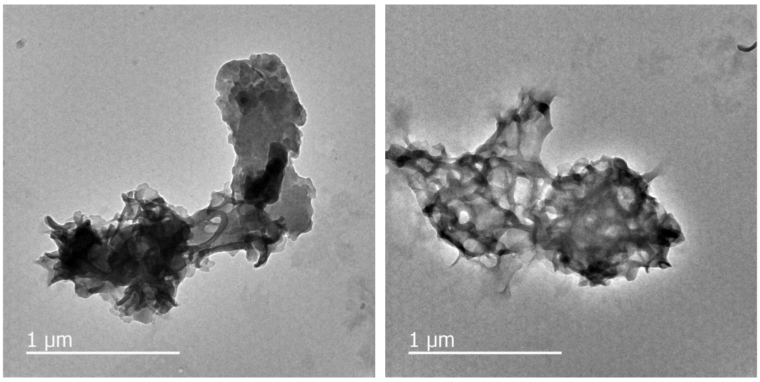


Fig. A.13 Samples of TEM images collected for T_5=1875 K that shows a compact carbonaceous lump with not clear boundaries between primary particles.

A.3 Soot formation at high fuel loadings

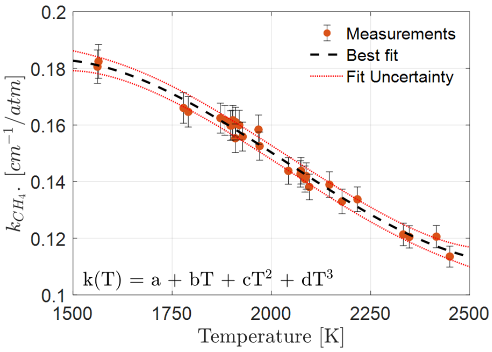


Fig. A.14 Measurements of temperature-dependent CH4 absorption coefficient near 3.366 μm. The fit coefficients are a = -0.3845, b = 9.618×10^{-4}, c = -5.169×10^{-7}, d = 8.467×10^{-11} for T in K

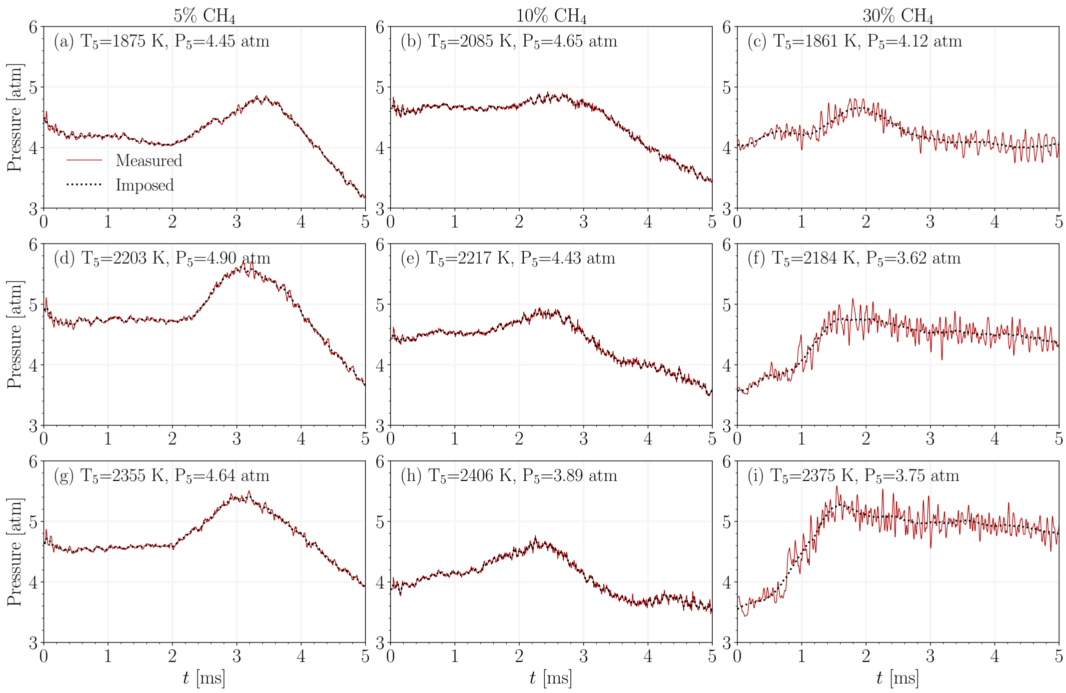


Fig. A.15 The measured pressure trace and the filtered profile imposed in the model for (a, d, g) 5%, (b, c, h) 10%, and 30% CH_4 pyrolysis at $T_5\approx$ 1900, 2200, and 2400 K.

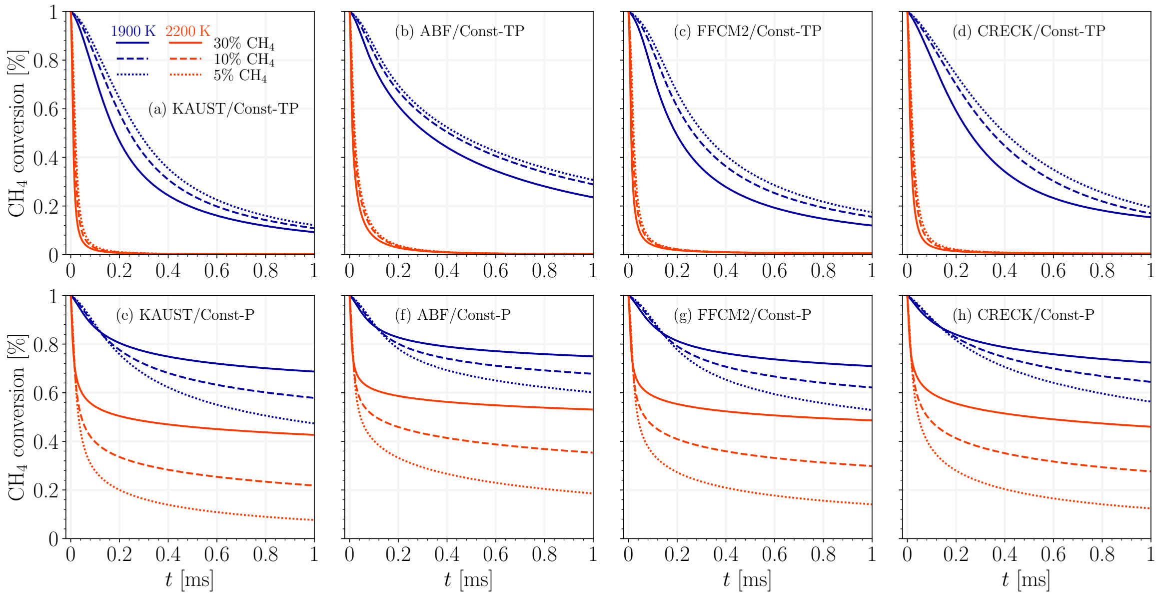


Fig. A.16 The CH_4 conversion predicted using various kinetic models in Const-TP and Const-P simulations.

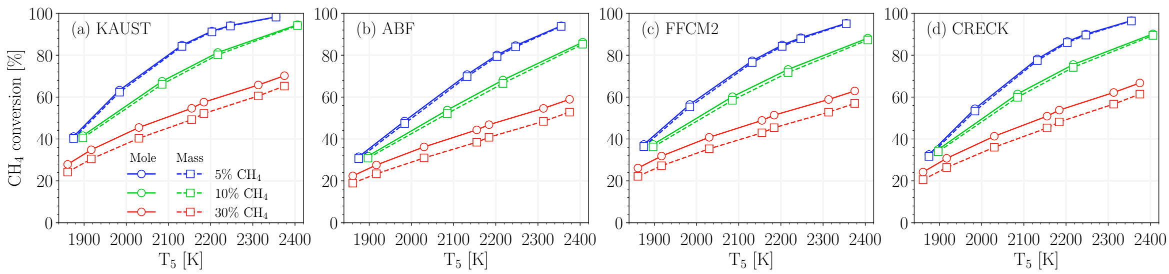


Fig. A.17 The CH_4 conversion at 1 ms computed using the method based on normalized mole fraction (circles) and carbon mass fraction (squares) predicted using various kinetic models.

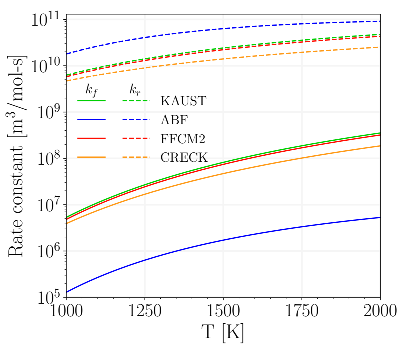


Fig. A.18 The comparison of forward and reverse rate constant of $C_2H_2 + CH_3 \leftrightarrow$ P-$C_3H_4 + H$.

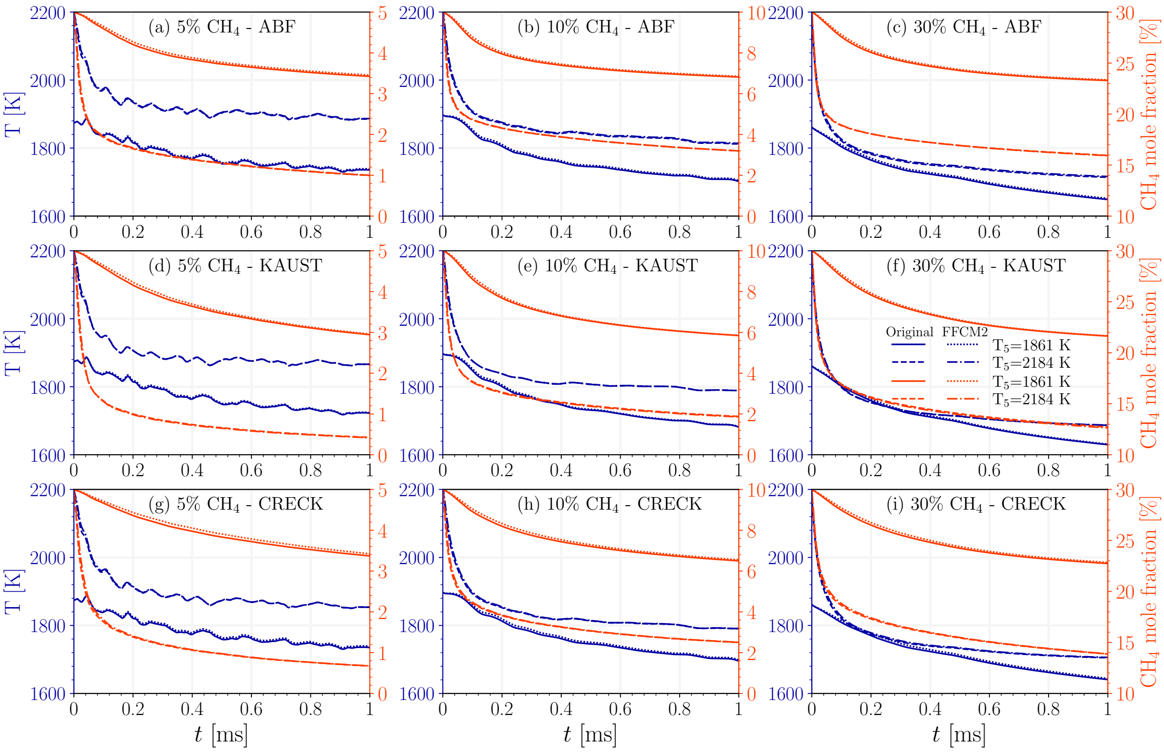


Fig. A.19 The comparison of temperature and CH_4 mole fraction predicted using ABF, KAUST and CRECK using original rates (dashed line) and the modified (dotted) rates for methane decomposition reactions, $CH_4 + (M) \longleftrightarrow CH_3 + H$ and $CH_4 + H \leftrightarrow CH_3 + H_2$. The modified rates were taken from FFCM2.

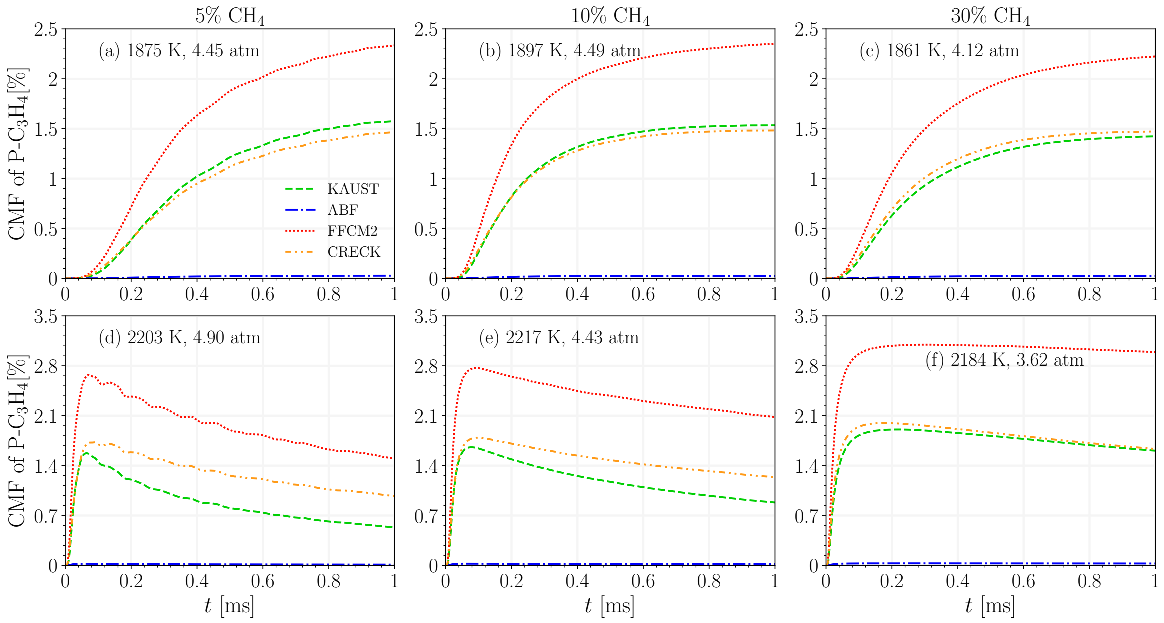


Fig. A.20 The carbon mass fraction of propyne (P-C_3H_4) predicted using various kinetic models for 5%, 10% and 30% CH_4 at T_5≈1900 K and 2200 K.

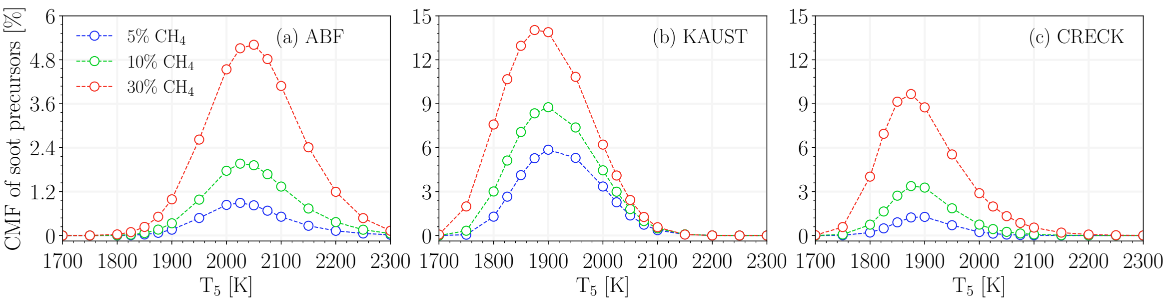


Fig. A.21 The CMF of soot precursors combined at 1 ms over 1700 K<T_5<2300 K predicted using (a) ABF, (b) KAUST, and (c) CRECK for 5%, 10%, and 30% loadings.

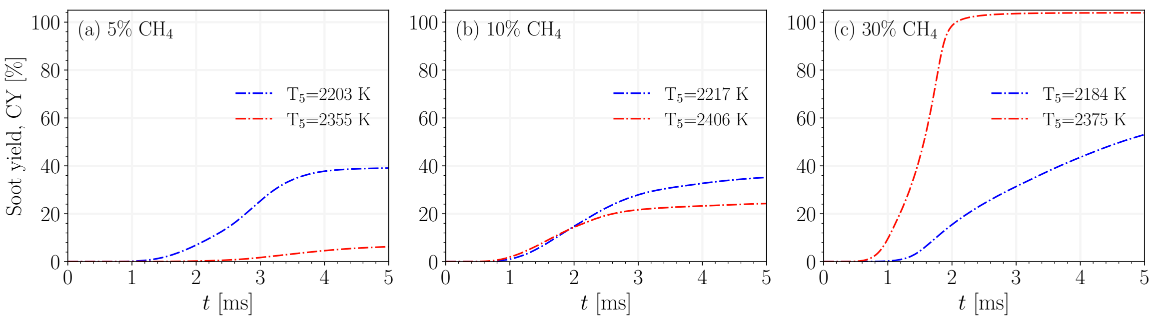


Fig. A.22 The time history of soot yield, CY, for (a) 5%, (b) 10%, and (c) 30% CH_4 pyrolysis predicted using CRECK at T_5≈2200 K and 2400 K.

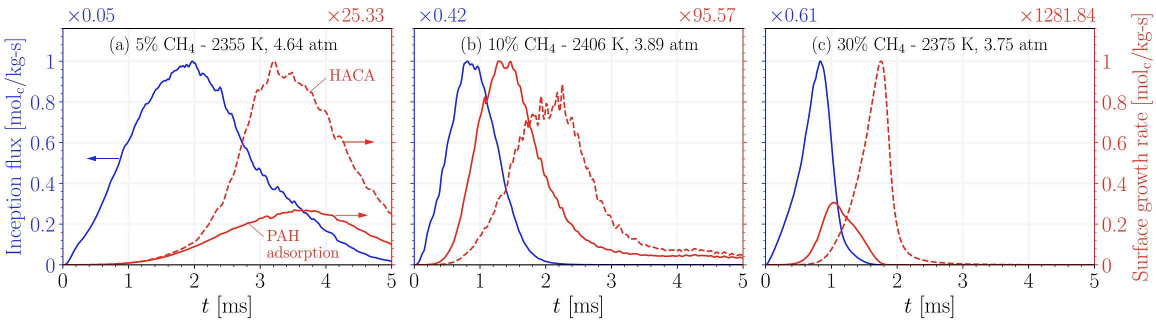


Fig. A.23 The soot inception flux and surface growth rates via PAH adsorption and HACA (a) 5%, (b) 10%, and (c) 30% CH_4 pyrolysis predicted using CRECK at $T_5 \approx 2400$ K.